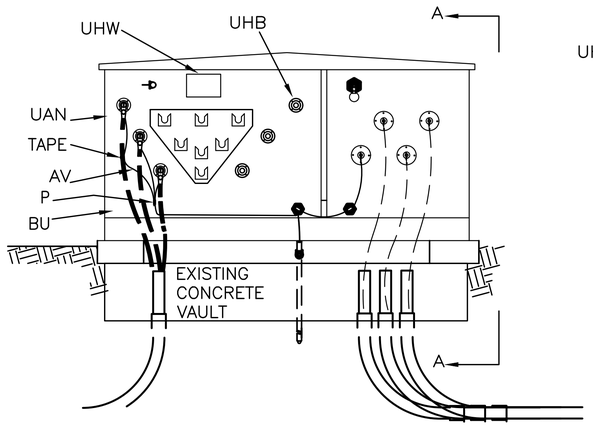
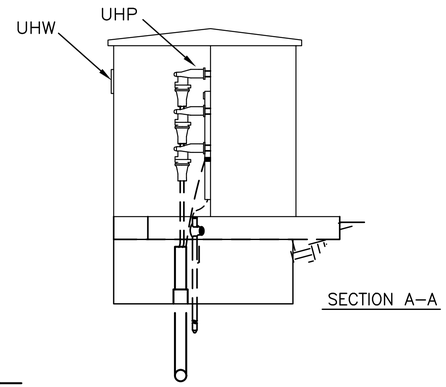




SECTION A–A

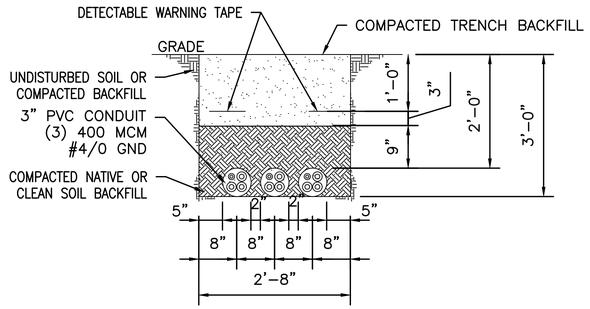
ITEM	QTY.	MATERIAL
P		CONNECTORS, AS REQUIRED
AV		JUMPERS, COPPER, AS REQUIRED
BU	2	CONNECTOR, EQUIPMENT GROUND
UAN	1	TRANSFORMER, PAD MOUNTED, THREE PHASE,
		WITH TWO LOAD BREAK BUSHINGS PER PHASE
		AND INTERNAL FUSE.
UHB	3	INSULATED COVERS
UHW	2	SIGNS, "DANGER" AND "WARNING"
UHP	3	ELBOW TERMINATION
		TAPE, AS REQUIRED

NOTES:

1. INSTALL "DANGER" SIGN ON TRANSFORMER INSIDE ENCLOSURE. INSTALL "WARNING " SIGN ON OUTSIDE SURFACE OF ENCLOSURE.

1 / 051E102 THREE–PHASE, LOOP FEED, PAD MOUNTED TRANSFORMER

NOT TO SCALE



NOTES:

1. TRENCH BACKFILL SHALL BE FREE FROM ROCK, GRAVEL, OR OTHER MATERIAL WHICH MIGHT DAMAGE THE CABLE JACKET. CLEANED SOIL BACKFILL WHEN USED SHALL CONTAIN NO SOLID MATERIAL LARGER THAN 1 INCH. THIS SOIL LAYER SHALL BE CAREFULLY COMPACTED SO THAT THE CABLE WILL NOT BE DAMAGE.

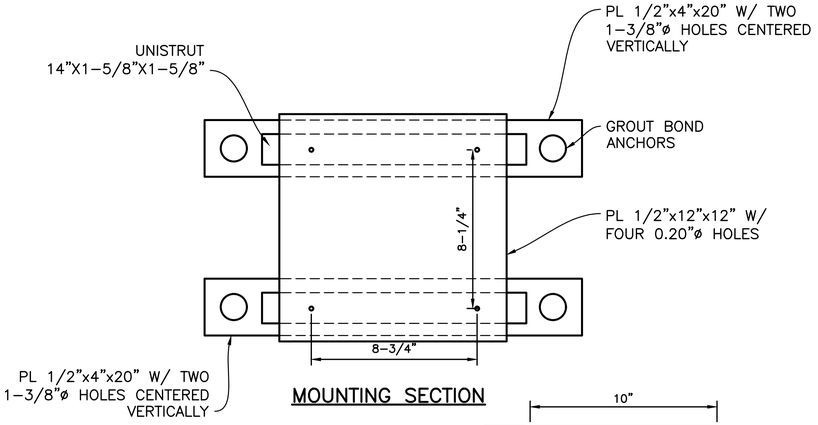
2 / 051E102 TYPICAL LOW–VOLTAGE ELECTRICAL TRENCH DETAIL

NOT TO SCALE

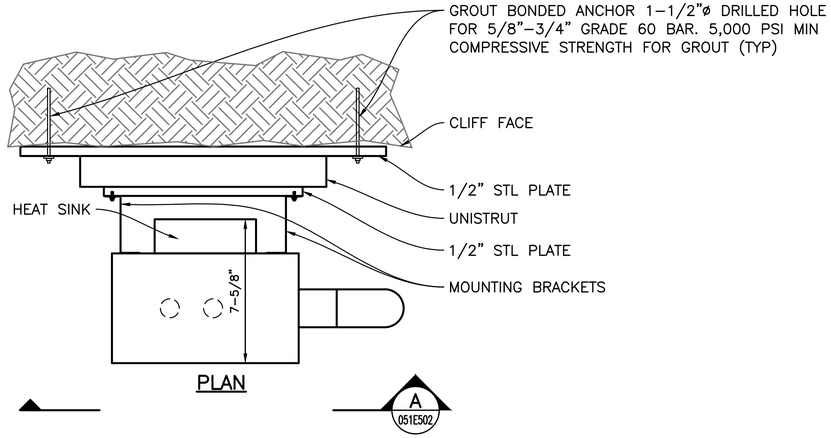
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 1/14/2026 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	T. VANG	051 E501	SITEWORK TRANSFORMER AND TRENCH DETAIL	113 182752
	CADD R.D. CLARK			PMIS/PKG NO. 318719
	TECH. REVIEW: P. DAVILA		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 31 OF 113
	DATE: JUNE 2026			

10/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS_ELEC\051 E501.dwg

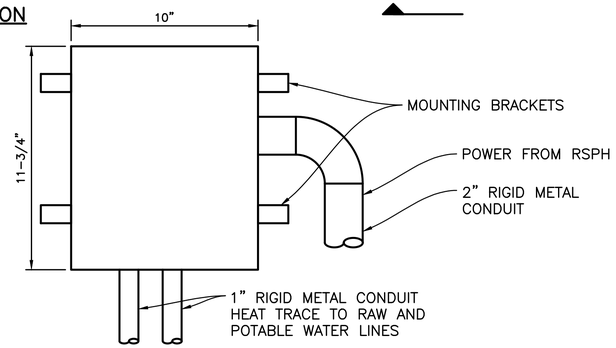


MOUNTING SECTION



PLAN

A
051E502



A
051E502
SECTION
NOT TO SCALE

1
051E502
HEAT TRACE CONTROL PANEL
NOT TO SCALE

GENERAL NOTES:

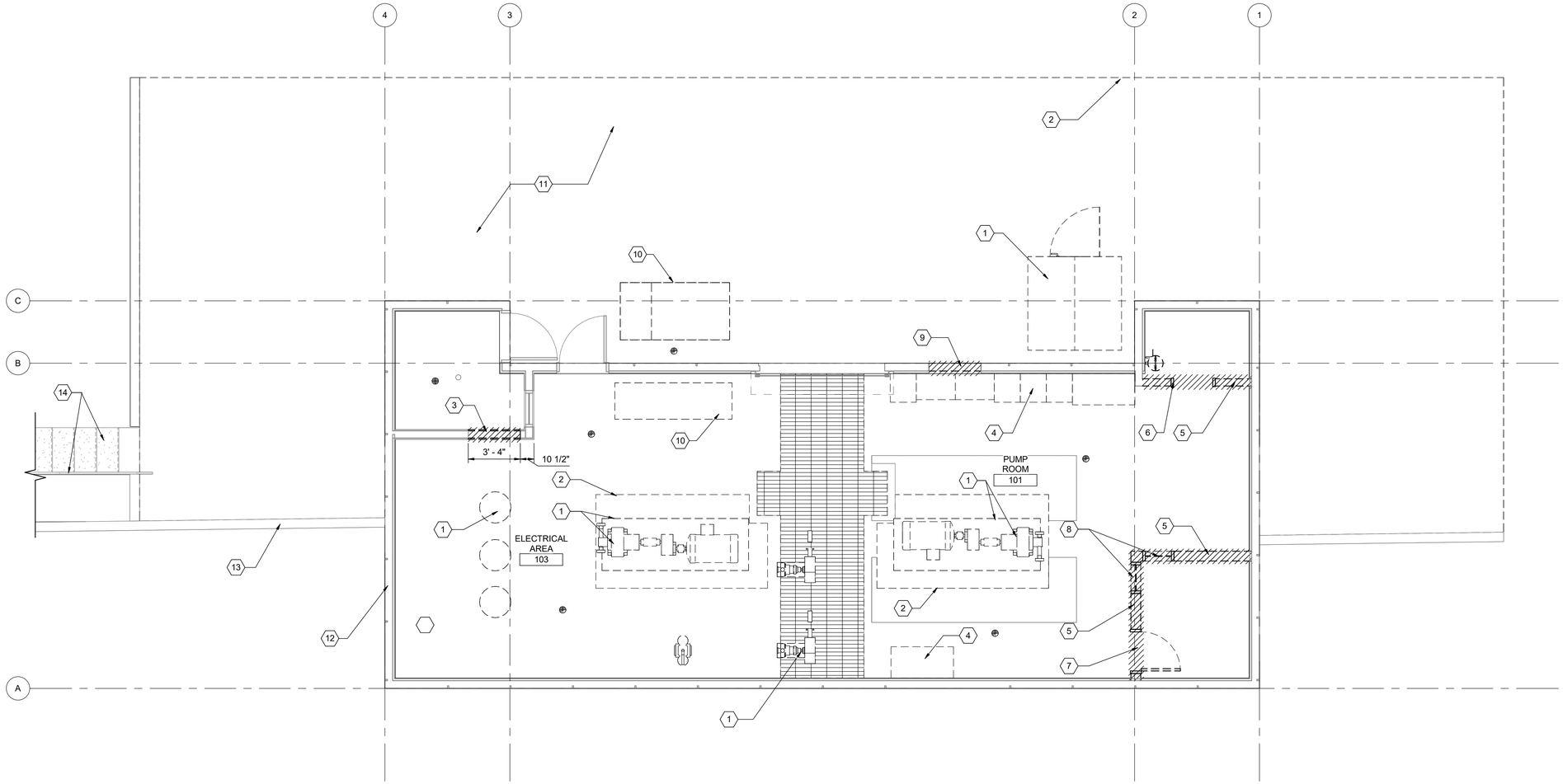
A. HEAT TRACE AND INSULATION SHALL BE INSTALLED ON EXPOSED PIPES FROM BOREHOLE EXITS TO THE POINT PIPES TRANSITION UNDERGROUND. APPROXIMATE EXTENT SHOWN BY HATCHING ON ELECTRICAL SITE PLAN 051-E101. REFER ALSO TO PIPING PLAN AND PROFILE ON 051-C104 AND SPECIFICATION SECTION 40 41 13.

B. DETAILS ON THIS SHEET INDICATE MOUNTING OF THE HEAT TRACE CONTROL PANEL NEAR THE BOREHOLE EXITS.

C. HEAT TRACE SYSTEM IS POWERED FROM ROARING SPRINGS PUMPHOUSE.

8/12/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10414450_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\051-E502.dwg

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD R.D. CLARK TECH. REVIEW: P. DAVILA DATE: JUNE 2026	SUB SHEET NO. 051 E502	TITLE OF SHEET SITEWORK ELECTRICAL DETAILS HEAT TRACE PANEL NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 32 OF 113



1 DEMOLITION FLOOR PLAN
052X101

GENERAL NOTES:

A. PROTECT WALLS, CEILINGS, FLOORS, AND OTHER EXISTING FINISH WORK THAT ARE TO REMAIN AND ARE EXPOSED DURING DEMOLITION.

B. CONTRACTOR TO PATCH AND REPAIR HOLES AND DAMAGED SURFACES CAUSED TO ADJACENT CONSTRUCTION BY DEMOLITION. WHERE REPAIRS TO EXISTING SURFACES ARE REQUIRED, PATCH TO PRODUCE SURFACES SUITABLE FOR NEW MATERIALS.

C. CONTRACTOR TO TRANSPORT DEMOLISHED MATERIALS OFF OWNER'S PROPERTY AND LEGALLY DISPOSE OF THEM.

D. ALL DEMOLITION WORK IS TO BE COORDINATED WITH NEW WORK. IF UNDOCUMENTED CONDITIONS EXIST THAT ARE AFFECTED BY NOTED DEMOLITION, CONTACT THE CONTRACTING OFFICER BEFORE PROCEEDING.

KEYNOTES: (#)

REFER TO PROCESS DEMOLITION DRAWINGS FOR EXTENTS OF PROCESS DEMOLITION.

1 REFER TO PROCESS DEMOLITION DRAWINGS FOR EXTENTS OF PROCESS DEMOLITION.

2 REFER TO STRUCTURAL DEMOLITION DRAWINGS FOR EXTENTS OF STRUCTURAL DEMOLITION.

3 DEMOLISH WALL AND PREPARE OPENING FOR NEW DOOR AND FRAME. SALVAGE ALL UNDAMAGED SOUND ATTENUATION PANELS FOR REUSE.

4 REFER TO ELECTRICAL DEMOLITION DRAWINGS FOR EXTENTS OF ELECTRICAL DEMOLITION.

5 DEMOLISH EXISTING WALL. SALVAGE ALL UNDAMAGED SOUND ATTENUATION PANELS FOR REUSE.

6 DEMOLISH EXISTING FRAME.

7 DEMOLISH EXISTING DOOR AND FRAME.

8 DEMOLISH EXISTING INTERIOR WINDOW.

9 CUT EXISTING PRECAST WALL AND PREPARE OPENING FOR NEW DOOR AND FRAME. SALVAGE ALL UNDAMAGED SOUND ATTENUATION PANELS FOR REUSE.

10 REFER TO MECHANICAL & PLUMBING DEMOLITION DRAWINGS FOR EXTENTS OF MECHANICAL AND PLUMBING DEMOLITION.

11 SEE STRUCTURAL FOR SLAB REPLACEMENT.

12 EXISTING PRECAST WALL TO REMAIN.

13 DEMOLISH RETAINING WALL; SEE STRUCTURAL

14 EXISTING STAIR AND HANDRAIL TO REMAIN.

LEGEND:

//// DEMO

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.16 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: E. BUTTMAN TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 X101	TITLE OF SHEET DEMOLITION RSPH ARCHITECTURAL DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 33 OF 113

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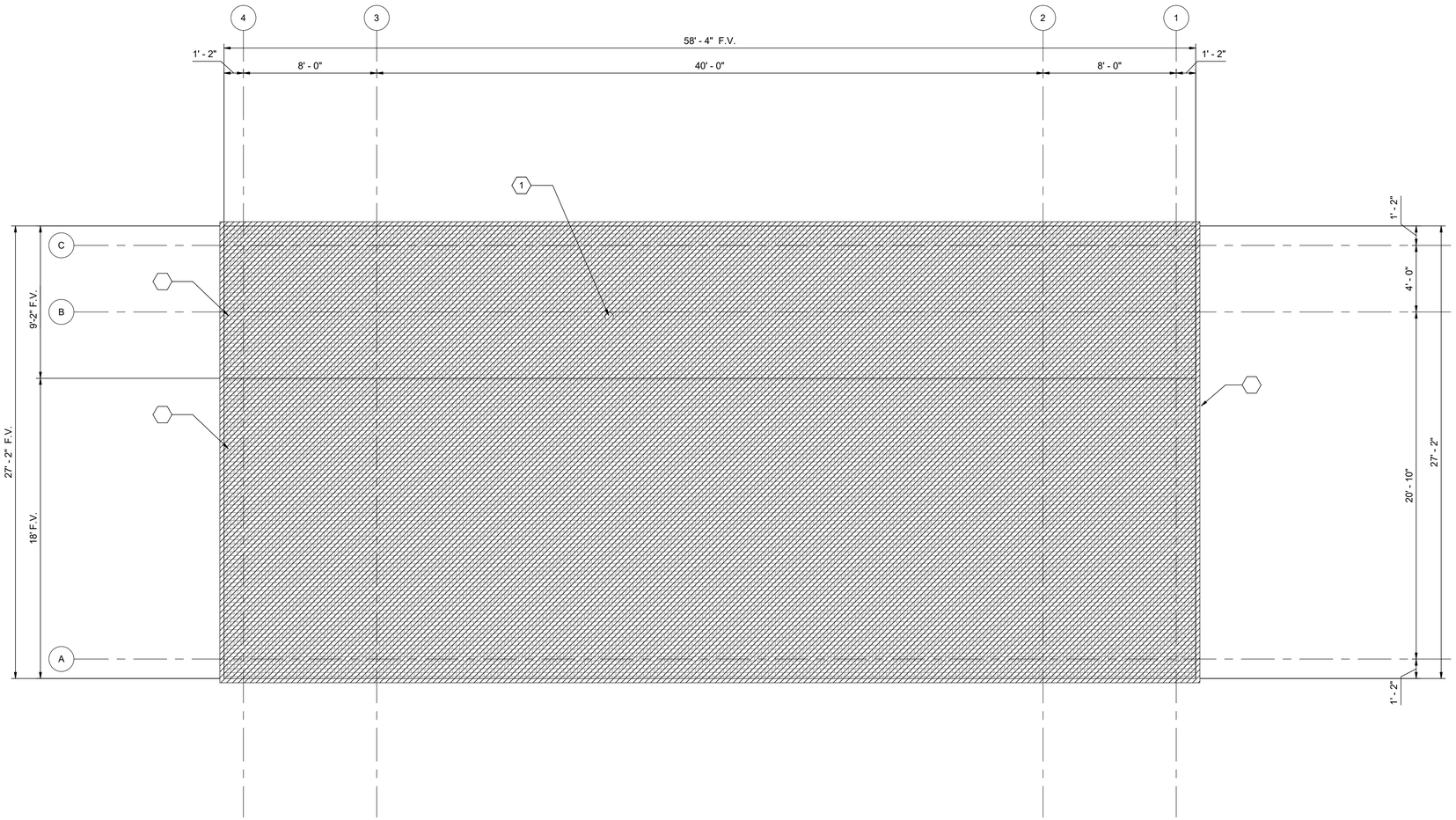

1 DEMOLITION ROOF PLAN
052X102

GENERAL NOTES:

A. PROTECT WALLS, CEILINGS, FLOORS, AND OTHER EXISTING FINISH WORK THAT ARE TO REMAIN AND ARE EXPOSED DURING DEMOLITION.

B. CONTRACTOR TO PATCH AND REPAIR HOLES AND DAMAGED SURFACES CAUSED TO ADJACENT CONSTRUCTION BY DEMOLITION. WHERE REPAIRS TO EXISTING SURFACES ARE REQUIRED, PATCH TO PRODUCE SURFACES SUITABLE FOR NEW MATERIALS.

C. CONTRACTOR TO TRANSPORT DEMOLISHED MATERIALS OFF OWNER'S PROPERTY AND LEGALLY DISPOSE OF THEM.

D. ALL DEMOLITION WORK IS TO BE COORDINATED WITH NEW WORK. IF UNDOCUMENTED CONDITIONS EXIST THAT ARE AFFECTED BY NOTED DEMOLITION, CONTACT THE ARCHITECT/ENGINEER BEFORE PROCEEDING.

E. SALVAGE EXISTING RADIO ANTENNAS AND TURN OVER TO CONTRACTING OFFICER'S REPRESENTATIVE.

KEYNOTES: (#)

<varies>

1 REFER TO MECHANICAL & PLUMBING DEMOLITION DRAWINGS FOR EXTENTS OF MECHANICAL AND PLUMBING DEMOLITION.

LEGEND:

//// DEMO

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 4/19/2026 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	E. BUTTMAN	052 X102	DEMOLITION RSPH ARCHITECTURAL ROOF DEMOLITION PLAN	113 182752
	CADD: E. BUTTMAN			PMIS/PKG NO. 318719
	TECH. REVIEW: S. HEANEY			SHEET 34 OF 113
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_A.rvt
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GENERAL NOTES

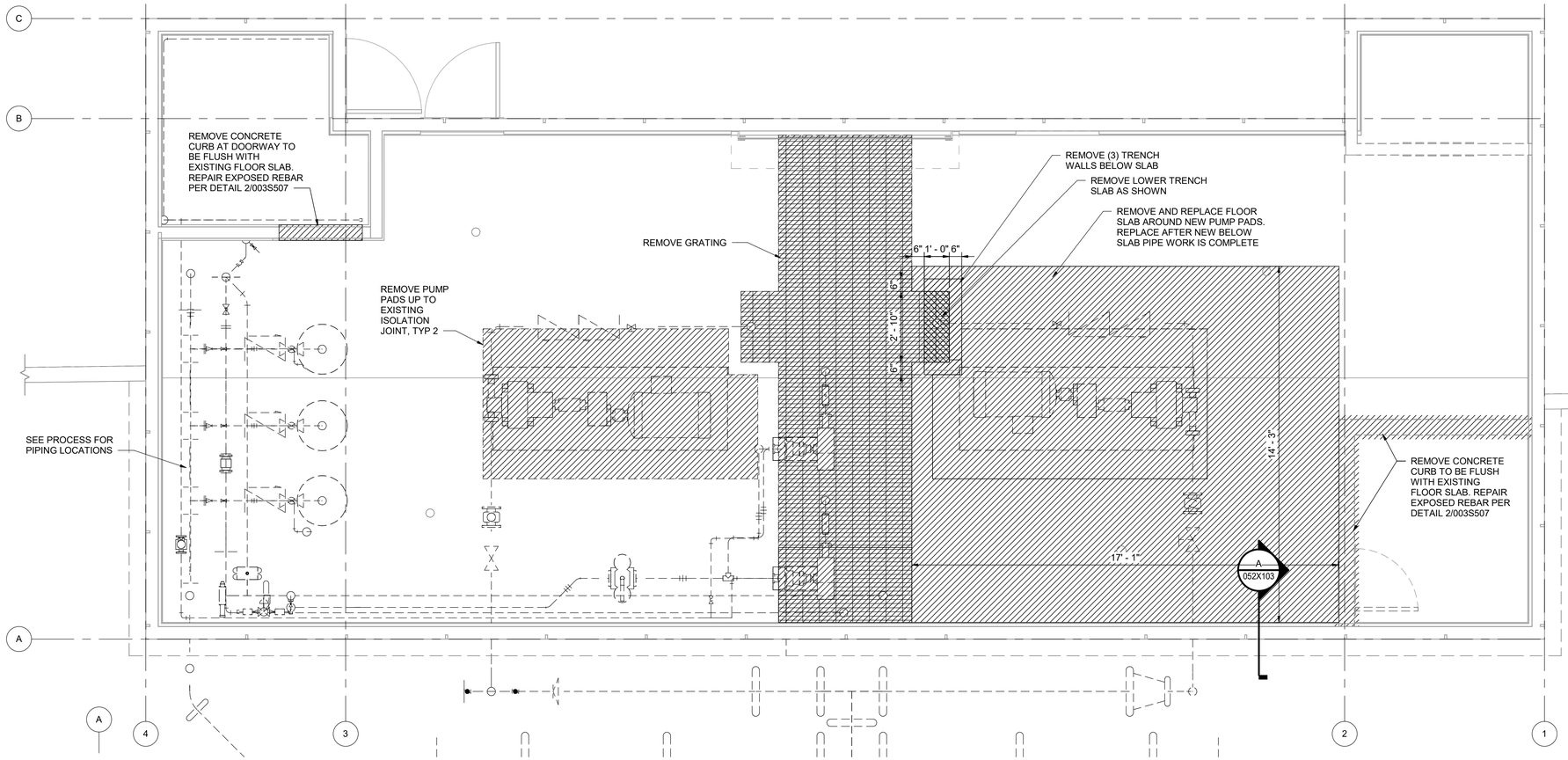
A. AT ALL CONCRETE SURFACES ADJUST TO CONCRETE DEMOLITION, REPAIR CONCRETE SURFACE WITH SIKA-REPAIR 223 OR SIMILAR AND APPLY SIKA ARMATEC 110 OR SIMILAR AT ALL EXPOSED REBAR ENCOUNTERED.

B. SEE DISCIPLINE SPECIFIC DEMOLITION SHEETS FOR EXTENTS OF ARCHITECTURAL / PROCESS / MECHANICAL / ELECTRICAL DEMOLITION.

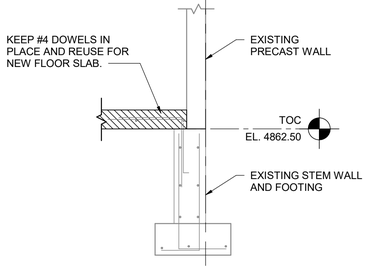
C. ALL DEMOLITION WORK IS TO BE COORDINATED WITH NEW WORK. IF UNDOCUMENTED CONDITIONS EXIST THAT ARE AFFECTED BY NOTED DEMOLITION, CONTACT THE CONTRACTING OFFICER BEFORE PROCEEDING.

LEGEND:

//// DEMO



1 052X103 DEMOLITION PLAN - STRUCTURAL
SCALE (A)



A 052X103 SLAB DEMOLITION AT WALL SECTION
SCALE (B)

SCALE (B) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JOSE CORONADO Date Signed 4/6/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 X103	TITLE OF SHEET DEMOLITION RSPH DEMOLITION PLAN STRUCTURAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 35 OF 113

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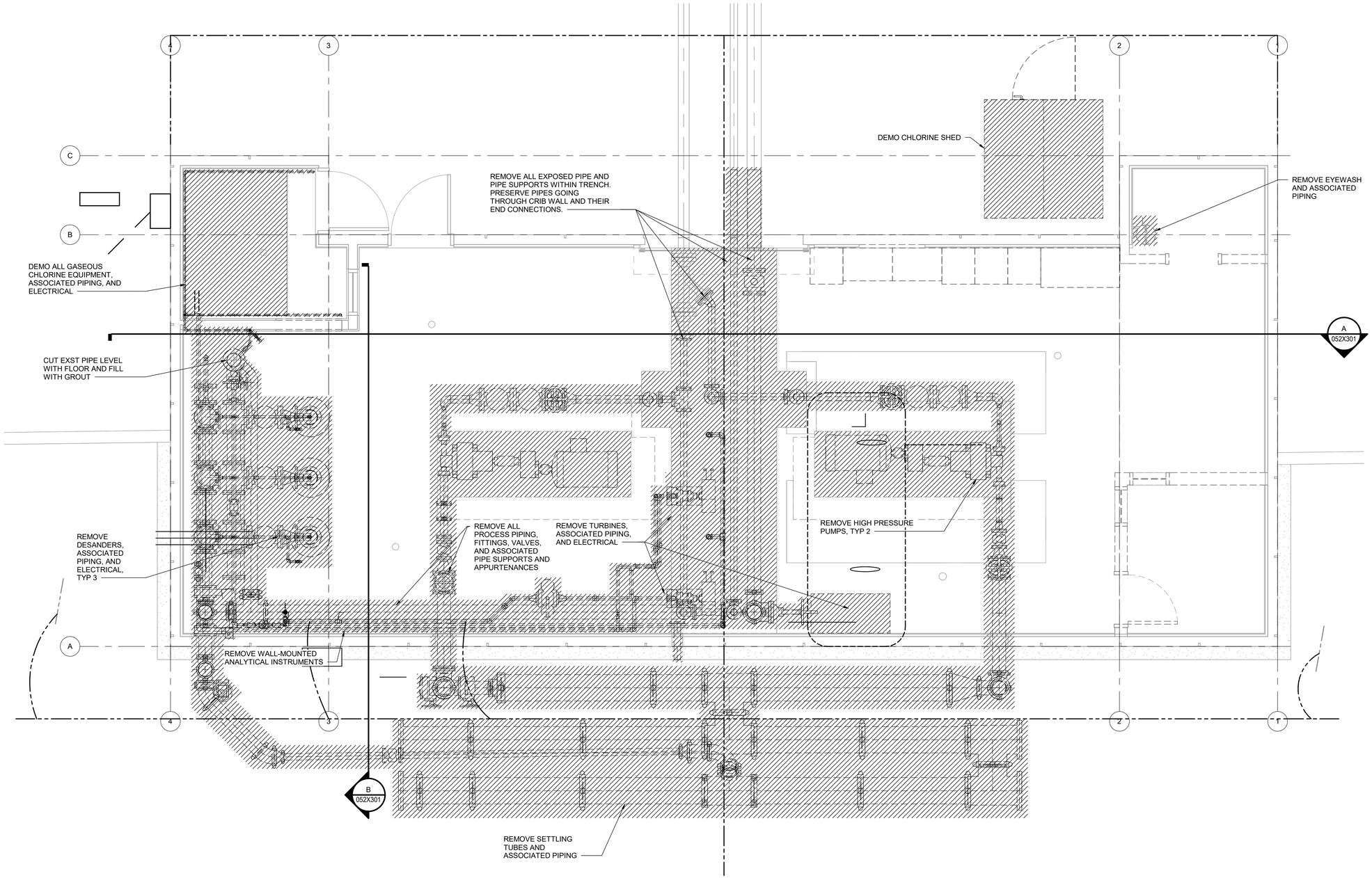

LEGEND:

//// DEMO

1 / 052X102 DEMOLITION PLAN

SCALE A 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAFTED: P. VELEZ TECH. REVIEW: R. MCCANN DATE: MAY 2026	SUB SHEET NO. 052 X104	TITLE OF SHEET DEMOLITION RSPH DEMOLITION PLAN PROCESS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 36 OF 113

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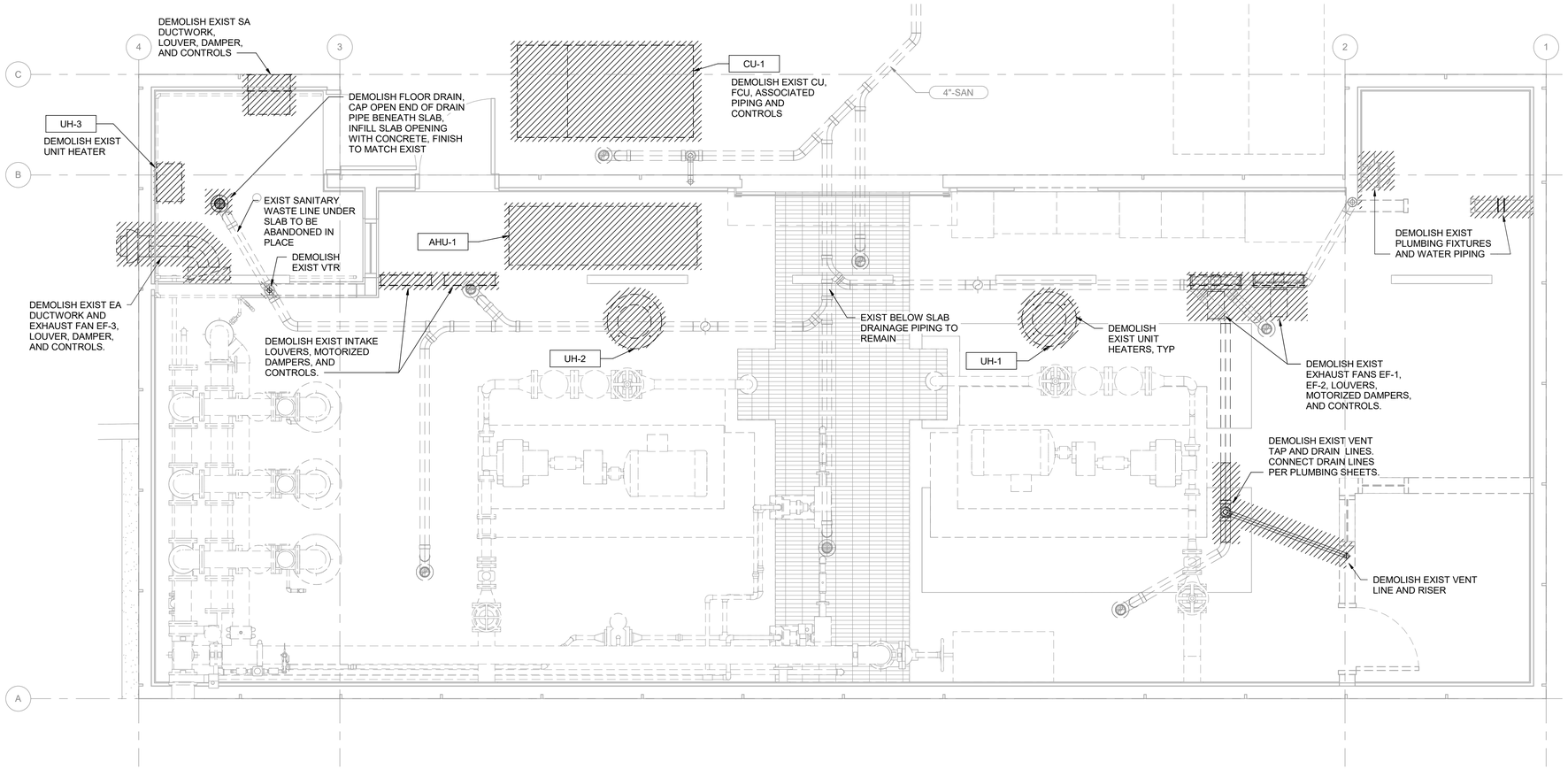
LEGEND:

//// DEMOLITION



1 / 052X105 DEMOLITION PLAN - MECHANICAL

SCALE (A)

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED:
A. PUTMAN
CADD
B. BAXTER
TECH. REVIEW:
H. PEDRAZA
DATE:
MAY 2026

SUB SHEET NO.
052
X105

TITLE OF SHEET
DEMOLITION RSPH
DEMOLITION PLAN
MECHANICAL

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.
113
182752

PMIS/PKG NO.
318719

SHEET
37 OF 113

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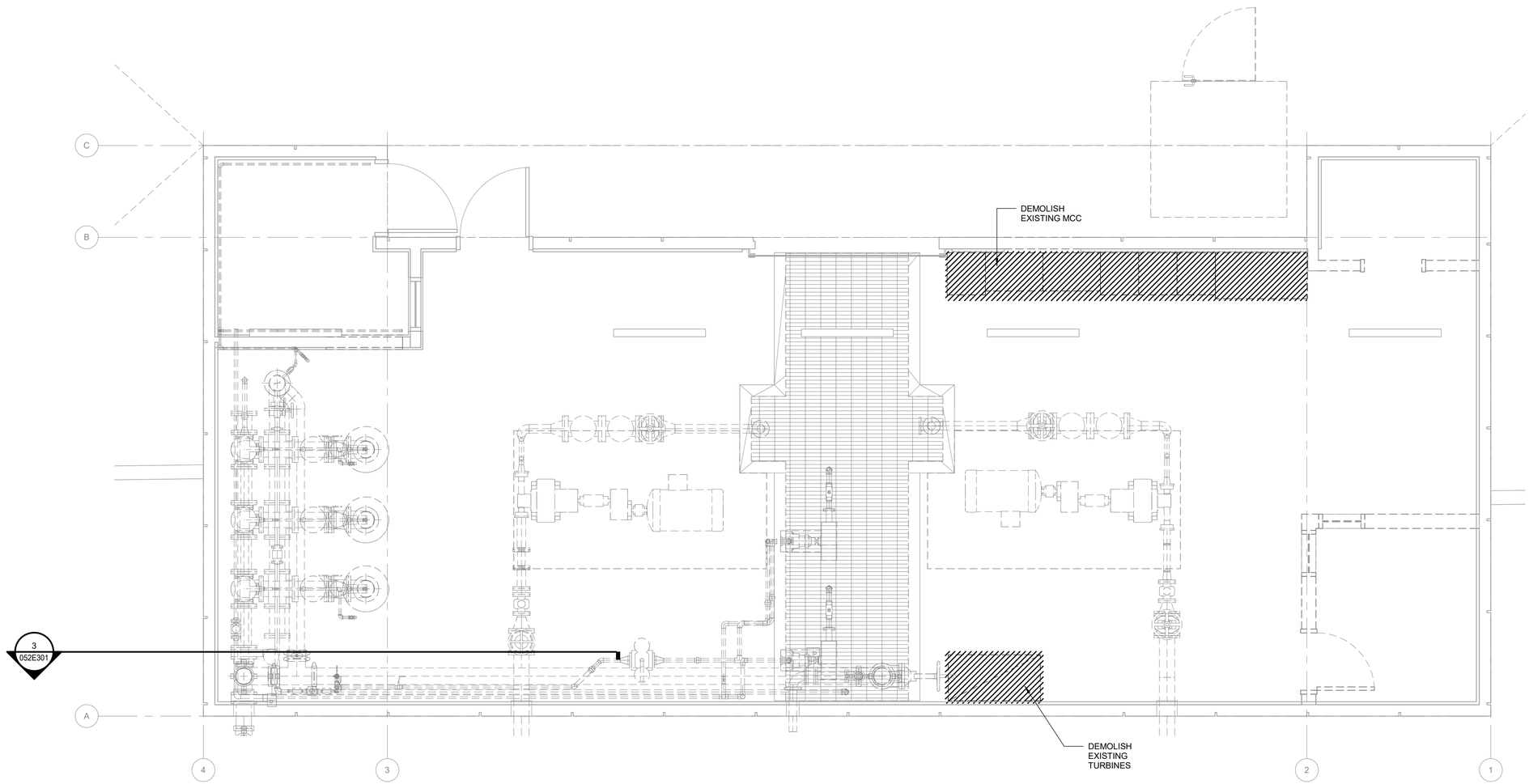

GENERAL NOTES

A. SALVAGE EXISTING CAMERAS AND TURN OVER TO CONTRACTING OFFICER'S REPRESENTATIVE.

B. VERIFY ALL EXISTING CONDITIONS PRIOR TO BEGINNING ANY DEMOLITION WORK.

C. DEMO ALL EXISTING CONDUCTORS, ABOVE GROUND CONDUIT AND RACEWAY, AND ELECTRICAL EQUIPMENT.

D. INSPECT ALL BELOW-GRADE CONDUITS PRIOR TO RE-USE.

E. INDIVIDUAL MCC BUCKETS, MOTORS, VFDS, ETC., ARE NOT SHOWN.

F. CONTRACTOR SHALL VERIFY ALL DIMENSIONS, LAYOUT, AND CONNECTION DETAILS AS REQUIRED TO SAFELY DEMOLISH THE EQUIPMENT.

G. DISCONNECT AND MAKE SAFE ALL ELECTRICAL SYSTEMS, EQUIPMENT, AND DEVICES SCHEDULED FOR DEMOLITION BEFORE REMOVAL.

H. COORDINATE ALL SHUTDOWNS WITH OWNER/FACILITY PERSONNEL TO PREVENT SERVICE INTERRUPTIONS TO ACTIVE SYSTEMS.

I. MAINTAIN AND PROTECT EXISTING SERVICES NOT SCHEDULED FOR REMOVAL THROUGHOUT DEMOLITION ACTIVITIES.

J. REMOVE ALL ELECTRICAL EQUIPMENT, LIGHTING, RECEPTACLES, SWITCHES AND WIRING.

K. REMOVE ALL ABOVE-GRADE CONDUIT BACK TO THE NEAREST REMAINING ACCESSIBLE JUNCTION BOX.

L. CAP ALL BELOW-GRADE CONDUITS NOT REMOVED IN A SAFE AND ACCESSIBLE LOCATION.

M. PATCH AND REPAIR ALL SURFACES AFFECTED BY DEVICE, CONDUIT, OR EQUIPMENT REMOVAL. MATCH ADJACENT FINISHES.

N. COORDINATE DEMOLITION WORK WITH OTHER TRADES TO AVOID CONFLICTS AND ENSURE PROPER SEQUENCING, ESPECIALLY FOR PHASED CONSTRUCTION.

O. COMPLY WITH ALL APPLICABLE CODES AND SAFETY STANDARDS, INCLUDING NEC AND OSHA REQUIREMENTS.

P. PROVIDE TEMPORARY POWER AND LIGHTING AS REQUIRED TO MAINTAIN SAFE AND CONTINUOUS OPERATIONS DURING DEMOLITION.

LEGEND

//// DEMO

1 / 052X106 DEMOLITION PLAN - ELECTRICAL

SCALE (A)

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 2/18/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 X106	TITLE OF SHEET DEMOLITION RSPH DEMOLITION PLAN ELECTRICAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 / 182752 PMIS/PKG NO. 318719 SHEET 38 OF 113

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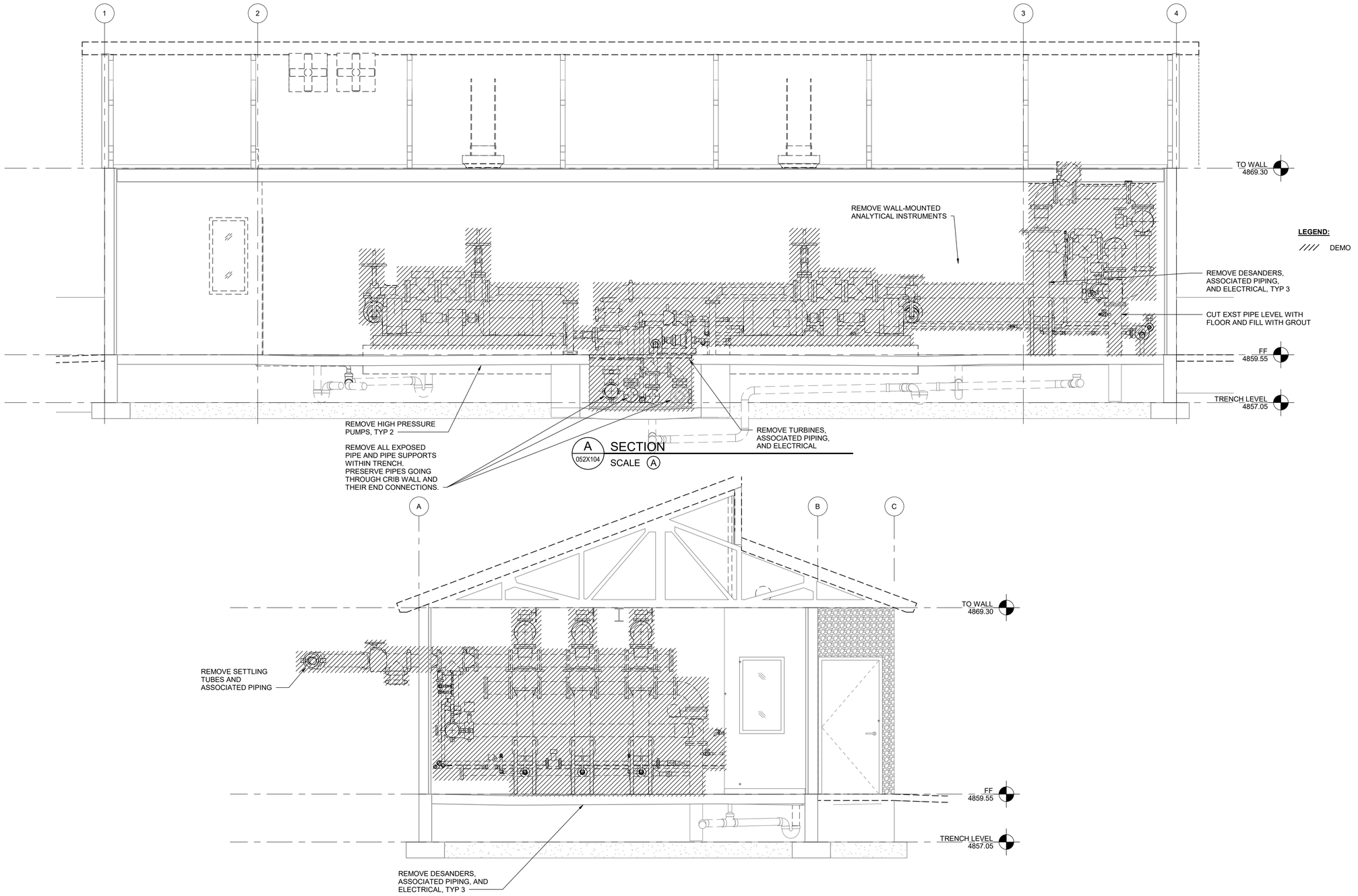

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Registered Professional Engineer (Civil)
CERTIFICATE NO.
76139
NICHOLAS
HUMPHREY
5-1-2026
Date Signed
ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	D. ZINCHENKO	052 X301	DEMOLITION RSPH DEMOLITION SECTIONS PROCESS
	CADD P. VELEZ		
	TECH. REVIEW: R. MCCANN		
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 / 182752
PMIS/PKG NO.	318719
SHEET	39 OF 113

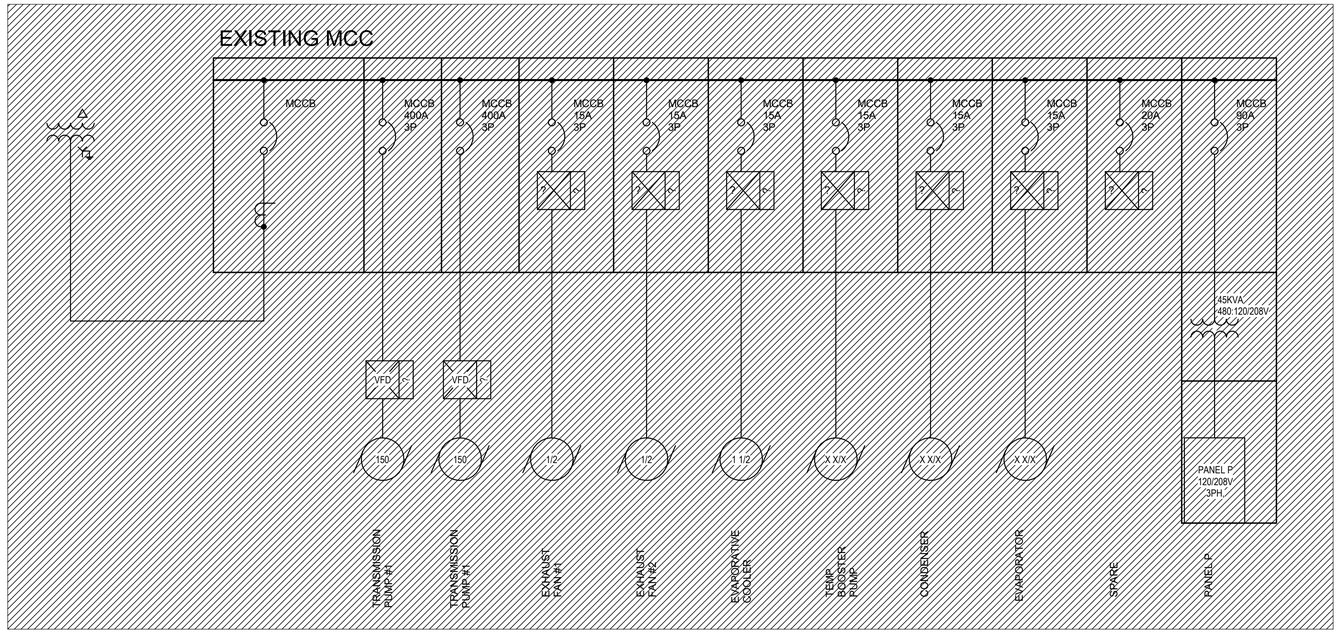
GENERAL NOTES:

A. ACCESSORY ITEMS NOT SHOWN.

B. DEMOLISH ALL ELECTRICAL CIRCUITS ASSOCIATED WITH EQUIPMENT AND DEVICES SHOWN ON OTHER DISCIPLINE DEMOLITION DRAWINGS.

LEGEND

//// DEMO



1 / 052X601 ROARING SPRINGS MCC DEMOLITION

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 2/4/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 X601	TITLE OF SHEET DEMOLITION RSPH DEMOLITION MCC ONELINE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 113

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_050_GENERAL.rvt
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GENERAL NOTES:

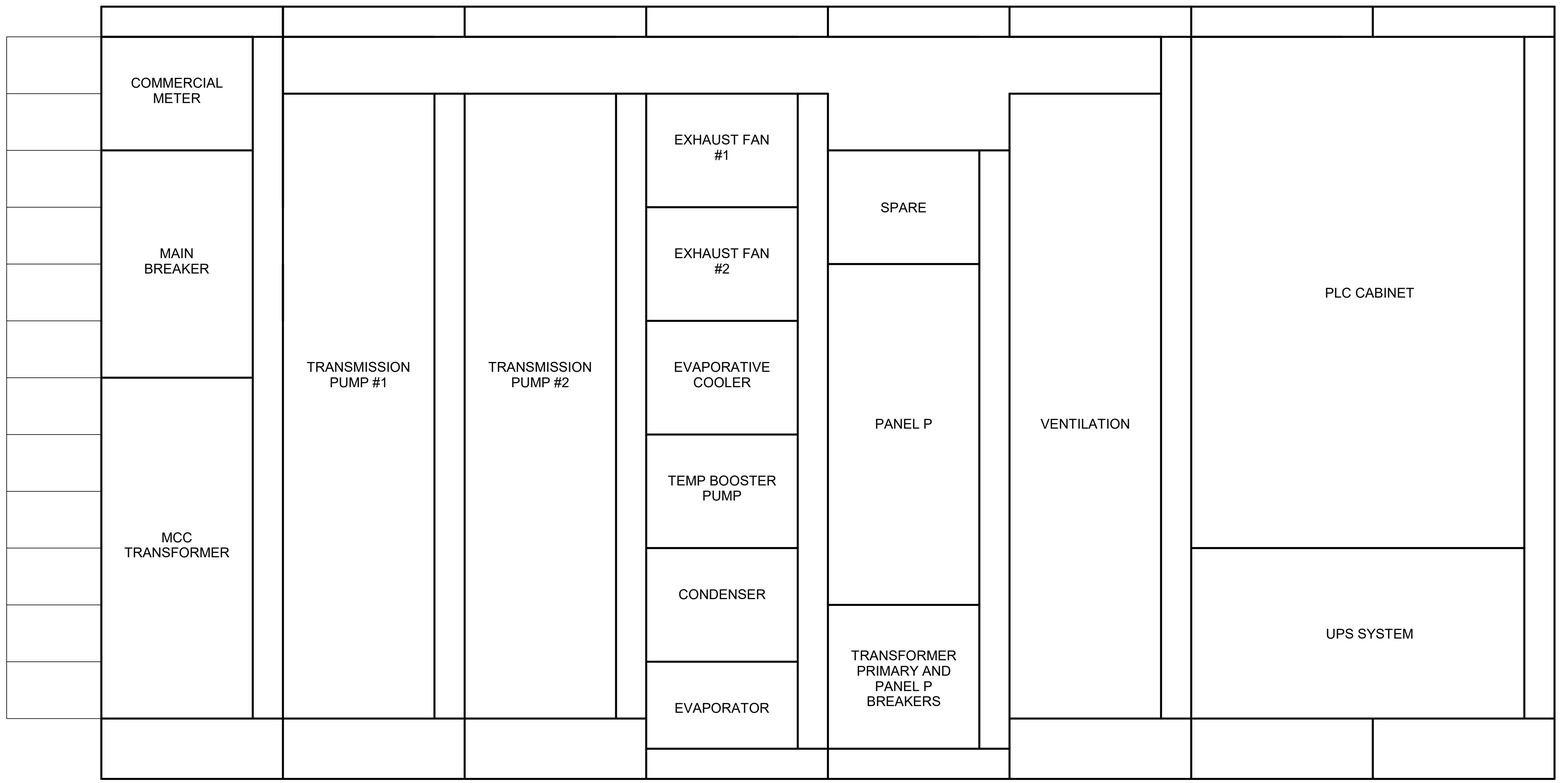
A. ACCESSORY ITEMS NOT SHOWN.

B. DEMOLISH ALL ELECTRICAL CIRCUITS ASSOCIATED WITH EQUIPMENT AND DEVICES SHOWN ON OTHER DISCIPLINE DEMOLITION DRAWINGS.

LEGEND

//// DEMO

EXISTING MCC ELEVATION



1 052X602 ROARING SPRINGS MCC ELEVATION DEMOLITION

NOT TO SCALE

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REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 04/17/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	T. MOORE CADD S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	052 X602	DEMOLITION RSPH DEMOLITION MCC ELEVATION NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 41 OF 113

CODE REVIEW ANALYSIS

1.0 INTRODUCTION

THE FOLLOWING CODE REVIEW NARRATIVE IS PROVIDED TO SERVE AS A BASIS OF UNDERSTANDING FOR THE DEVELOPMENT OF THE DESIGN, DRAWINGS AND SPECIFICATIONS FOR MODIFICATIONS TO THE EXISTING NORTH RIM ROARING SPRINGS PUMPHOUSE (RSPH) IN THE GRAND CANYON (GRCA) NATIONAL PARK. THE FOLLOWING ANALYSIS REVIEWS THE APPLICABLE BUILDING AND LIFE SAFETY CODES.

THE AUTHORITY HAVING JURISDICTION (AHJ) IS THE NATIONAL PARKS SYSTEM (NPS), INCLUDING THE DENVER SERVICE CENTER (DSC).

1.1 DESIGN CODES

BUILDING CODE	2024 INTERNATIONAL BUILDING CODE
EXISTING BUILDING CODE	2024 INTERNATIONAL EXISTING BUILDING CODE
FIRE CODES	2024 INTERNATIONAL FIRE CODE
	NPS STRUCTURAL FIRE REFERENCE MANUAL 58
MECHANICAL CODE	2024 INTERNATIONAL MECHANICAL CODE
PLUMBING CODE	2024 INTERNATIONAL PLUMBING CODE
FUEL GAS CODE	2024 INTERNATIONAL FUEL GAS CODE
ENERGY CONSERVATION CODES	2024 INTERNATIONAL ENERGY CONSERVATION CODE
	2016 ASHRAE 90.1
ELECTRICAL CODE	2023 NATIONAL ELECTRICAL CODE (NFPA 70)
WILDLAND-URBAN INTERFACE	2024 INTERNATIONAL WILDLAND-URBAN INTERFACE CODE (IWUIC)
STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS	2022 NFPA 13
STANDARD FOR THE INSPECTION, TESTING, AND MAINTENANCE OF WATER-BASED FIRE PROTECTION SYSTEMS	2023 NFPA 25
NATIONAL FIRE ALARM CODE	2022 NFPA 72

CLASSIFICATION OF WORK - EXISTING NORTH RIM ROARING SPRINGS PUMPHOUSE PUMPHOUSE (RSPH)

INTERNATIONAL EXISTING BUILDING CODE (IEBC) - CLASSIFICATION OF WORK

SUMMARY OF SCOPE OF WORK:
NORTH RIM ROARING SPRINGS PUMPHOUSE PUMPHOUSE (RSPH):
- MAIN ROOM:
REMOVE ALL PROCESS EQUIPMENT AND PIPING; REMOVE ALL ELECTRICAL EQUIPMENT AND EXPOSED CONDUIT; REMOVE STORAGE ROOM WALLS; REPLACE HVAC SYSTEM COMPONENTS; INSTALL NEW HIGH-PRESSURE PUMPS AND AUTO-FILTERS AND PIPING; INSTALL NEW FINISHED WATER BOOSTER SYSTEM AND PIPING; INSTALL NEW DILUTE SODIUM HYPOCHLORITE DOSING SYSTEM AND PIPING; INSTALL NEW MCC AND ASSOCIATED ELECTRICAL DRIVES; INSTALL NEW ESEW AND HOT WATER HEATER; INSTALL NEW SEPARATION CURTAIN B/W PUMP AREA AND ELECTRICAL AREA; INSTALL NEW EGRESS DOOR TO THE EXTERIOR OF THE BUILDING; MODIFY STRUCTURAL CONCRETE TO ALLOW FOR PROCESS UPGRADES
- CHLORINE ROOM WILL BE MODIFIED INTO A NEW OPERATIONS ROOM:
REMOVE ALL CHLORINE CYLINDERS; REMOVE ALL PROCESS AND HVAC EQUIPMENT; INSTALL NEW DOOR INTO MAIN ROOM; INSTALL PLC, CONTROL PANELS, AND HMI; INSTALL NEW HVAC EQUIPMENT WITHIN ROOM;
- IMMEDIATE VISCINITY OF BUILDING AND BUILDING STRUCTURE:
REMOVE CHLORINE CYLINER STORAGE SHED, ALL CHLORINE CYLINDERS, AND APPURTENANCES; INSTALL NEW 5,000-GAL WATER TANK NEXT TO BUILDING. POTABLE WATER WILL BE USED FOR PROVIDING WATER AT MANZANITA AND COTTONWOOD. 4,000-GAL WILL BE DEDICATED TO A FIRE WATER SUPPLY FOR A FIRE SPRINKLER SYSTEM AT ANOTHER INNER CANYON LOCATION; REPLACE OUTDOOR LIGHTING FIXTURES; REPLACE WOOD SHINGLE ROOF WITH METAL ROOF; REPLACE HVAC CONDENSER

IEBC CLASSIFICATION OF WORK:
THE ABOVE SCOPE OF WORK IS CLASSIFIED BY IEBC CHAPTER 5 AS AN ALTERATION.
- NO CHANGES TO STRUCTURAL BEARING WALLS/MEMBERS ARE EXPECTED.
- THE SCOPE OF WORK DOES NOT AFFECT THE LEVEL OF FIRE PROTECTION PROVIDED.
- THE BUILDING IS NOT REQUIRED BY THE IEBC TO BE BROUGHT UP TO THE CURRENTLY ADOPTED EDITION OF IBC REQUIREMENTS, HOWEVER ALL NEW WORK WILL MEET THE IBC.

IWUIC APPLICABILITY AND RETROACTIVITY:
- THE EXISTING NORTH RIM ROARING SPRINGS PUMP STATION BUILDING PREDATES THE ADOPTION OF IWUIC. THE BUILDING AND ITS EXISTING CONDITIONS ARE ALLOWED TO HAVE THEIR USE OR OCCUPANCY CONTINUED, IF SUCH CONDITION, USE OR OCCUPANCY WAS LEGAL AT THE TIME OF THE ADOPTION OF IWUIC WITH NO ADDITIONAL RETROACTIVITY OF IWUIC.

CODE DATA TABLE

2.0 GRAND CANYON NATIONAL PARK - EXISTING NORTH RIM ROARING SPRINGS PUMPHOUSE [52]

2.1 DESCRIPTION

THE FOLLOWING CODE REVIEW NARRATIVE IS PROVIDED TO SERVE AS A BASIS OF UNDERSTANDING FOR THE DEVELOPMENT OF THE DESIGN, DRAWINGS AND SPECIFICATIONS FOR THE EXISTING NORTH RIM ROARING SPRINGS PUMP STATION IN THE GRAND CANYON (GRCA) NATIONAL PARK.

2.2 GENERAL BUILDING DATA SUMMARY

OCCUPANCY CLASSIFICATION(S)	MODERATE HAZARD INDUSTRIAL GROUP F-1
BUILDING AREA	APPROXIMATELY 1,330 SF
FIRE AREA	APPROXIMATELY 1,330 SF
HEIGHT	13'-4" AVERAGE BUILDING HEIGHT (1 STORY ABOVE GRADE)
TYPE OF CONSTRUCTION	II-B (NON-COMBUSTIBLE)
FIRE SEPARATION DISTANCE	MORE THAN 30'-0"
FIRE PROTECTION SYSTEMS	FIRE EXTINGUISHERS FIRE HYDRANT (EXISTING NON-CONFORMING CONDITION, TO REMAIN DUE TO THE EXTREME REMOTE LOCATION ON THE SIDE OF THE GRAND CANYON. THERE ARE NO EXISTING FIRE HYDRANTS OR FIRE PROTECTION WATER SUPPLY TO THE SITE)
OCCUPANT LOAD (TOTAL)	14 (SEE 2.10 FOR CALCULATION)

2.3 USE & OCCUPANCY CLASSIFICATION:		IBC	IFC
OCCUPANCY GROUP(S):	MODERATE HAZARD INDUSTRIAL GROUP F-1	§306	-
ACCESS. OCCUPANCY (<10%)	N/A	§304	-
CLASSIFICATION OF HAZARDS	CORROSIVE	§307	-
HAZARDOUS MATERIALS	SODIUM HYPOCHLORITE	§307	-
2.4 SPECIAL REQUIREMENTS:		IBC	IFC
AREAS WITH HAZARDOUS MATERIALS	PUMP ROOM	§414	-
EMERGENCY EYE WASH / EMERGENCY SHOWER (ESEW) STATIONS	ESEW: SODIUM HYPOCHLORITE	[29 CODE OF FEDERAL REGULATIONS, OSHA 1910.151]	-
2.4.1 HAZARDOUS MATERIALS CLASSIFICATIONS:			
MATERIAL NAME AND QUANTITY PRESENT; LOCATION AND ANY RELEVANT DETAILS	HAZARD CLASSIFICATIONS (MAXIMUM ALLOWABLE QUANTITY, OR MAQ); IF OVER MAQ, OCCUPANCIES RESULTING	CONCENTRATION	C.A.S. # (HMEX)
SODIUM HYPOCHLORITE; LESS THAN 50 GALLONS; PUMP ROOM 101	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLONS MAQ); EXCEEDING MAQ CREATES A HIGH-HAZARD GROUP H-4	12.5% SOLUTION	7681-52-9

CODE DATA TABLE (CONTINUED)

2.5 BUILDING HEIGHTS & AREAS:		IBC	IFC
ACTUAL HEIGHT & AREA	13'-4" AND ONE STORY TOTAL AREA OF 1,330 SF	-	-
ALLOWABLE AREA (SUM OF RATIOS)	UNSPRINKLERED GROUP F-1: 15,500 SF	TABLE 506.2	
ALLOWABLE HEIGHT	UNSPRINKLERED GROUP F-1: 55'-0" AND 2 STORIES	TABLE 504.3, AND TABLE 504.4	-
MEZZANINE AREA (%)	N/A	§505.2	-
EQUIPMENT PLATFORM AREA (%)	N/A	§505.3	-
FRONTAGE INCREASE	NOT NEEDED NOR TAKEN	§506.3	-
UNLIMITED AREA BUILDINGS	N/A	§507	-
MIXED-USE & OCCUPANCY	SINGLE USE GROUP F-1	§508	-
INCIDENTAL USES	N/A	§509	-
SPECIAL PROVISIONS	N/A	§510	-
2.6 TYPES OF CONSTRUCTION:		IBC	IFC
CONSTRUCTION TYPE	II-B	§602	-
GENERAL BUILDING FIRE-RESISTANCE RATINGS (PRIMARY STRUCTURAL FRAME, BEARING AND NONBEARING WALLS AND PARTITIONS, FLOOR, ROOF)	0-HR	TABLE 601	-
FIRE SEPARATION DISTANCE (FSD)	MORE THAN 30 FT	TABLE 705.5	-
FIRE-RESISTANCE RATING REQUIREMENTS FOR EXTERIOR WALLS BASED UPON FSD	0-HR	TABLE 705.5	-
2.7 FIRE & SMOKE PROTECTION:		IBC	IFC
SHAFT ENCLOSURES FIRE-RESISTANCE RATINGS	N/A	§707.3.1, AND §713.4	-
VERTICAL EXIT ENCLOSURE FIRE-RESISTANCE RATINGS	N/A	§707.3.2, AND §1023.1	-
OCCUPANCY SEPARATIONS	N/A (OCCUPANCY SEPARATIONS)	§508, AND TABLE 508.4	-
OPENING PROTECTIVES	N/A (NO PROTECTED OPENINGS)	§716	-
DUCT AND AIR TRANSFER OPENINGS	N/A (NO FIRE-RATED BARRIERS, PARTITIONS, ASSEMBLIES, ETC.)	§717	-
UNPROTECTED EXTERIOR WALL OPENINGS	NO LIMIT	§705.8.1 EXCEPTION 2, AND TABLE 705.9	-
CORRIDORS	N/A (NO CORRIDORS)	§1020	-

CODE DATA TABLE (CONTINUED)

2.8 INTERIOR FINISHES:		IBC	IFC
CEILING & WALL (ROOMS AND ENCL. SPACES)	UNSPRINKLERED GROUP F-1: CLASS A, B OR C	TABLE 803.13, IEBC 702, 802, AND 903.3	-
CEILING & WALL (EXIT ENCLOSURES)	N/A (NO EXIT ENCLOSURES)		
CEILING & WALL (CORRIDORS)	N/A (NO EXIT CORRIDORS)		
FLOOR	UNSPRINKLERED GROUP F-1: DOC FF-1 "PILL TEST" (CPSC 16 CFR PART 1630) OR WITH ASTM D2859	§804.4, IEBC 702.2, 802.4, AND 903.3	
2.9 FIRE PROTECTION SYSTEMS:		IBC	IFC
AUTOMATIC SPRINKLER SYSTEM	NOT REQUIRED FOR GROUP F-1 OCCUPANCIES UNDER 12,000 SF	§903.2.4	§903.2.4
PORTABLE FIRE EXTINGUISHERS	REQUIRED; PROVIDED THROUGHOUT	§906.1	§906.1
FIRE ALARM AND DETECTION	NOT REQUIRED; SMOKE DETECTORS PROVIDED IN ELECTRICAL ROOMS ARE FOR ASSET PROTECTION ONLY. NOT FOR FIRE & LIFE SAFETY. SEE SHEET 052E131.	§907.2.4	§907.2.4
FIRE-FLOW REQUIREMENT	1,500 GPM (@20 PSI) FOR 2 HOURS	-	§B105, AND TABLE B105.1(2)
MINIMUM NUMBER OF FIRE HYDRANTS	1 HYDRANT IS REQUIRED (EXISTING NON-CONFORMING CONDITION)	-	TABLE C102.1
2.10 MEANS OF EGRESS:		IBC	IFC
CUMMULATIVE OCCUPANT LOAD	14 OCCUPANTS	§1004, AND TABLE 1004.5	-
INDUSTRIAL	1,330 SF @ 1:100 SF (GROSS) = 14		
EGRESS CAPACITY PER OCCUPANT LOAD	0.3 (STAIRS) & 0.2 (OTHER)	§1005.3	-
EXIT ACCESS - COMMON PATH	UNSPRINKLERED GROUP F-1: 75 FT	TABLE 1006.2.1	-
EXIT ACCESS - TRAVEL DISTANCE	UNSPRINKLERED GROUP F-1: 200 FT	TABLE 1017.2	-
MIN. EXIT DOORS	2 FOR MOST SPACES SEE LIFE SAFETY PLANS FOR EXITING	§1006	-
ACCESSIBLE MEANS OF EGRESS	N/A	§1009	-
EXIT DISCHARGE; ACCESS TO PUBLIC WAY	PUBLIC WAY	§1028.5	-
2.11 ACCESSIBILITY:		IBC	IFC
EQUIPMENT SPACES	EQUIPMENT SPACES ARE EXEMPT	§1103.2.9 [2010 ADA §203.5]	-

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. COOPER CADD J. BELON TECH. REVIEW: M. SHUMPERT DATE: MAY 2026	SUB SHEET NO. 052 F001	TITLE OF SHEET RSPH CODE ANALYSIS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 42 OF 113	

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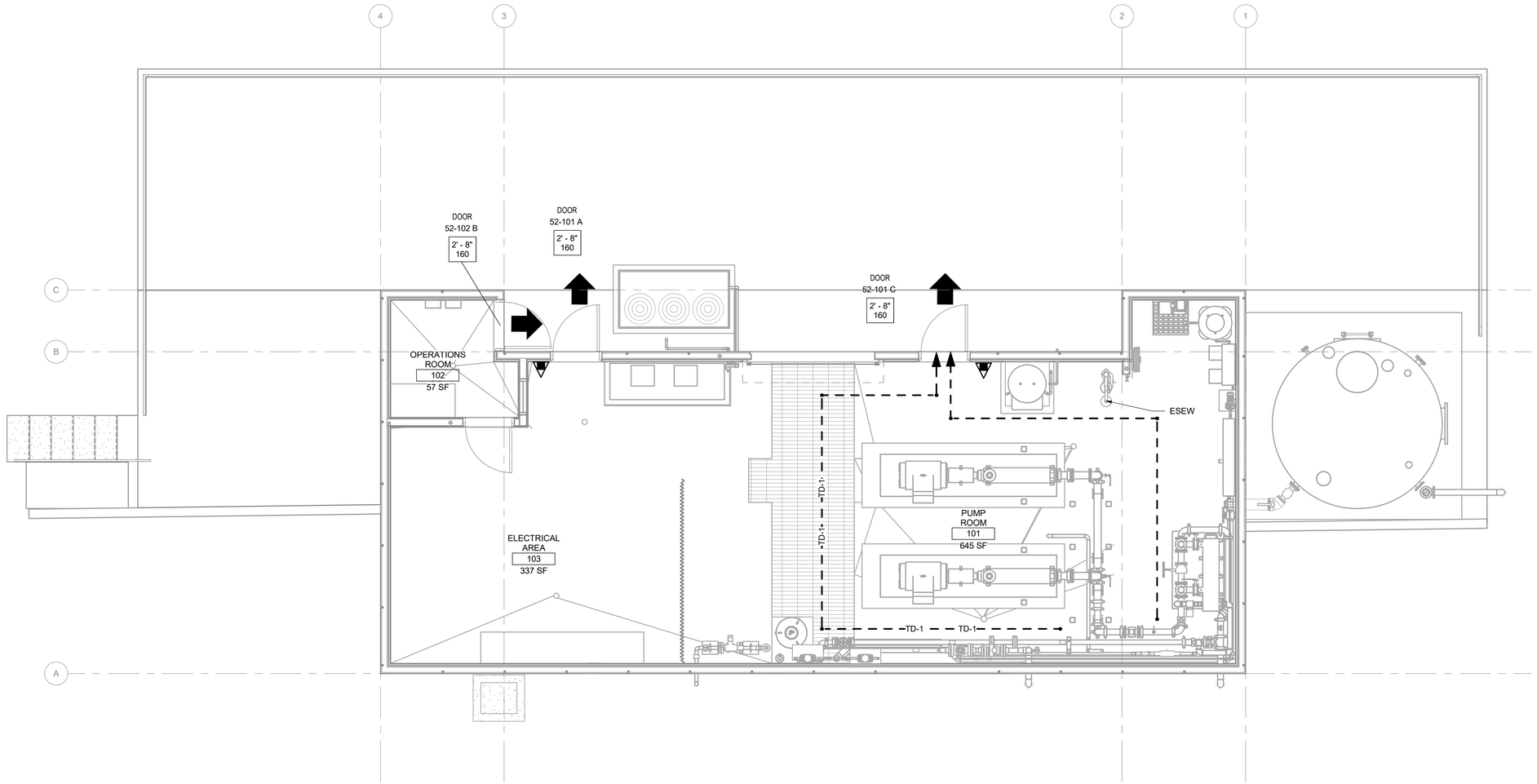

1 LIFE SAFETY PLAN
052F101

LIFE SAFETY NOTES:

1. FOR EXIT LIGHTS AND EMERGENCY LIGHTING, SEE ELECTRICAL LIGHTING DRAWINGS (E SERIES).

LIFE SAFETY SYMBOLS:

EXIT DOOR TAG

A — CLEAR DOOR WIDTH [FEET - INCHES]
B — OCCUPANT LOAD CAPACITY [NO.OF PEOPLE]

OTHERS

FIRE EXTINGUISHER

LINE TYPES FOR MEANS OF EGRESS:

COMPONENT — SYMBOL

TRAVEL DISTANCE — TD

EXIT DISCHARGE

EXIT DISCHARGE ARROW

MEANS OF EGRESS WIDTH CAPACITY			
	PROVIDED EGRESS CAPACITY - LEVEL COMPONENTS & RAMPS		
EXIT ID	EGRESS CAPACITY FACTOR	CLEAR WIDTH	OCCUPANT LOAD CAPACITY
52-101 A	0.2	2' - 8"	160
52-101 C	0.2	2' - 8"	160
52-102 B	0.2	2' - 8"	160

MEANS OF EGRESS ARRANGEMENT DISTANCE SCHEDULE - BUILDING SUMMARY		
	EGRESS DISTANCE	
EGRESS PATH ID	PROVIDED	MAXIMUM PERMITTED
TD-1	48' - 0"	200' - 0"
TD-2	30' - 6"	200' - 0"

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

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Registered Professional Engineer (Not Architect) CERTIFICATE NO. 67918 STUART GRAHAM Signed 5-1-26 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. COOPER	052 F101	RSPH LIFE SAFETY PLAN	113 182752
	CADD J. BELON			PMIS/PKG NO. 318719
	TECH. REVIEW: M. SHUMPERT			SHEET 43 OF 113
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

GENERAL NOTES

A. REFER TO SHEET 050S001 FOR GENERAL STRUCTURAL NOTES.

B. REFER TO 003S501-003S507 FOR STRUCTURAL STANDARD DETAILS.

C. COORDINATE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR LOCATIONS OF PIPING AND OTHER EQUIPMENT.

D. (*) INDICATES EXISTING DIMENSIONS TO BE FIELD VERIFIED BY CONTRACTOR.

E. EXISTING BUILDING INFORMATION IS AS INDICATED ON THE AS-BUILT DRAWINGS.

F. REFER TO STANDARD DETAILS IN SERIES 003 IN VOLUME 0 REFERENCED HEREIN.

KEYNOTES (#)

1 SHALLOW EQUIPMENT SUPPORT PAD PER DETAIL 2/003S501.

2 1 3/4" REMOVABLE ALUMINUM GRATING. MCNICHOLS SWAGE LOCKED SAFE-T-GRID OR EQUAL. REFER TO STRUCTURAL STANDARD DETAIL 3/003S502 FOR GRATING SUPPORT.

3 INDUSTRIAL CURTAIN PER ARCH.

4 MONORAIL BEAM ABOVE.

5 CONCRETE PIPE SUPPORT PER PROCESS DETAIL 3/005D501.

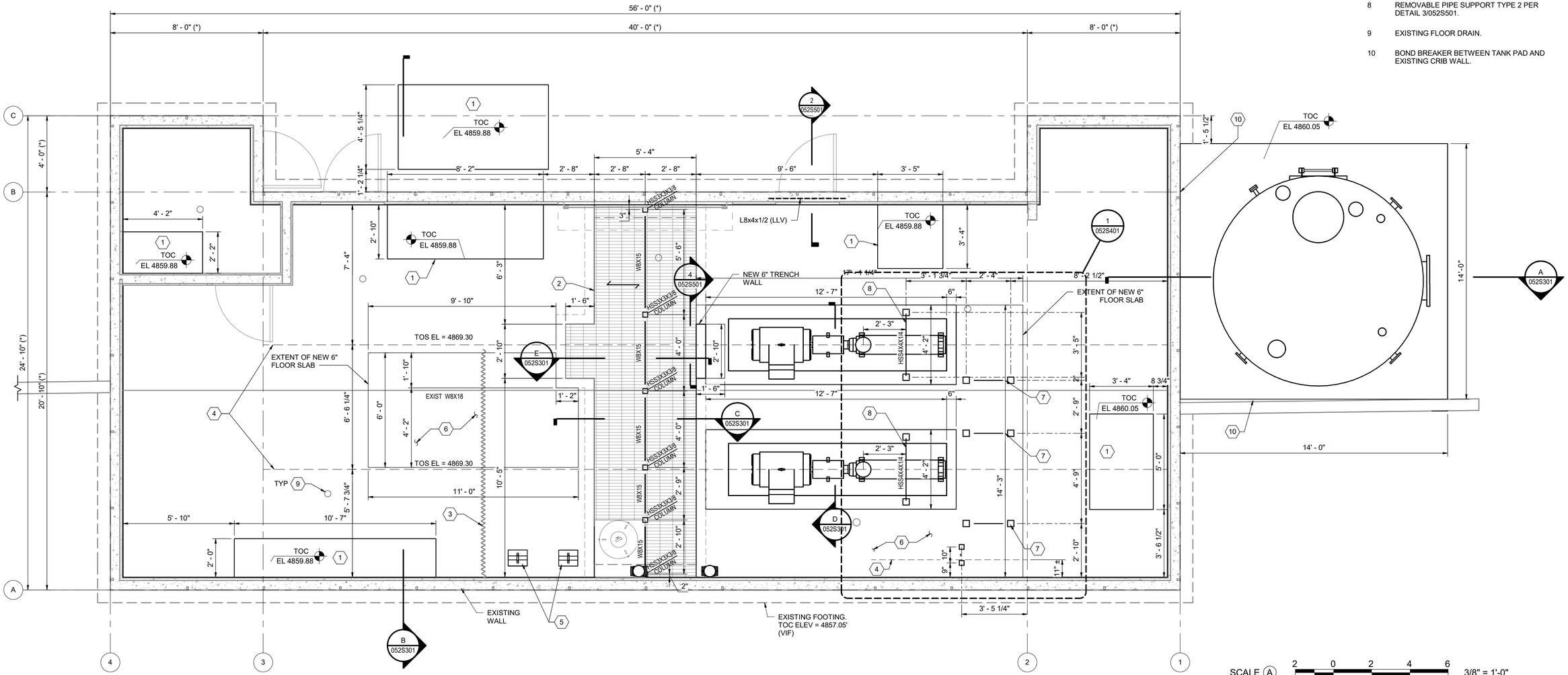
6 NEW 6" FLOOR SLAB W/ #4@12" EW. SEE STRUCTURAL DEMOLITION PLAN FOR EXTENTS OF SLAB REPLACEMENT.

7 PIPE SUPPORT TYPE 1 PER DETAIL 2/052S501.

8 REMOVABLE PIPE SUPPORT TYPE 2 PER DETAIL 3/052S501.

9 EXISTING FLOOR DRAIN.

10 BOND BREAKER BETWEEN TANK PAD AND EXISTING CRIB WALL.



1 052S101 FLOOR PLAN
SCALE (A)

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 S101	TITLE OF SHEET RSPH FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 44 OF 113

REGISTERED PROFESSIONAL ENGINEER 85509 JOSE CORONADO CERTIFICATE NO. Date Signed 04/17/2026 ARIZONA U.S.A.

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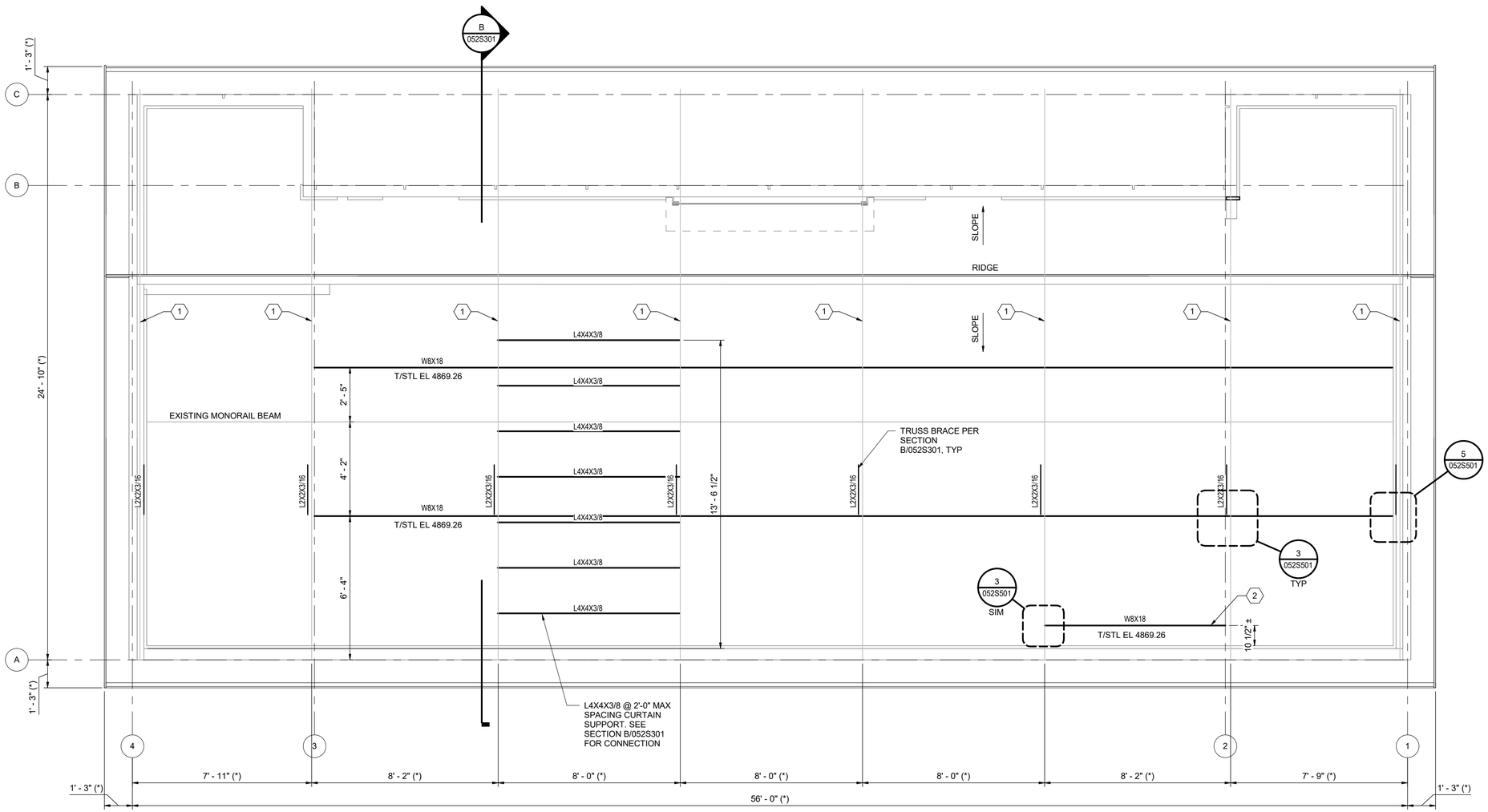

1 052S102 ROOF FRAMING PLAN
SCALE A

GENERAL NOTES

A. REFER TO SHEET 050S001 FOR GENERAL STRUCTURAL NOTES.

B. COORDINATE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR LOCATIONS OF PIPING AND OTHER EQUIPMENT.

C. (*) INDICATES EXISTING DIMENSIONS TO BE FIELD VERIFIED BY CONTRACTOR.

D. EXISTING BUILDING INFORMATION IS AS INDICATED ON THE AS-BUILT DRAWINGS.

E. REFER TO STANDARD DETAILS IN SERIES 003 IN VOLUME 0 REFERENCED HEREIN.

KEYNOTES (#)

1 EXISTING ROOF TRUSS.

2 ALIGN BEAM WITH PROCESS VALVES V5-2031 AND CKV-52333. PROVIDE MONORAIL HOIST ON BEAM.

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85599 JOHN CORONADO Date Signed 04/17/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN CADD R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 S102	TITLE OF SHEET RSPH ROOF FRAMING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 45 OF 113

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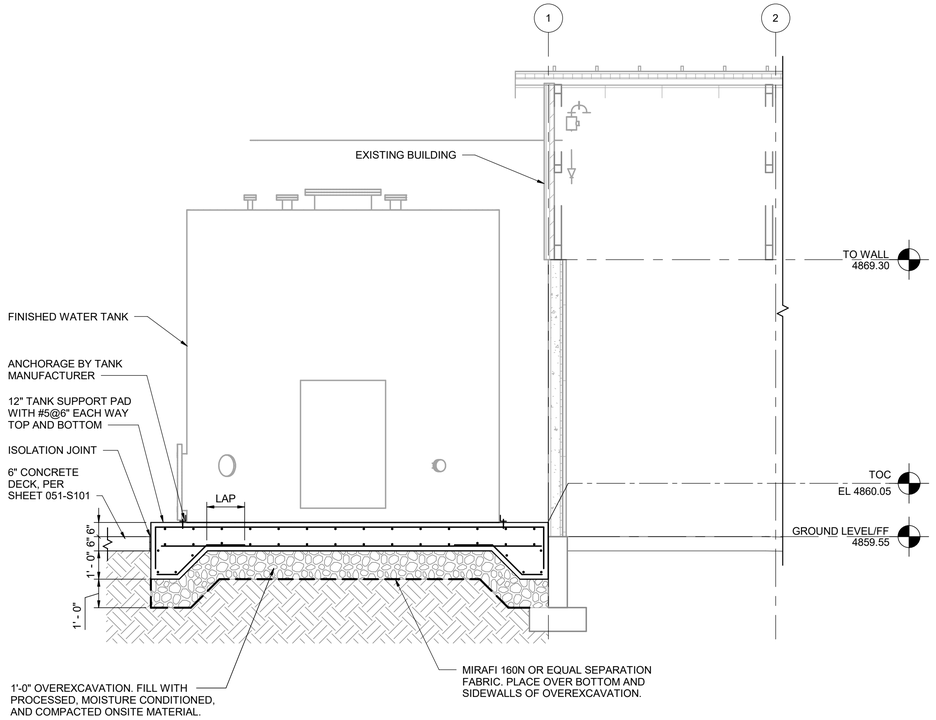
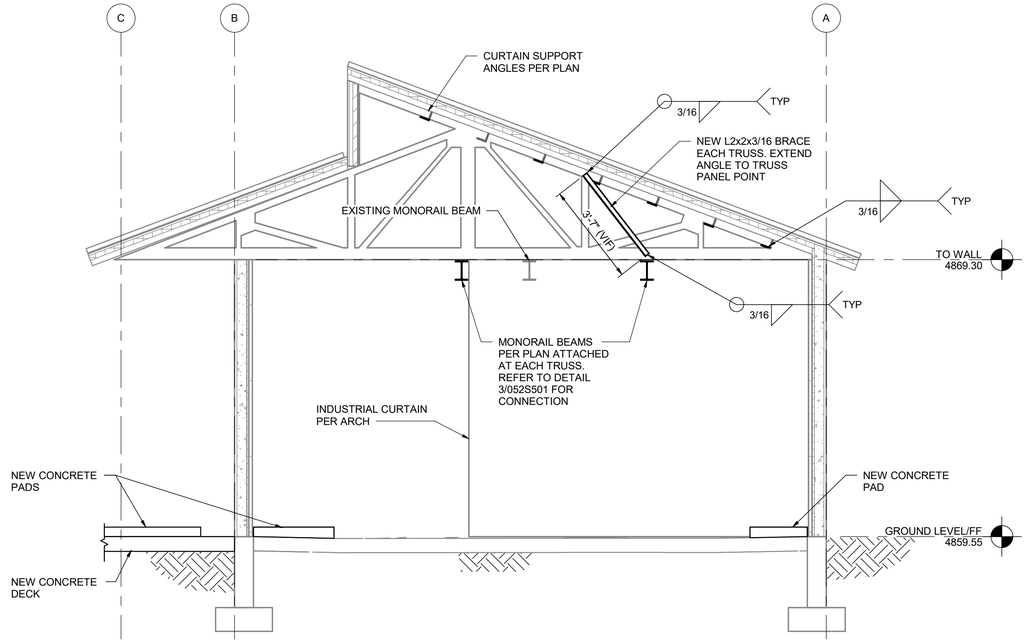
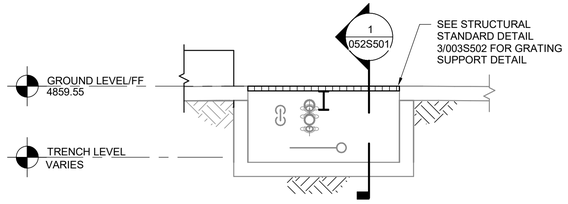


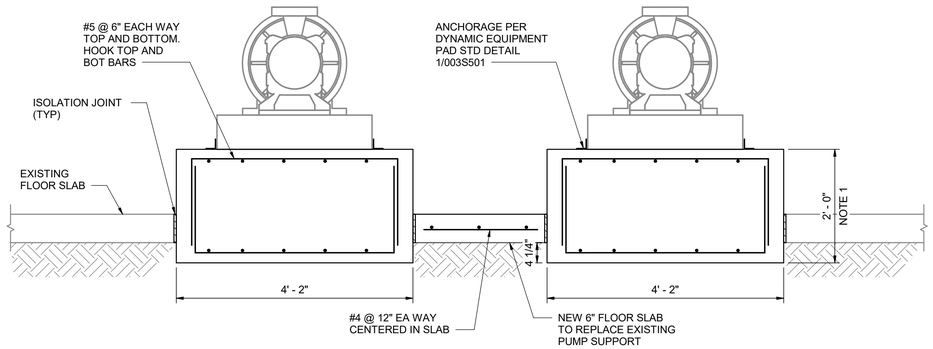
A 052S101 FINISHED WATER TANK SUPPORT PAD SECTION
SCALE A



B 052S101 SECTION
SCALE A



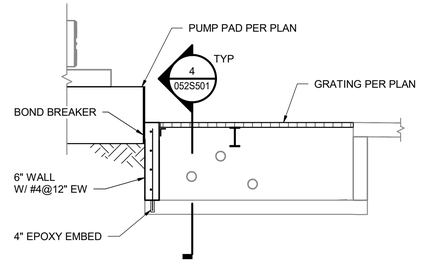
C 052S101 TRENCH SECTION
SCALE A



NOTES:

1. VERIFY FINAL PAD HEIGHT BASED ON PROCESS EQUIPMENT.

D 052S101 PUMP SUPPORT PAD SECTION
SCALE B



E 052S101 TRENCH FILL SECTION
SCALE A

SCALE B 1 0 1 2 3 — SCALE OF FEET — 3/4" = 1'-0"

SCALE A 2 0 2 4 6 — SCALE OF FEET — 3/8" = 1'-0"

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 85509
ZACHARY J. CORONADO
Date Signed 05/15/2026
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 S301	TITLE OF SHEET RSPH SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 46 OF 113

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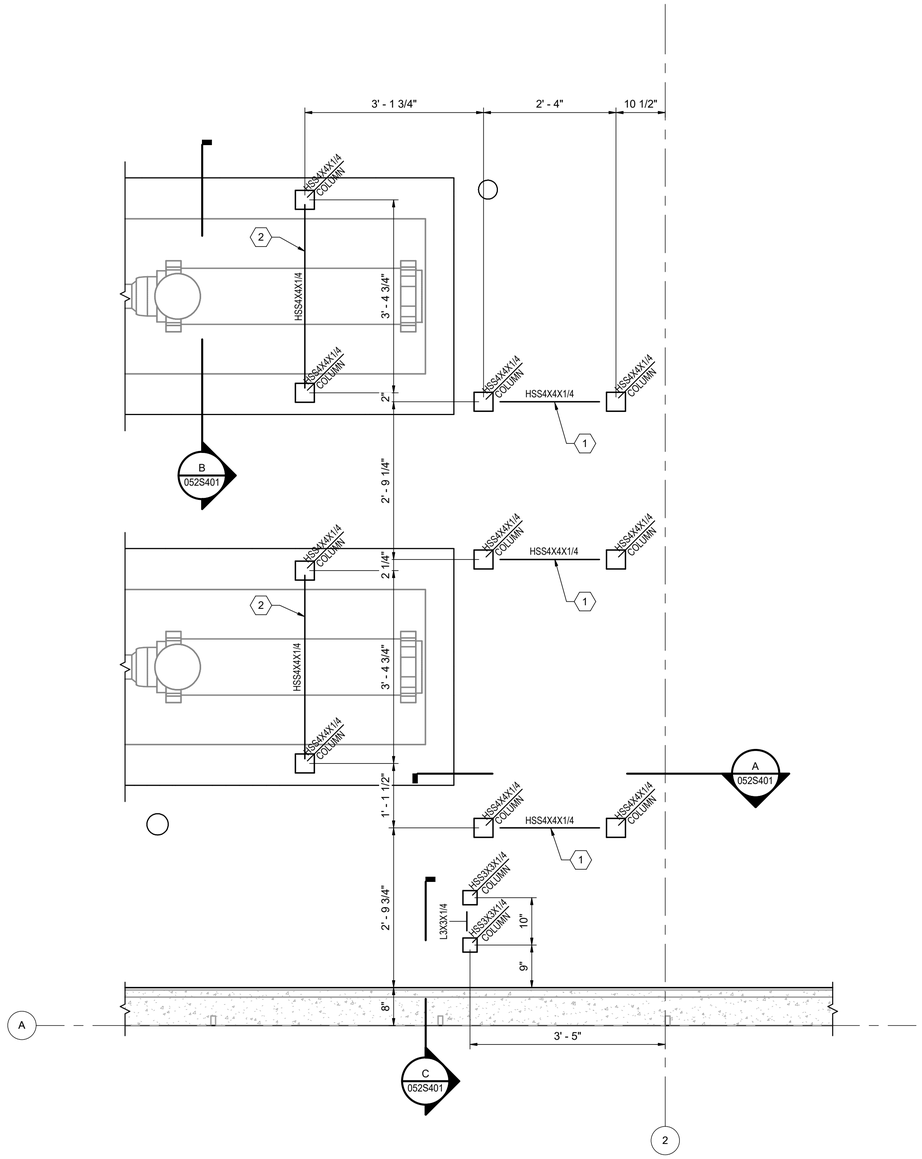
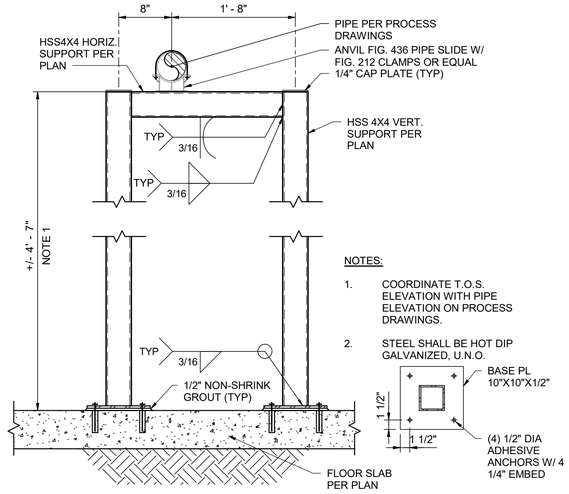


1 052S401 ENLARGED FLOOR PLAN

SCALE A

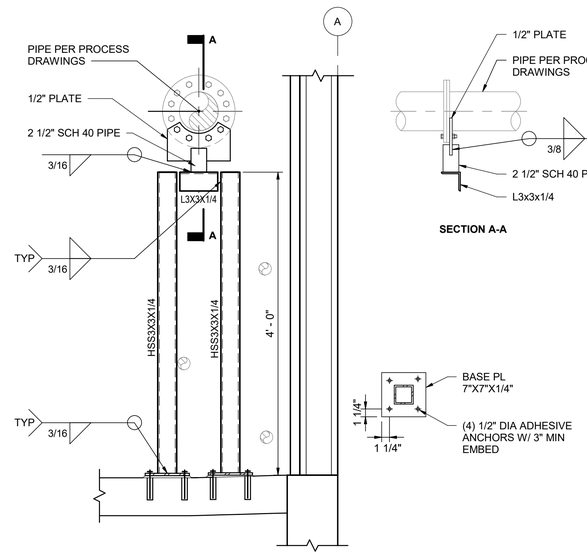


A 052S401 PIPE RACK DETAIL (TYPE 1)

NOT TO SCALE

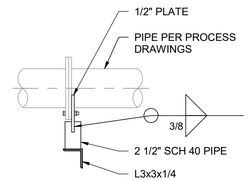
B 052S401 PIPE RACK DETAIL(TYPE 2 REMOVABLE)

NOT TO SCALE



C 052S401 PIPE RACK DETAIL (TYPE 3)

NOT TO SCALE



SECTION A-A

GENERAL NOTES

A. REFER TO SHEET 050S001 FOR GENERAL STRUCTURAL NOTES.

B. COORDINATE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR LOCATIONS OF PIPING AND OTHER EQUIPMENT.

C. (*) INDICATES EXISTING DIMENSIONS TO BE FIELD VERIFIED BY CONTRACTOR.

D. EXISTING BUILDING INFORMATION IS AS INDICATED ON THE AS-BUILT DRAWINGS.

E. REFER TO STANDARD DETAILS IN SERIES 003 IN VOLUME 0 REFERENCED HEREIN.

KEYNOTES (#)

1 PIPE SUPPORT TYPE 1 PER DETAIL 2/052S501.

2 REMOVABLE PIPE SUPPORT TYPE 2 PER DETAIL 3/052S501.

SCALE A 1 0 1 2 3 3/4" = 1'-0"

SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 CORONADO Date Signed 04/27/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN CADD R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 S401	TITLE OF SHEET RSPH ENLARGED FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 47 OF 113

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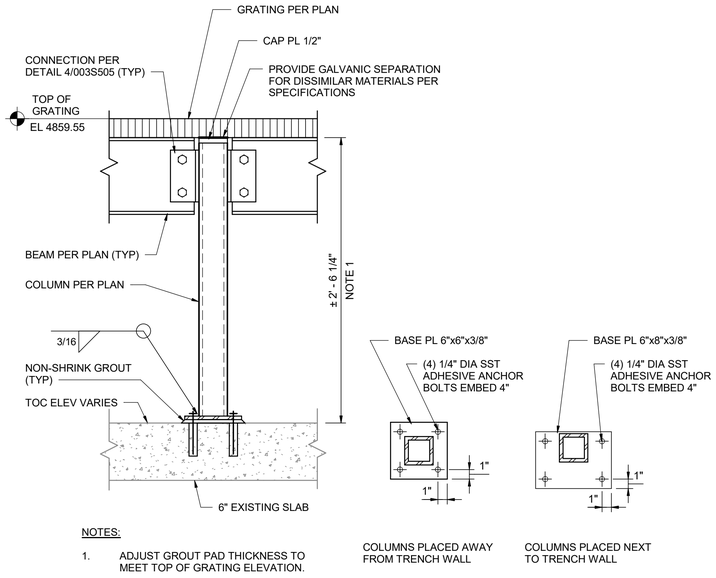
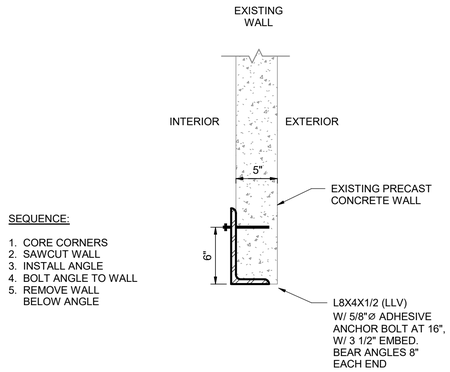
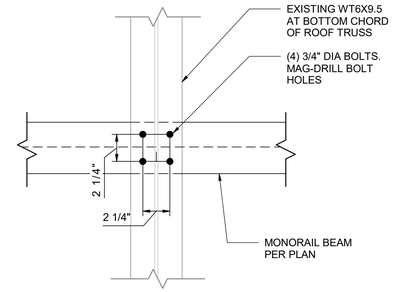


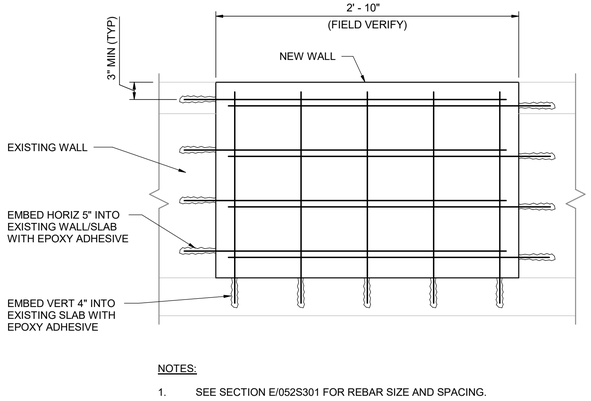
1 052S501 GRATING SUPPORT POST DETAIL
NOT TO SCALE



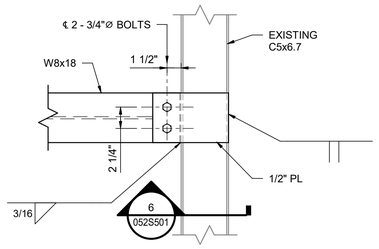
2 052S501 LINTEL ABOVE DOOR
NOT TO SCALE



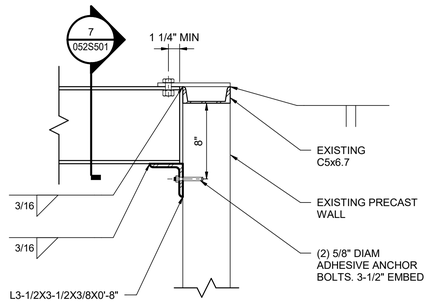
3 052S501 MONORAIL BEAM CONNECTION
NOT TO SCALE



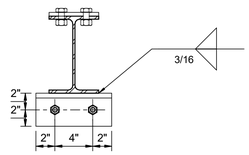
4 052S501 NEW WALL
NOT TO SCALE



5 052S501 CONNECTION DETAIL
NOT TO SCALE



6 052S501 CONNECTION DETAIL
NOT TO SCALE



7 052S501 CONNECTION DETAIL
NOT TO SCALE

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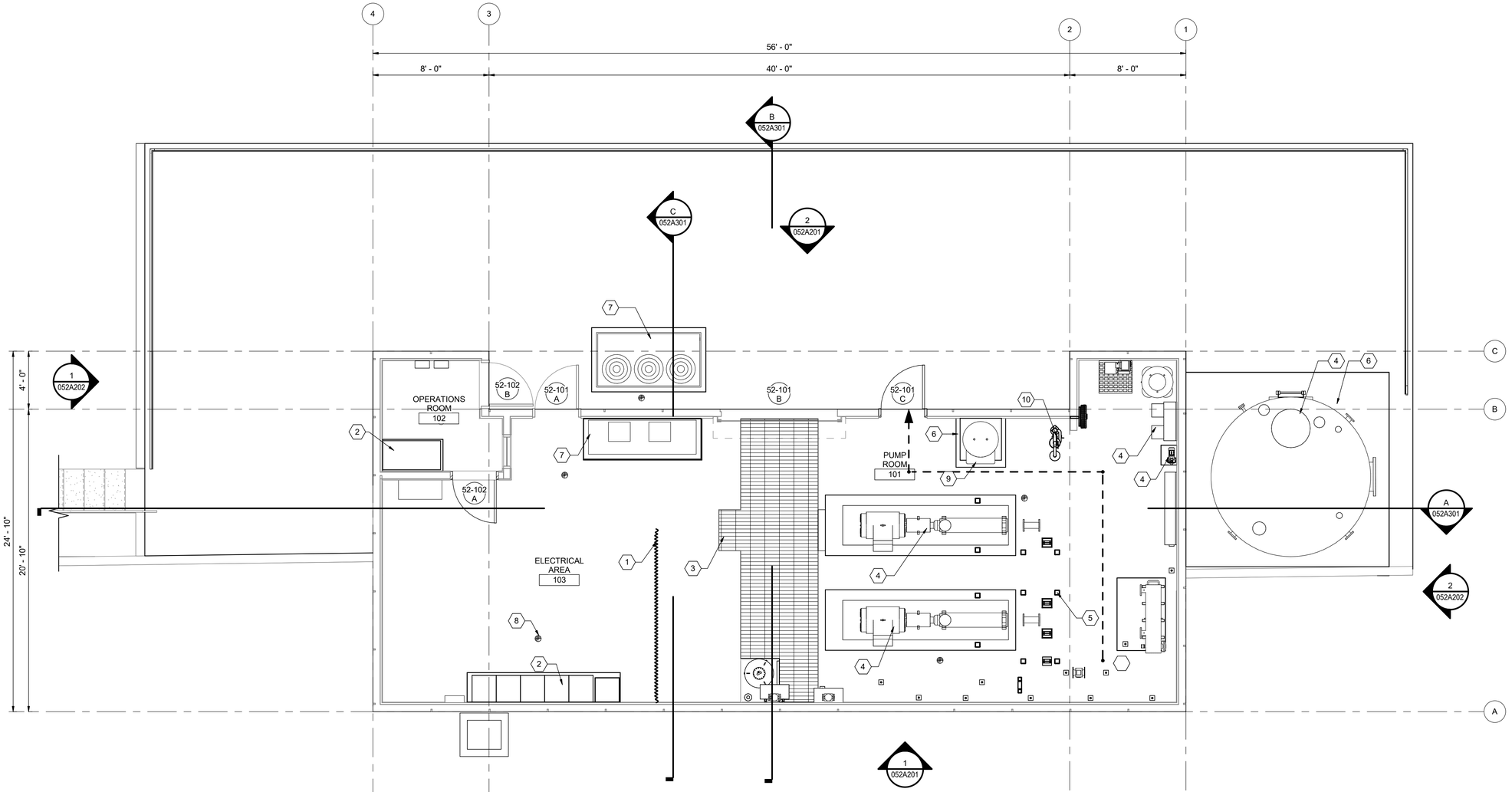
REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 85509 — J. CORONADO — Date Signed 04/27/2026 — ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 052 S501	TITLE OF SHEET RSPH DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 48 OF 113



1 052A101 FLOOR PLAN
SCALE (A)

GENERAL NOTES

A. SEE SERIES 004 SHEETS (VOL 1) FOR STANDARD DETAILS REFERENCED HEREIN.

B. SEE SHEET 004A509 FOR TYPICAL WALL TYPES TAGGED AT PLANS.

C. SEE SHEET 004A601 FOR DOOR AND FRAME SCHEDULES, ELEVATIONS AND NOTES.

D. SEE SHEET 004A602 AND SPECIFICATIONS FOR MATERIAL FINISH SCHEDULES AND NOTES.

E. SEE 052F SERIES SHEETS FOR CODE ANALYSIS, LIFE SAFETY PLANS AND FIRE EXTINGUISHER LOCATIONS.

F. PROVIDE IN-WALL BLOCKING IN WALLS FOR WALL-MOUNTED ITEMS, INCLUDING CABINETS, SHELVING, TOILET & BATH ACCESSORIES, & SIMILAR ITEMS

G. ALL EXISTING ITEMS ARE HALF TONE, NEW ITEMS ARE BOLD, TAGGED, OR NOTED.

H. ALL DIMENSIONS ARE ACTUAL AND ARE TO FACE OF MASONRY, FACE OF METAL STUD, CENTERLINE OF COLUMN OR CENTERLINE OF BEAM UNLESS NOTED OTHERWISE. ALL DIMENSIONS ARE FOR BIDDING PURPOSES ONLY. ACTUAL FIELD DIMENSIONS SHALL BE VERIFIED PRIOR TO SUBMITTAL OF SHOP DRAWINGS, ORDERING RELATED MATERIALS AND PERFORMING DEMOLITION OR CONSTRUCTION WORK. ALL DIMENSIONS ALIGNING WITH OR RELATED TO EXISTING CONDITIONS ARE TO BE LAID OUT PRIOR TO COMMENCING WORK AND VERIFIED.

I. VERIFY MECHANICAL AND ELECTRICAL DRAWINGS FOR SCOPE AND INTERFACE. CONTRACTOR SHALL COORDINATE LOCATION FOR ALL MECHANICAL AND ELECTRICAL ITEMS WITH GENERAL CONSTRUCTION. REVIEW ANY DISCREPANCIES WITH THE CONTRACTING OFFICER PRIOR TO INSTALLATION AND/OR FABRICATION.

J. CONTRACTOR SHALL FIELD ROUTE PIPING / CONDUIT AS NECESSARY TO COORDINATE WITH EQUIPMENT LOCATIONS. SEE P&ID DRAWINGS FOR ALL PROCESSING PIPING FITTINGS, VALVES, AND OTHER PIPING APPURTENANCES AND LOCATIONS.

K. PLUMBING PIPES AND PROCESS PIPES SHALL NOT BE INSTALLED ABOVE ELECTRICAL PANELS OR CONTROL PANELS.

L. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

M. LOCATE INTERIOR CONTROL JOINTS WHERE SHOWN ON THE PLANS AND STANDARDS IDENTIFIED IN SPECIFICATIONS

N. ALL DIMENSIONS ALIGNING WITH OR RELATED TO EXISTING CONDITIONS ARE TO BE LAID OUT PRIOR TO COMMENCING WORK AND VERIFIED BY CONTRACTOR.

KEYNOTES: (#)

PROCESS EQUIPMENT; SEE PROCESS

1 INDUSTRIAL CURTAIN. REFER TO SHEET 052A501 DIVIDER DETAILS.

2 ELECTRICAL EQUIPMENT; SEE ELECTRICAL

3 METAL GRATING

4 PROCESS EQUIPMENT; SEE PROCESS

5 STEEL FRAMING, SEE STRUCTURAL

6 EQUIPMENT PAD; SEE STRUCTURAL TYP

7 MECHANICAL EQUIPMENT; SEE MECHANICAL

8 FLOOR DRAIN; SEE PLUMBING

9 PLUMBING EQUIPMENT; SEE PLUMBING

10 EYEWASH; SEE PLUMBING

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 2026.04.16 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	E. BUTTMAN	052 A101	RSPH FLOOR PLAN	113 182752
	CADD D. SPERRY			PMIS/PKG NO. 318719
	TECH. REVIEW: S. HEANEY			SHEET 49 OF 113
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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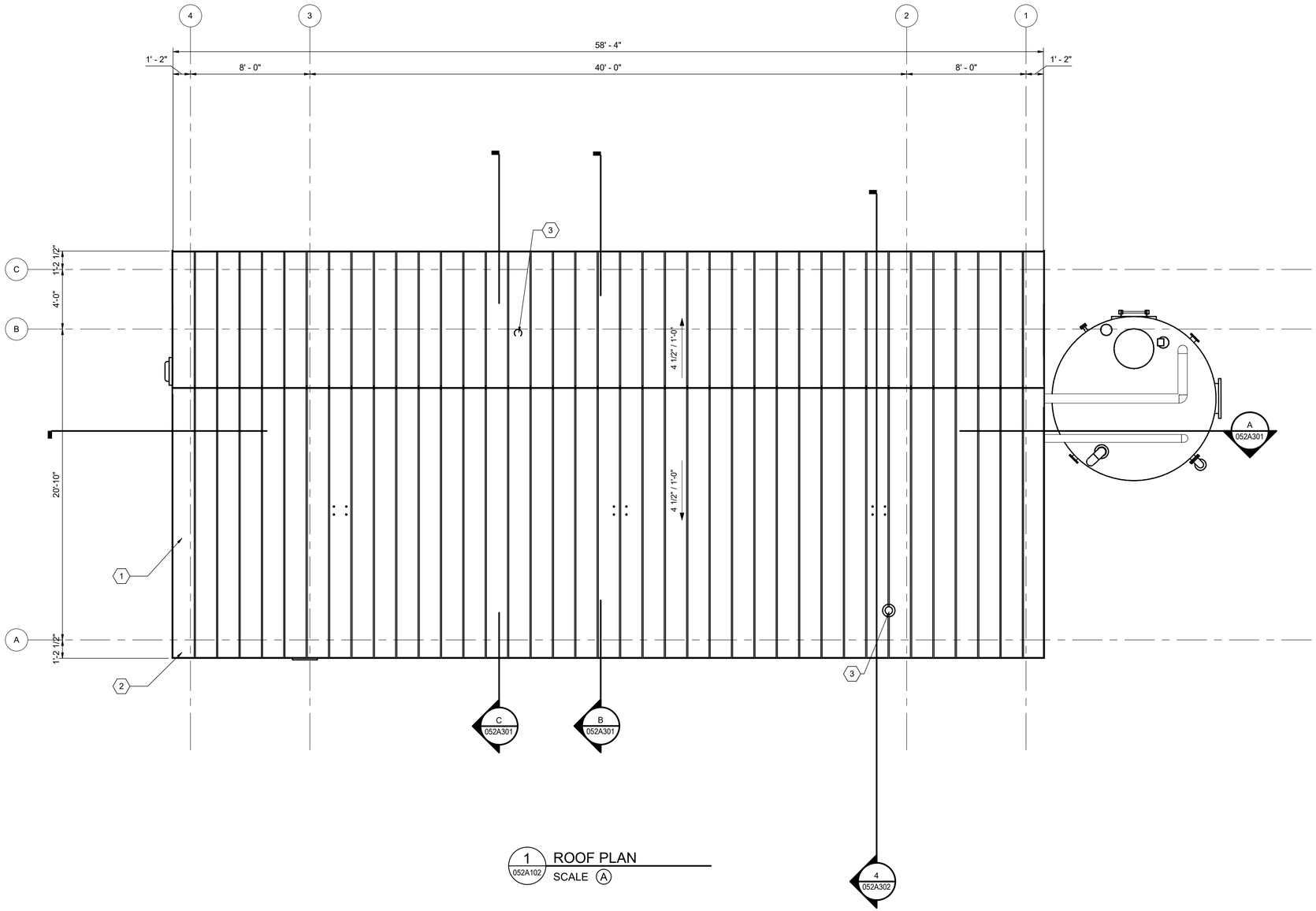

1 052A102 ROOF PLAN
SCALE (A)

GENERAL ROOF PLAN NOTES

A. SEE SERIES 004 SHEETS (VOL 1) FOR STANDARD DETAILS REFERENCED HEREIN.

B. PROVIDE ICE AND WATER SHEILD PROTECTION AT ALL EAVES, VALLEYS, PENETRATIONS, AND ROOF EDGES TO 36" MIN AT EA SIDE.

C. COORDINATE ALL ROOF PENETRATIONS THAT REQUIRE FLASHING; SEE MECHANICAL.

D. PROVIDE ALL FLASHING, JOINTS, DRIPS, TRIM, CAPS, AND COPINGS REQUIRED AS A COMPONENT OF THE ROOF ASSEMBLY.

E. REFER TO ENGINEERING DISCIPLINE DOCUMENTS FOR ADDITIONAL EQUIPMENT, PIPE, CONDUIT, VALVES, ETC...

F. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

G. INSTALL ALL ROOFING ASSEMBLIES IN ACCORDANCE WITH PROJECT SPECIFICATIONS, DETAILS, AND MANUFACTURER'S STANDARD INSTALLATION REQUIREMENTS.

H. SEE ELECTRICAL FOR LIGHTING PROTECTION SYSTEMS WHERE REQUIRED.

I. INSTALL CLASS -A ROOF ASSEMBLY PER 07 61 13.

KEYNOTES: (#)

1 STANDING SEAM METAL ROOF

2 METAL FASCIA

3 MECHANICAL EQUIPMENT; SEE MECHANICAL

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.02.12 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN DRAWN: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A102	TITLE OF SHEET RSPH ROOF PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 50 OF 113

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_A.rvt
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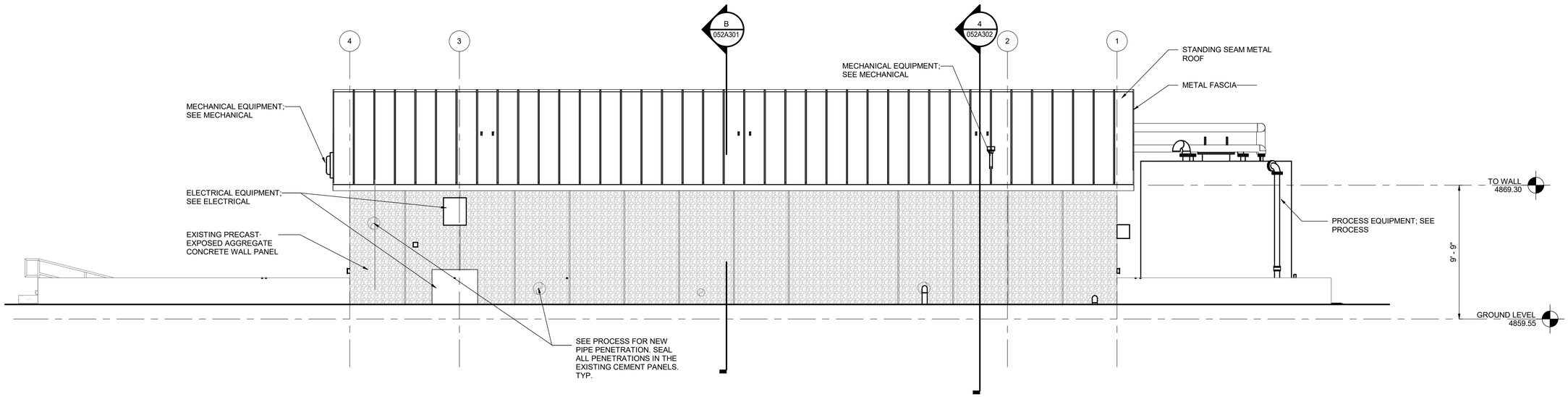
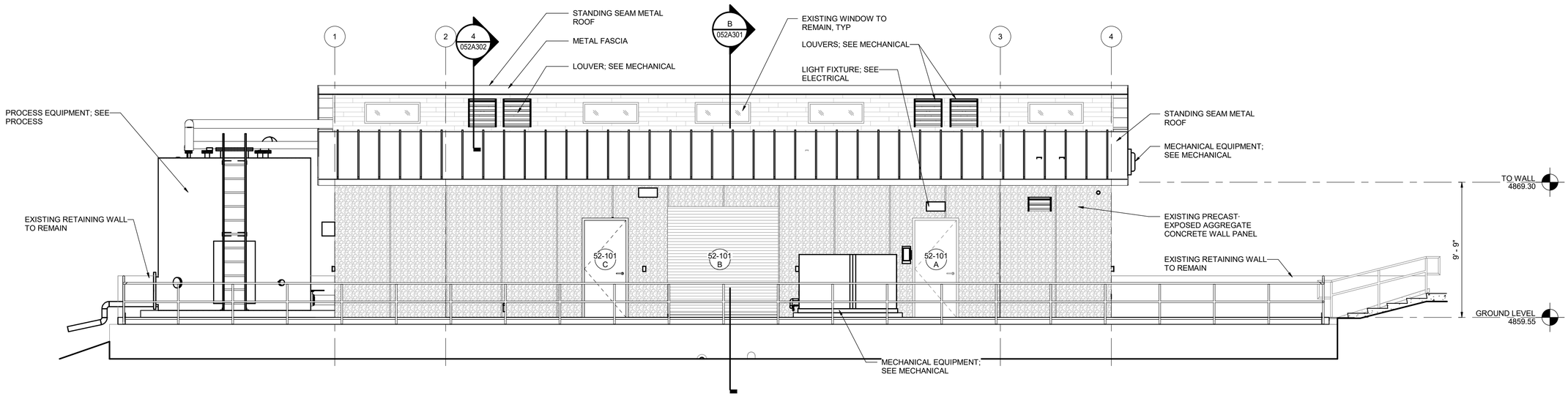


1 / 052A201 NORTH EXTERIOR ELEVATION

SCALE (A) NOTE: SEE ELECTRICAL FOR CAMERA LOCATIONS



2 / 052A201 SOUTH EXTERIOR ELEVATION

SCALE (A)

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.16 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN DRAWN: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A201	TITLE OF SHEET RSPH EXTERIOR ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	51 OF 113

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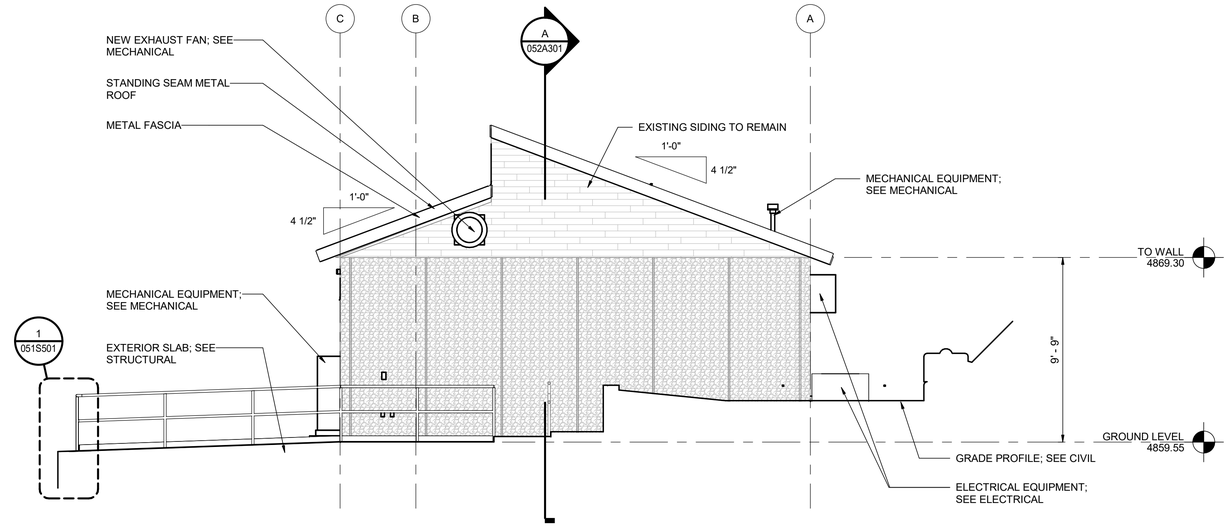
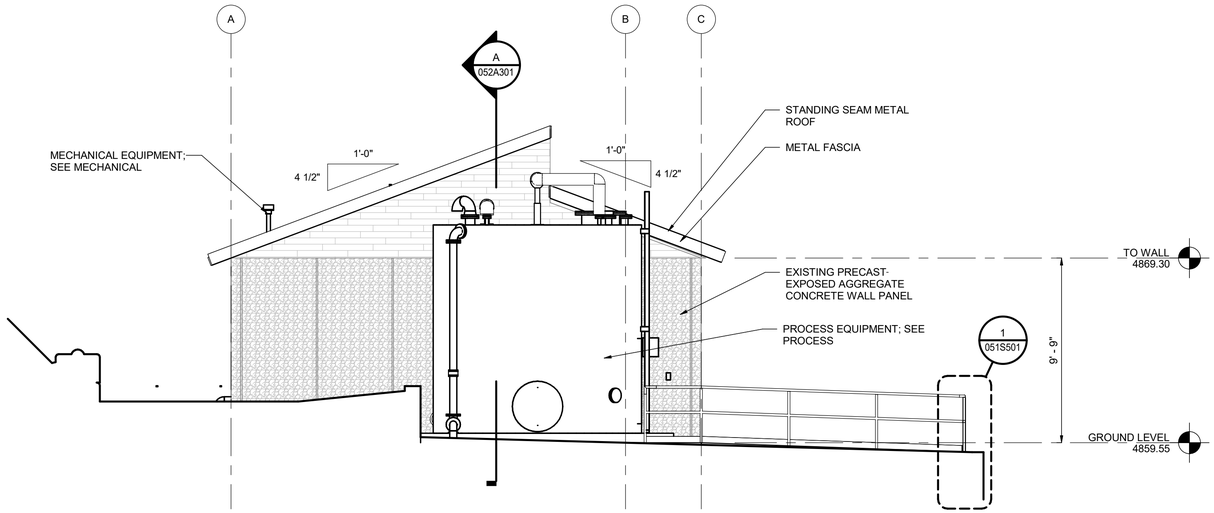


1 052A202 EAST EXTERIOR ELEVATION

SCALE (A)



2 052A202 WEST EXTERIOR ELEVATION

SCALE (A)

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 10/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN DRAWN: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A202	TITLE OF SHEET RSPH EXTERIOR ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752

PMIS/PKG NO. 318719

SHEET 52 OF 113

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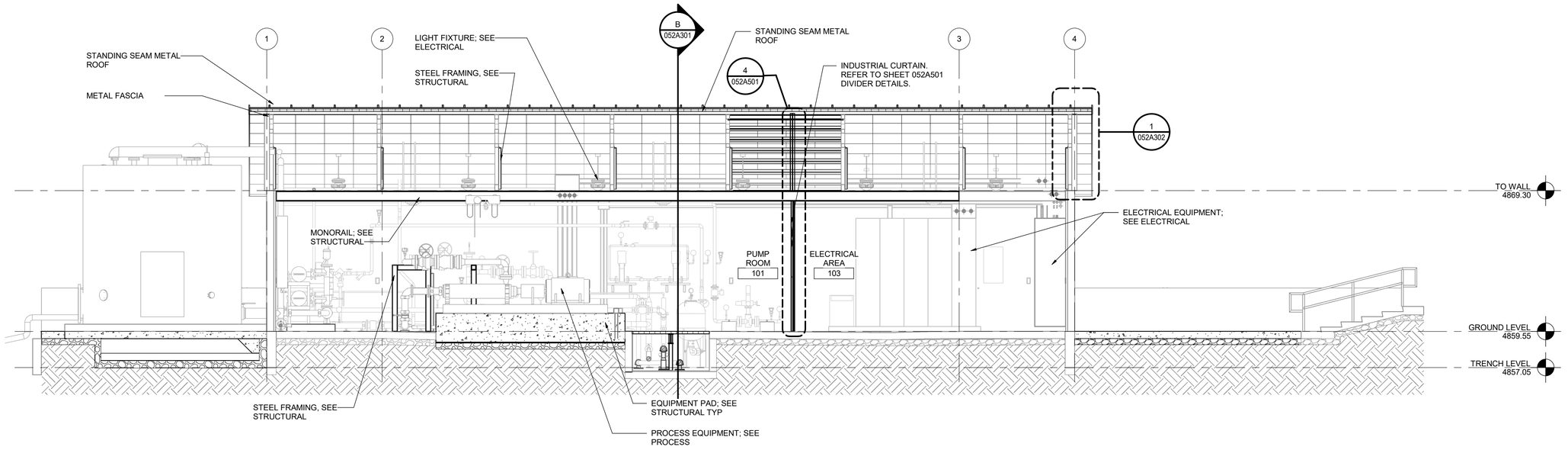
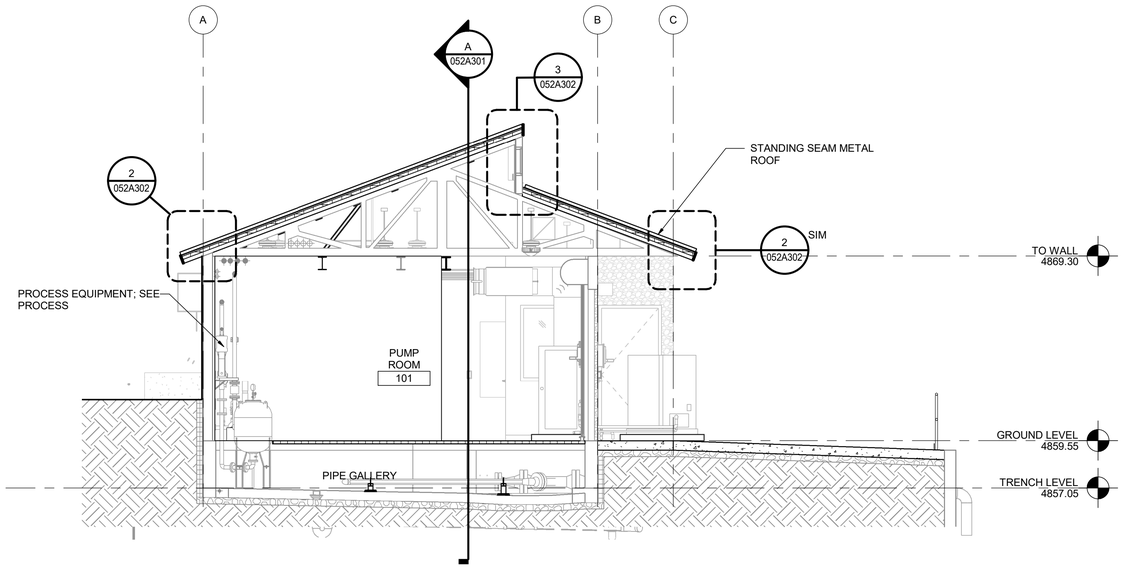
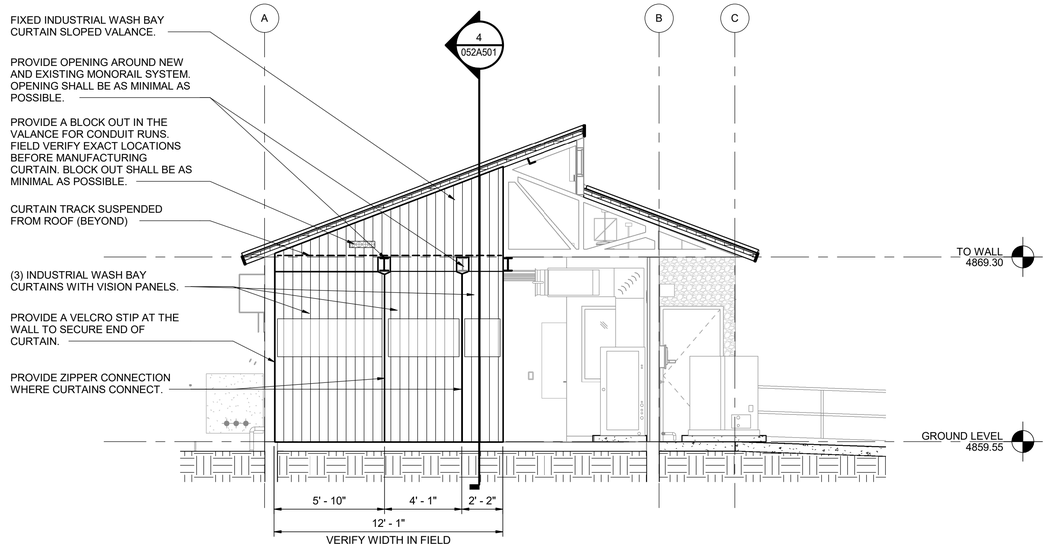


A 052A301 BUILDING SECTION
SCALE A



B 052A301 BUILDING SECTION
SCALE A



C 052A301 A301/C INDUSTRIAL CURTAIN SECTION
SCALE A

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 2026.10.09 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN CADD D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A301	TITLE OF SHEET RSPH BUILDING SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 53 OF 113

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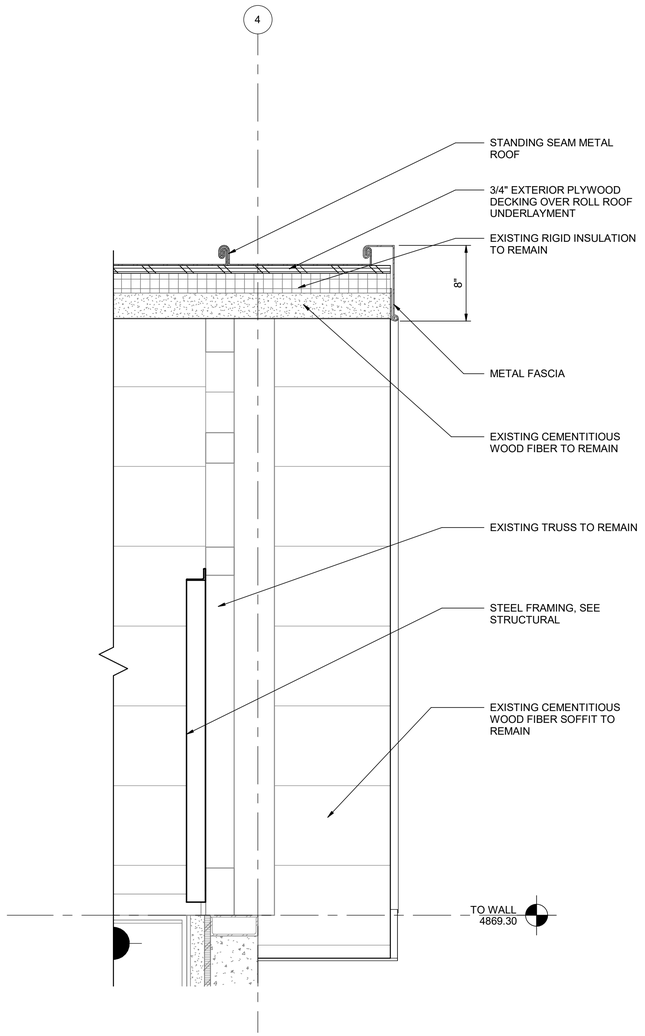
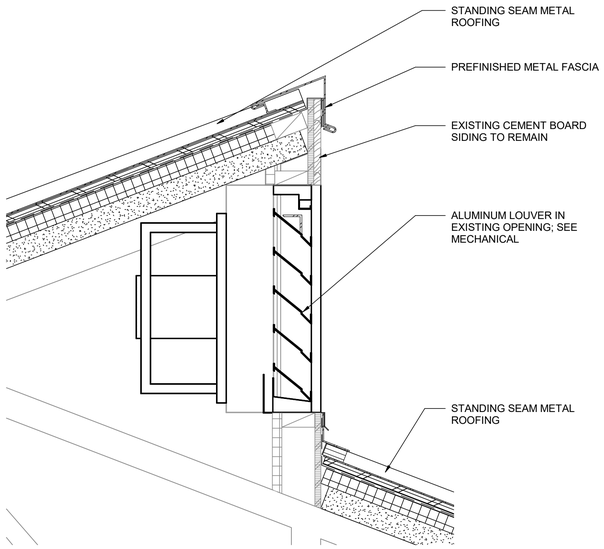
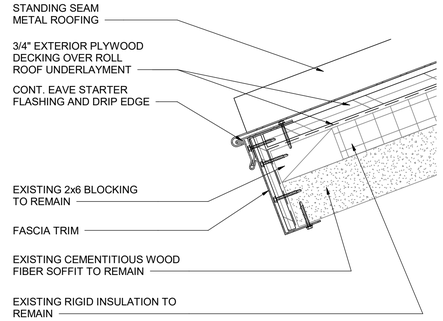


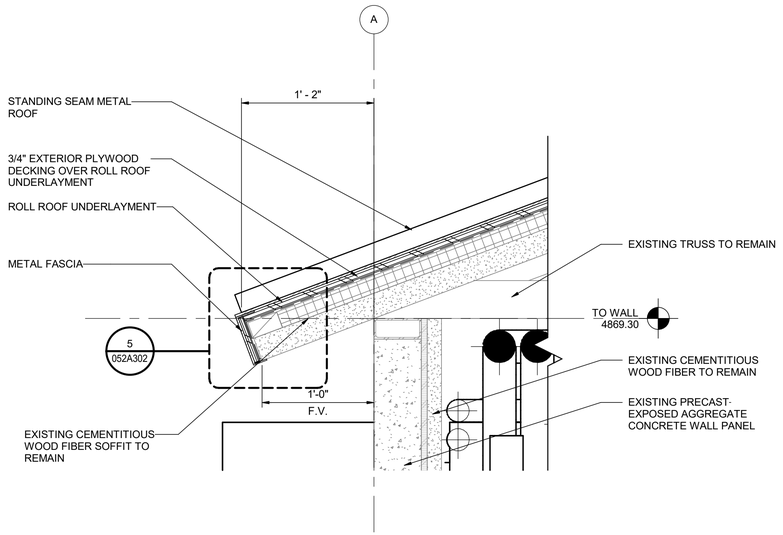
1 052A302 ROOF GABLE END
SCALE (A)



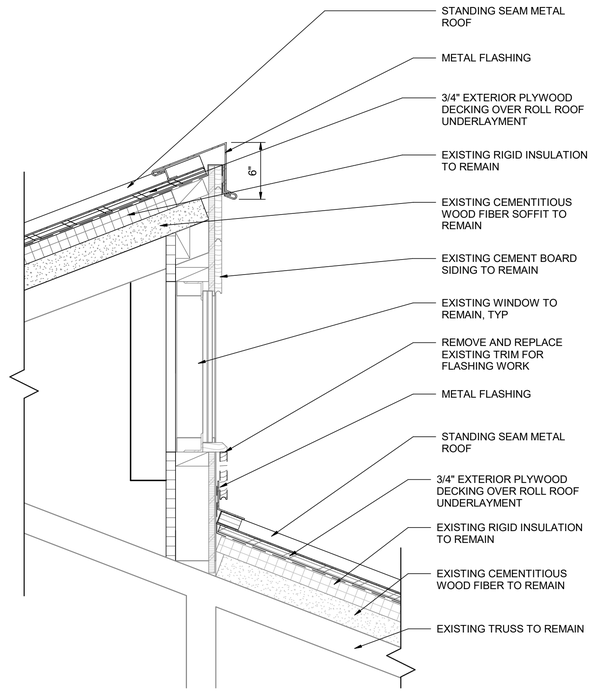
4 052A302 A302/4 LOUVER
SCALE (A)



5 052A302 A302/5 EAVE DETAIL
SCALE (B)



2 052A302 ROOF EAVE
SCALE (A)



3 052A302 ROOF CLERESTORY
SCALE (A)

SCALE (B) 6 3 0 6 — 3" = 1'-0"
SCALE OF FEET

SCALE (A) 12 6 0 12 — 1-1/2" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 04/09/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN CADD: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A302	TITLE OF SHEET RSPH WALL SECTION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 54 OF 113

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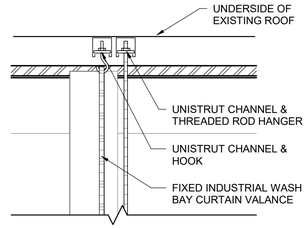
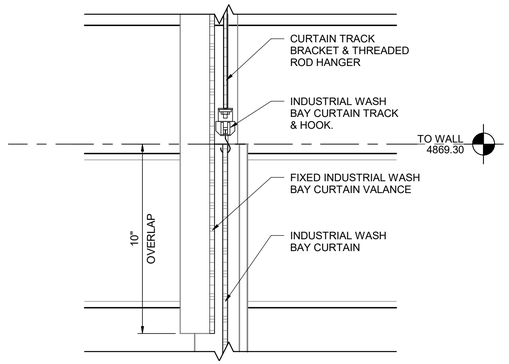
GENERAL NOTES

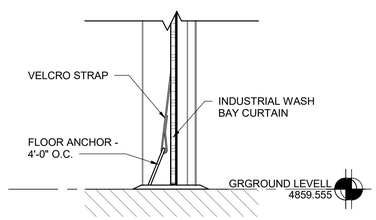
A. THIS SHEET IS SHOWING OPTION ONE FOR THE ROOM DIVIDER. SEE SHEET 052A502 FOR OPTION TWO.



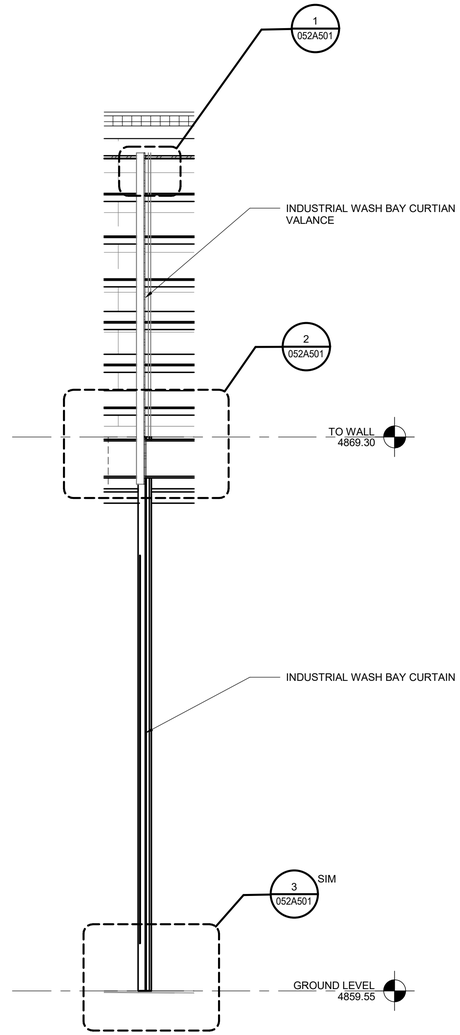
1 052A501 INDUSTRIAL CURTAIN ROOF ATTACHMENT
SCALE B



2 052A501 INDUSTRIAL CURTAIN - VALANCE/ CURTAIN OVERLAP
SCALE B



3 052A501 INDUSTRIAL CURTAIN - FLOOR ANCHORS
SCALE B



4 052A301 A501/4 INDUSTRIAL CURTAIN WALL SECTION
SCALE A

SCALE B 6 3 0 6 — 3" = 1'-0"
SCALE OF FEET

SCALE A 1 0 1 2 3 — 3/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 4/16/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN DRAWN: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A501	TITLE OF SHEET RSPH INDUSTRIAL CURTAIN DETAIL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 55 OF 113

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1 052A901 ISOMETRIC RENDERING

NOT TO SCALE

LICENSED ARCHITECT
CERTIFICATE NO.
85788
RICHARD MCKINLEY
DATE SIGNED 05/04/26
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: E. BUTTMAN DRAFTED: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 052 A901	TITLE OF SHEET RSPH 3D REPRESENTATION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 56 OF 113

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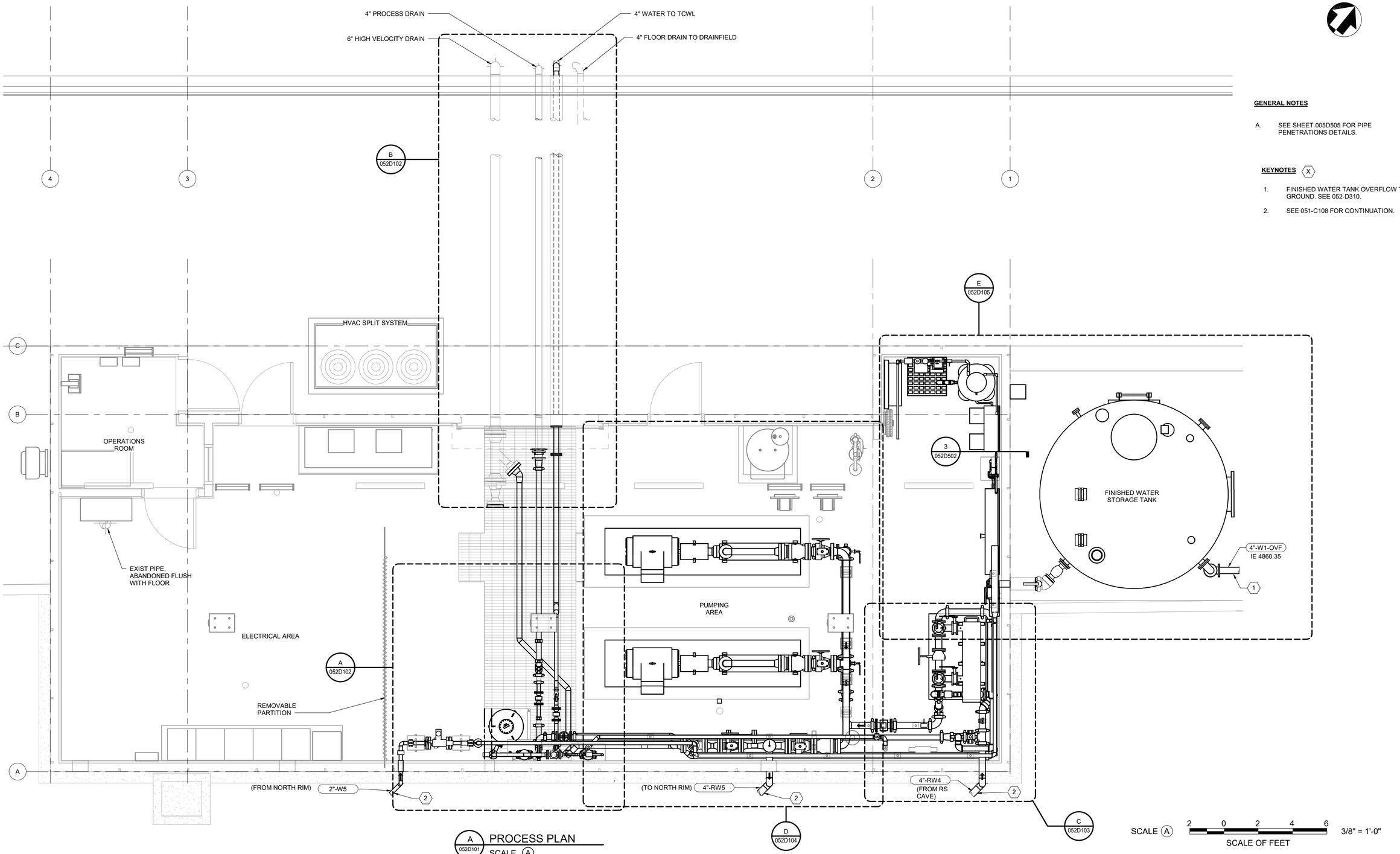

A 052D101 PROCESS PLAN

SCALE A

SCALE A 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATIONS DETAILS.

KEYNOTES (X)

1. FINISHED WATER TANK OVERFLOW TO GROUND. SEE 052-D310.
2. SEE 051-C108 FOR CONTINUATION.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D101	TITLE OF SHEET RSPH PROCESS FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 57 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_D.rvt
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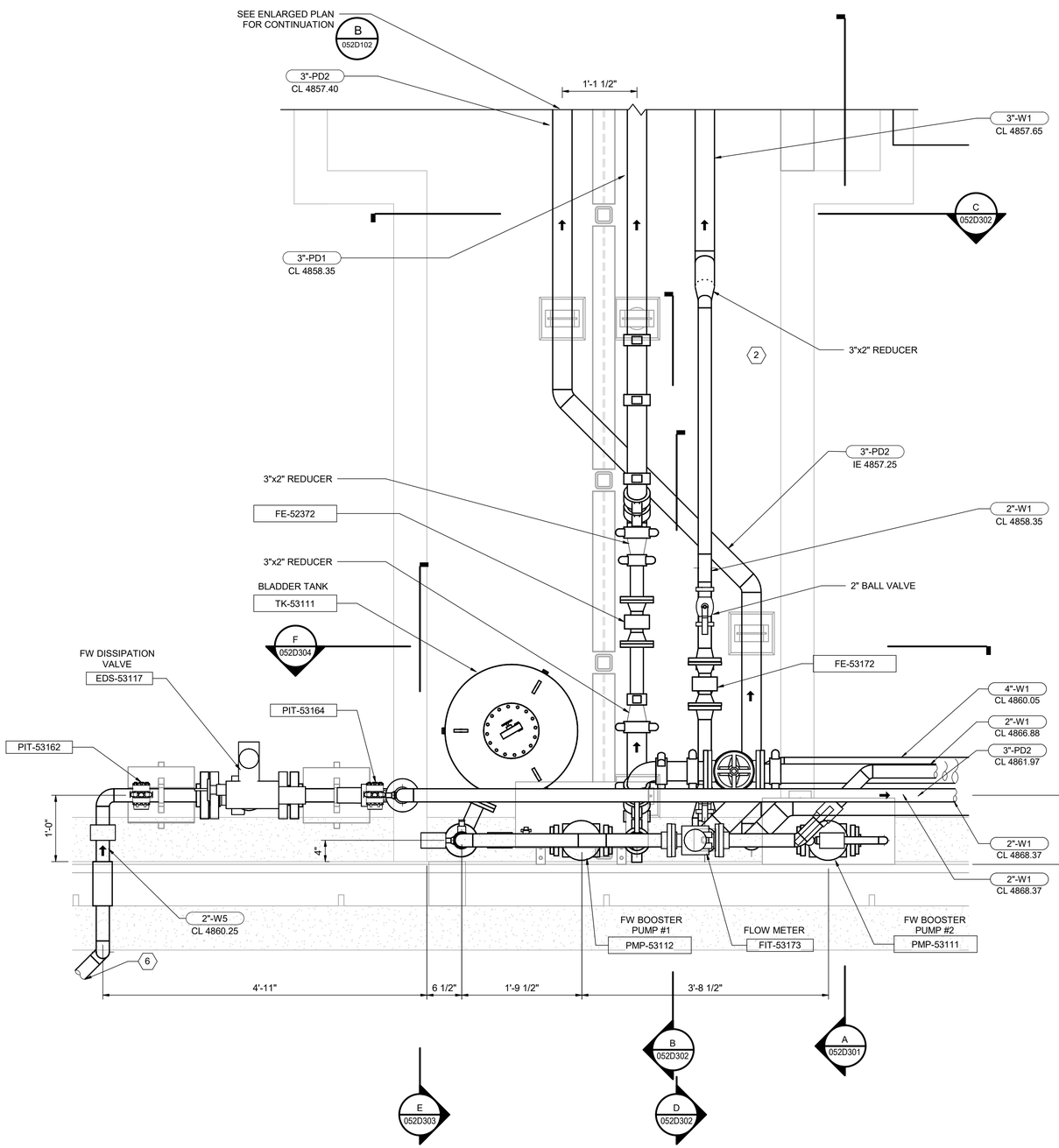
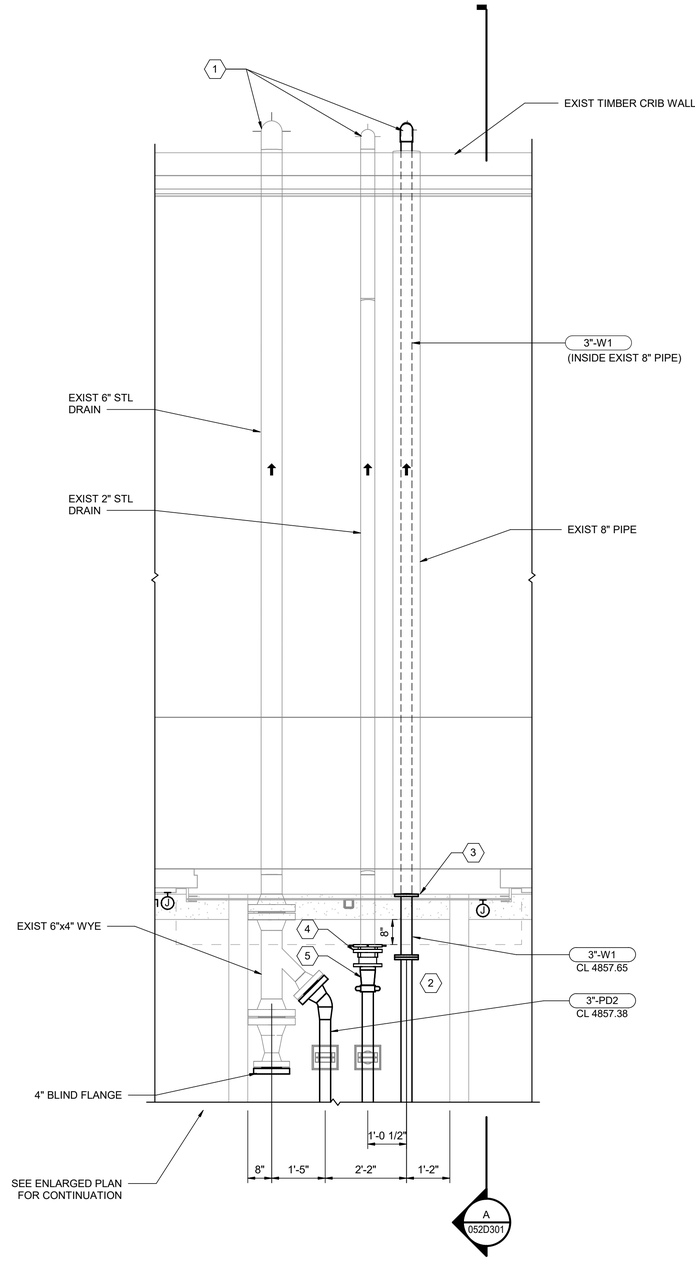


A 052D101 ENLARGED PLAN VIEW
SCALE (A)



B 052D101 ENLARGED PLAN VIEW
SCALE (B)

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATIONS DETAILS.

KEYNOTES (X)

1. SEE SHEET 051-C103 FOR CONTINUATION.
2. GRATING NOT SHOWN FOR CLARITY.
3. CUT FLUSH TO THE WALL.
4. 4" RESTRAINED FLANGE ADAPTOR.
5. 3"x4" ECC REDUCER, FLAT SIDE UP.
6. SEE SHEET 051-C108 FOR CONTINUATION.

SCALE (B) 2 0 2 4 — SCALE OF FEET — 1/2" = 1'-0"

SCALE (A) 1 0 1 2 — SCALE OF FEET — 1" = 1'-0"

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 9-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO CADD C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D102	TITLE OF SHEET RSPH PROCESS ENLARGED PLANS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 58 OF 113

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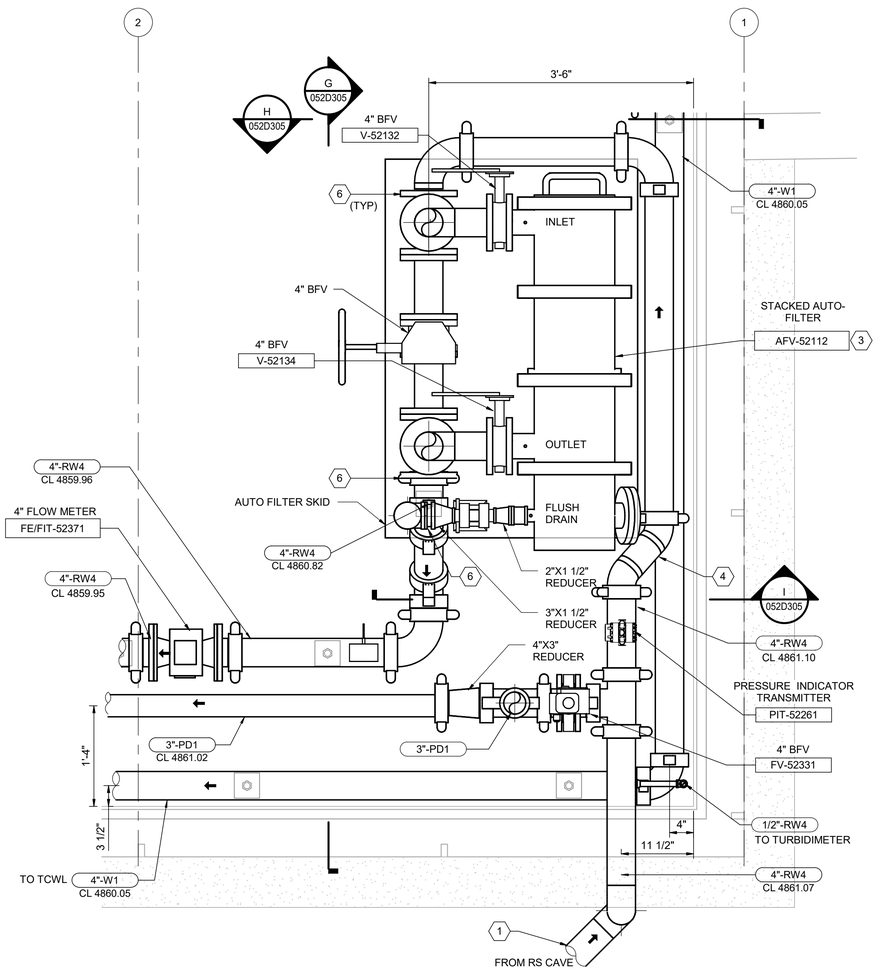
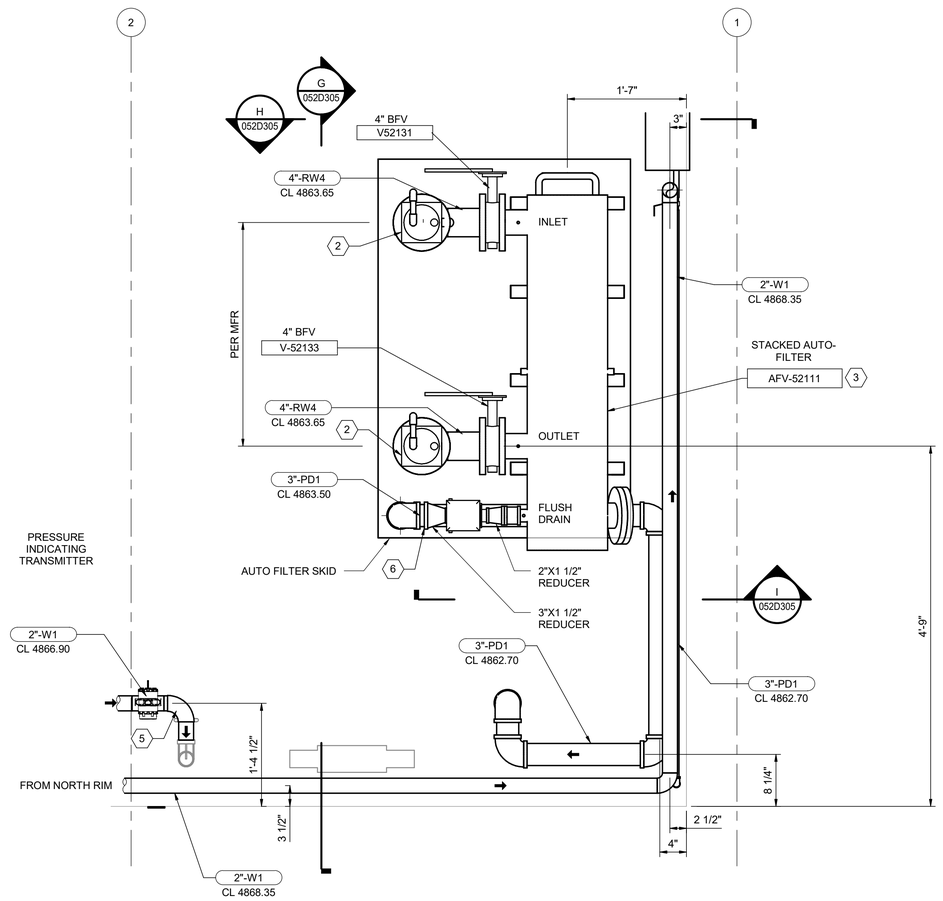


C 052D101 AUTO FILTER LOWER LEVEL PLAN VIEW AT ELEVATION 4859.55

SCALE A

PRESSURE
INDICATING
TRANSMITTER



C 052D101 AUTO FILTER UPPER LEVEL PLAN VIEW AT ELEVATION 4862.50

SCALE A

GENERAL NOTES

A. SEE SHEET 005D505 (VOL 0) FOR PIPE PENETRATIONS DETAILS.

B. DRAIN LINE FROM INSTRUMENTATION PANEL NOT SHOWN FOR CLIARITY. FIELD ROUTE AGAINST WALL AT ELEVATIONS SHOWN ON PROCESS SHEETS.

KEYNOTES (X)

1. SEE SHEET 051-C108 FOR CONTINUATION.
2. 1" AIR RELEASE VALVE WITH ORIFICE DIAMETER OF 1/16".
3. INSTALL FLANGE INSULATION KIT BETWEEN SS AUTOFILTER MANIFOLD AND CONNECTING CARBON STEEL PIPE AND FITTINGS.
4. INSTALL SHOP-WELDED DOUBLE ELBOW ASSEMBLY WITH GROOVED OR FLANGE ENDS TO OFFSET PIPE CENTERLINE ALIGNMENT HORIZONTALLY AND VERTICALLY.
5. TRANSITION PRESSURE BOOSTED W1 FROM PROCESS TO MECHANICAL.
6. LIMITS OF AUTO FILTER SKID PIPING CONNECTIONS.

SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D103	TITLE OF SHEET RSPH PROCESS ENLARGED PLANS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 59 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_D.rvt
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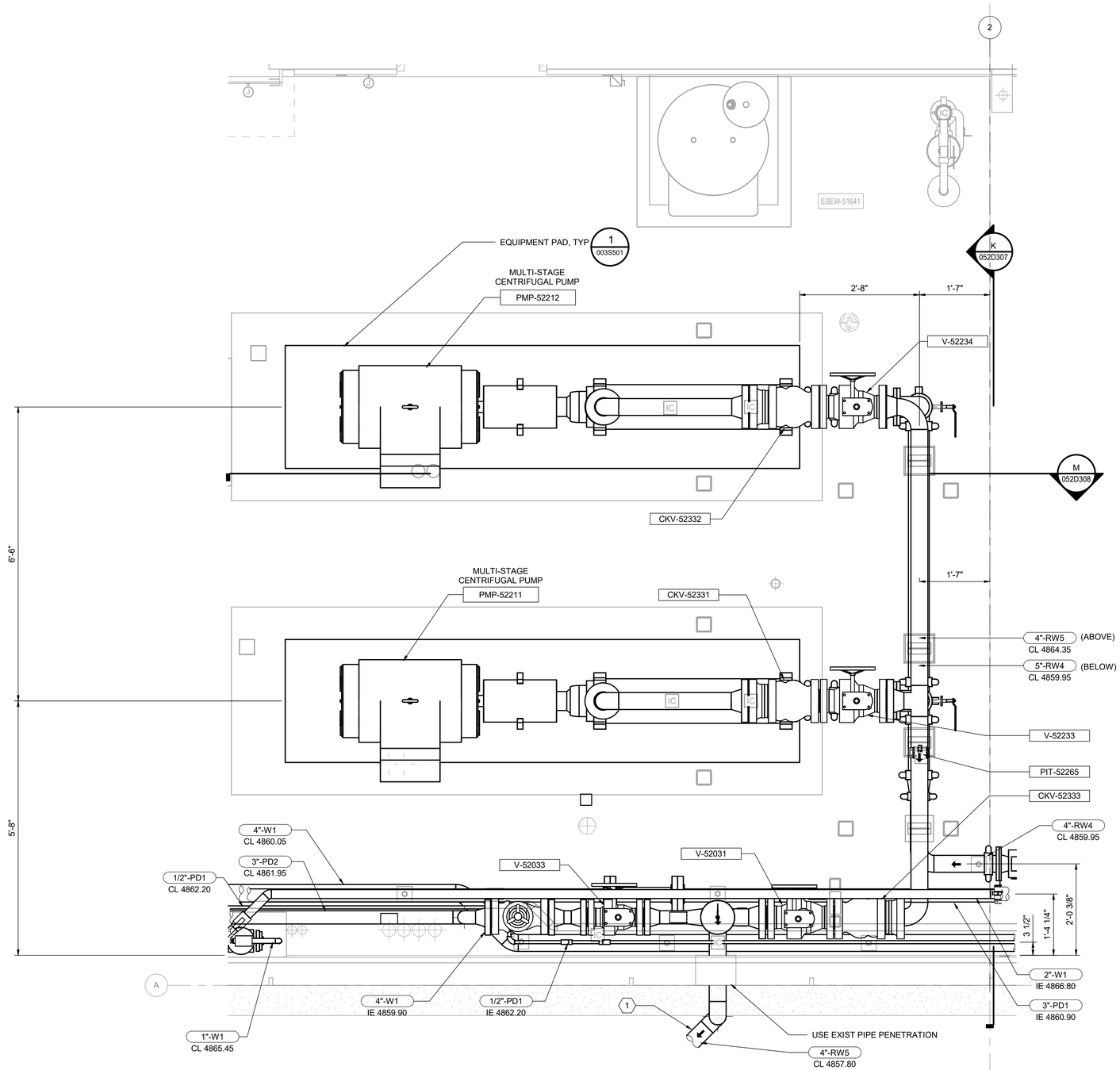

D 052D101 ENLARGED PLAN VIEW
SCALE A

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATIONS DETAILS.

B. COORDINATE RW PUMP CONDUIT CONCRETE PENETRATIONS WITH PUMP LOCATION AND MOTOR DIMENSIONS.

KEYNOTES X

1. SEE SHEET 051-C108 FOR CONTINUATION.

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5/1/2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

2 A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D104	TITLE OF SHEET RSPH PROCESS ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 60 OF 113

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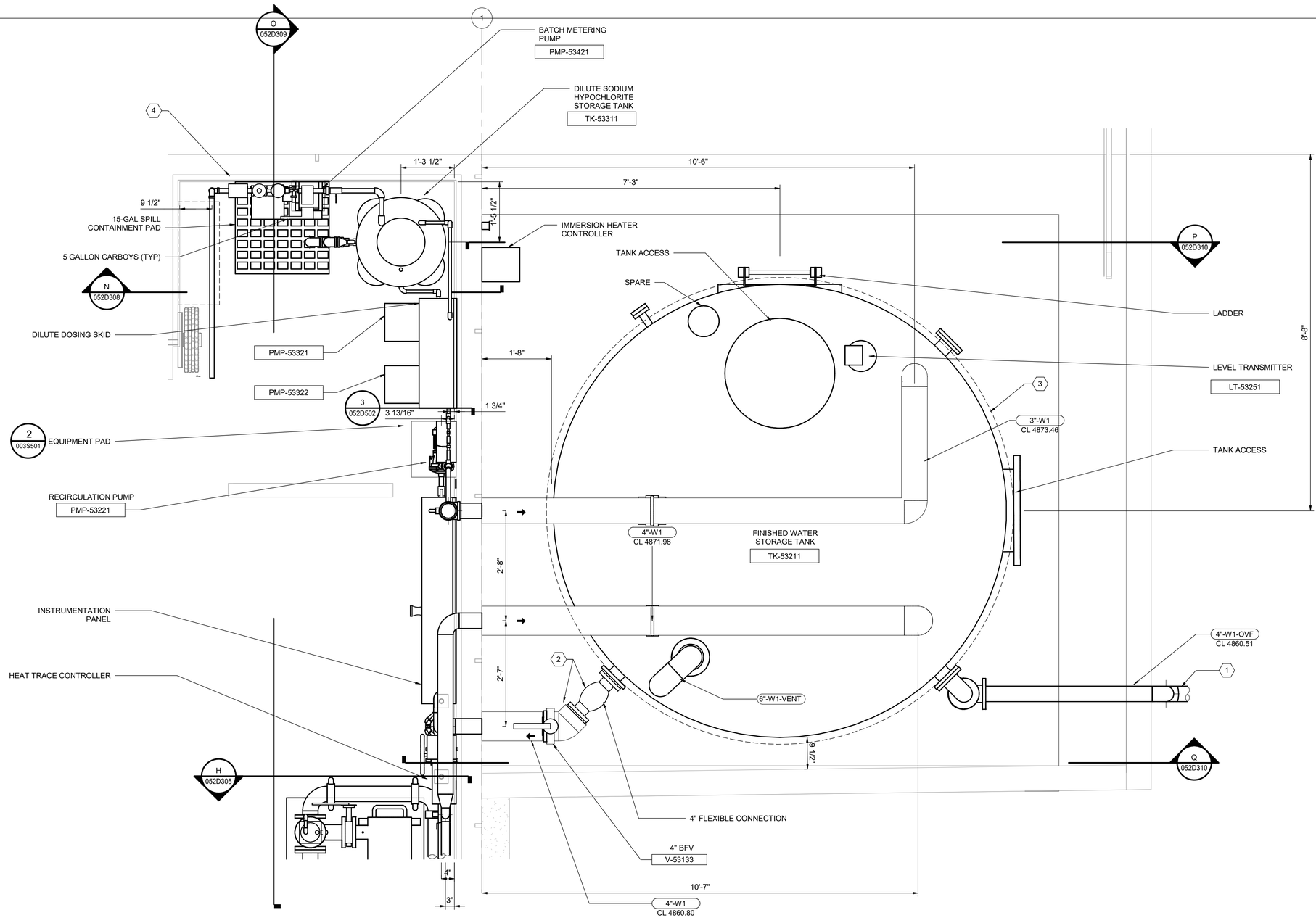

E 052D101 ENLARGED PLAN VIEW
SCALE A

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATIONS DETAILS.

KEYNOTES (X)

1. FINISHED WATER STORAGE TANK OVERFLOW TO GROUND. SEE 052D310.
2. INSULATION NOT SHOWN FOR CLARITY. INSULATE IN ACCORDANCE WITH SPEC SECTION 40 42 00.
3. LIMITS OF TANK INSULATION.
4. RESERVE 34" W x 28" H x 12" D SPACE FOR FUTURE WATER SOFTENER.

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 ARIZONA, U.S.A.
Nicholas Humphrey

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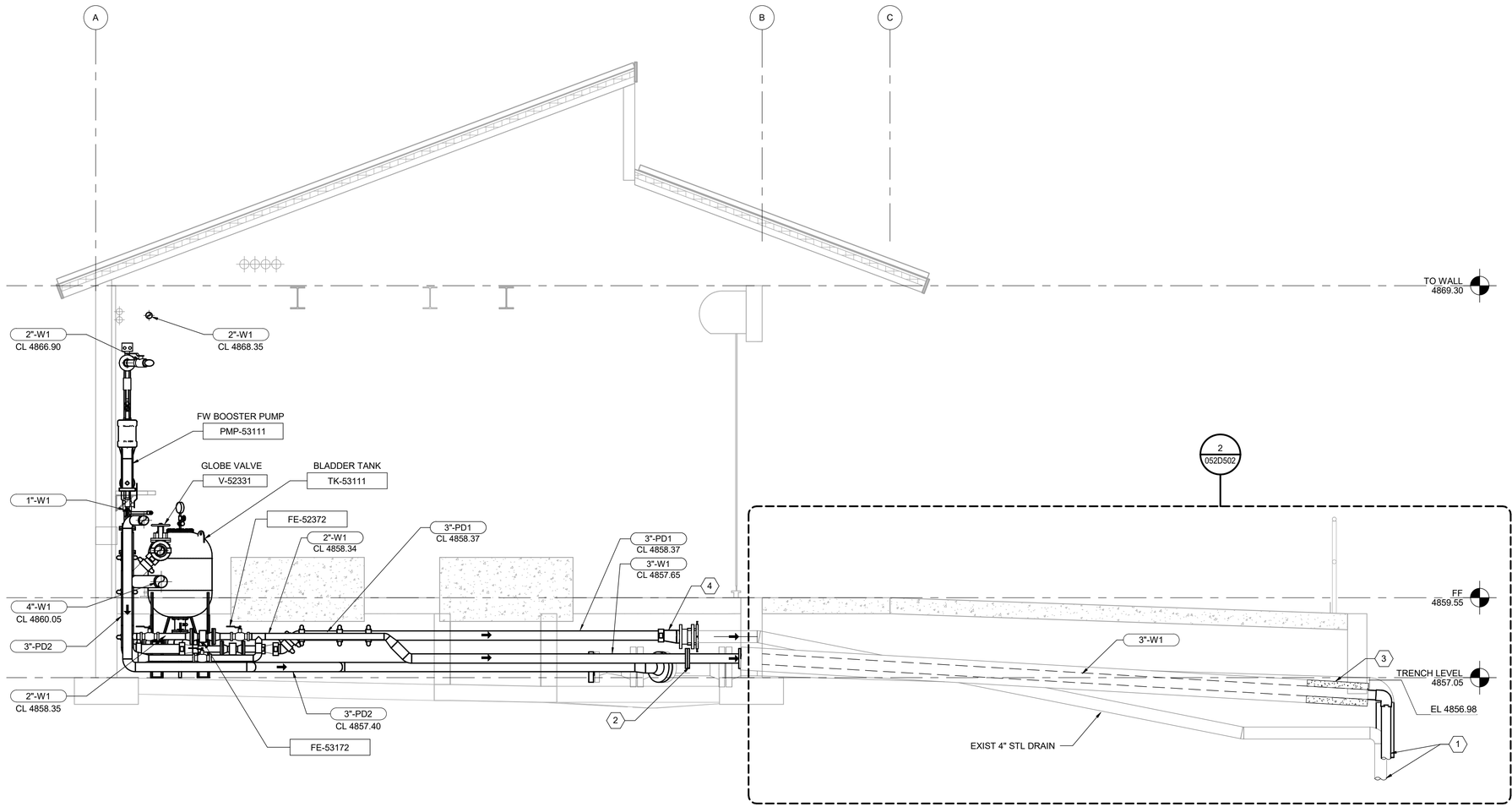
KEYNOTES (X)

1. SEE SHEET 051-C103 FOR CONTINUATION.
2. W-1 FLANGED TRANSITION POINT CS TO HDPE.
3. FILL MIN. 8" DEPTH OF ANNULAR SPACE WITH PIPE INSULATION. CAULK WITH SILICONE SEALENT.
4. 3"x4" ECC REDUCER, FLAT SIDE UP.



A 052D102 TRENCH PIPING SECTION
SCALE (A)

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO CADD C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D301	TITLE OF SHEET RSPH PROCESS SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 62 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_D.rvt
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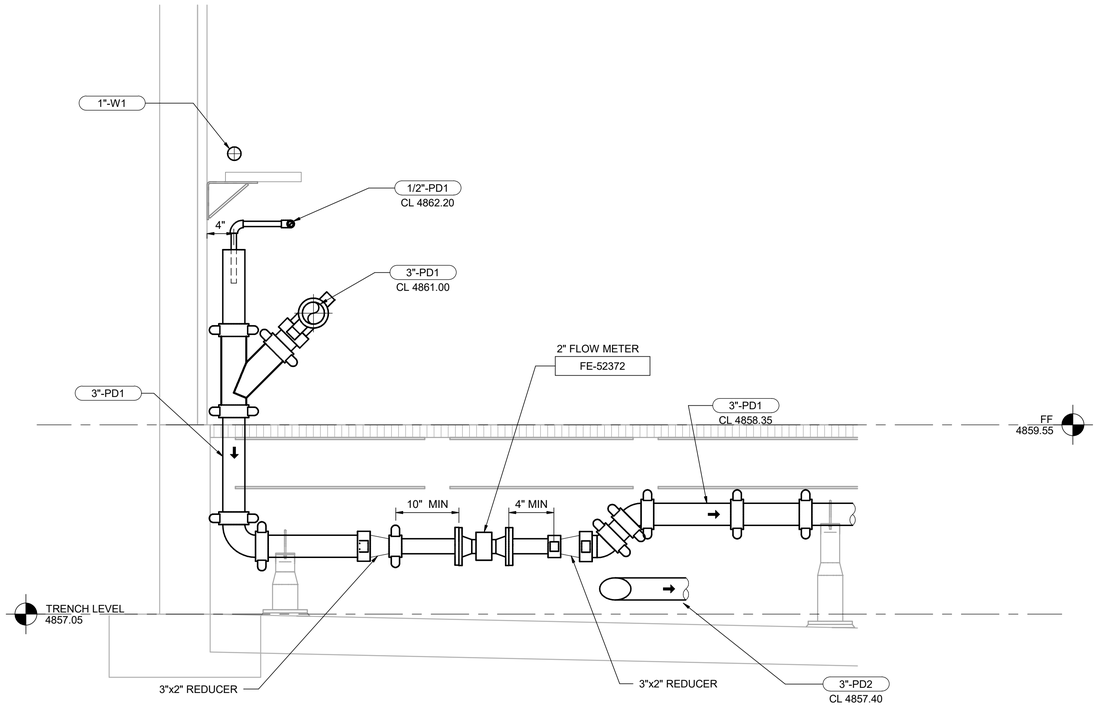
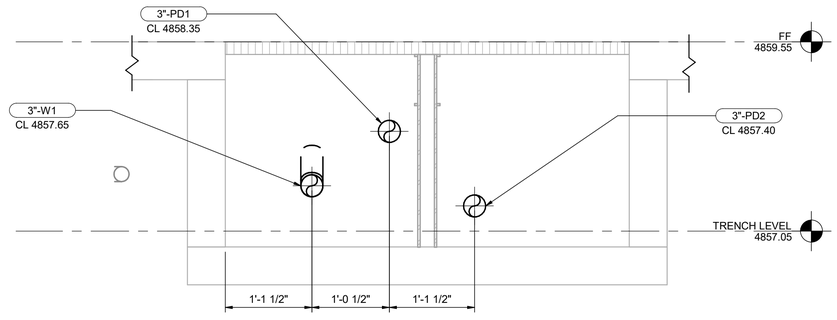


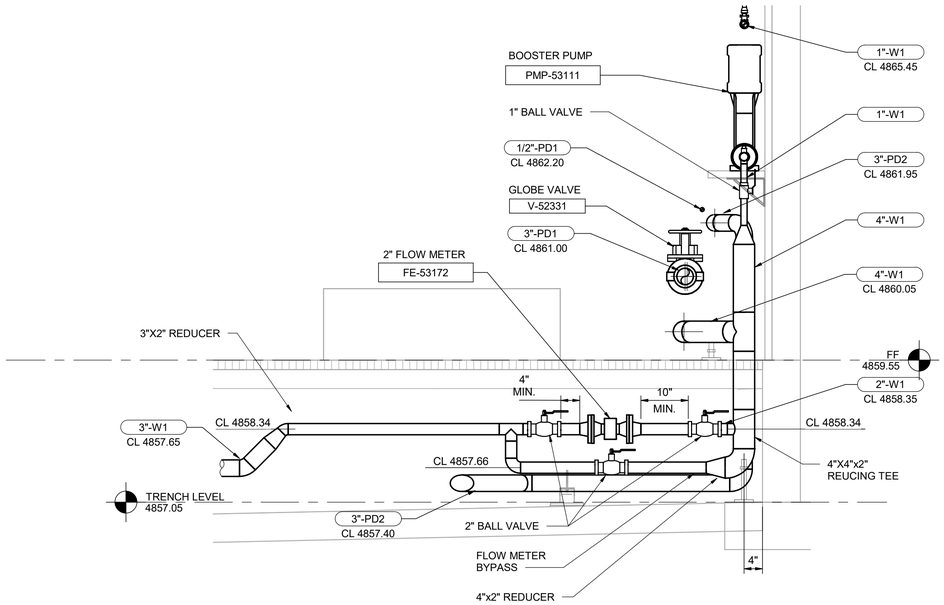
B 052D102 PROCESS DRAIN FLOW METER SECTION

SCALE A



C 052D102 TRENCH PIPING SECTION

SCALE A



D 052D102 FINISHED WATER FLOW METER SECTION

SCALE A

SCALE A 1 0 1 2 1" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO CADD C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D302	TITLE OF SHEET RSPH PROCESS SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752

PMIS/PKG NO. 318719

SHEET 63 OF 113

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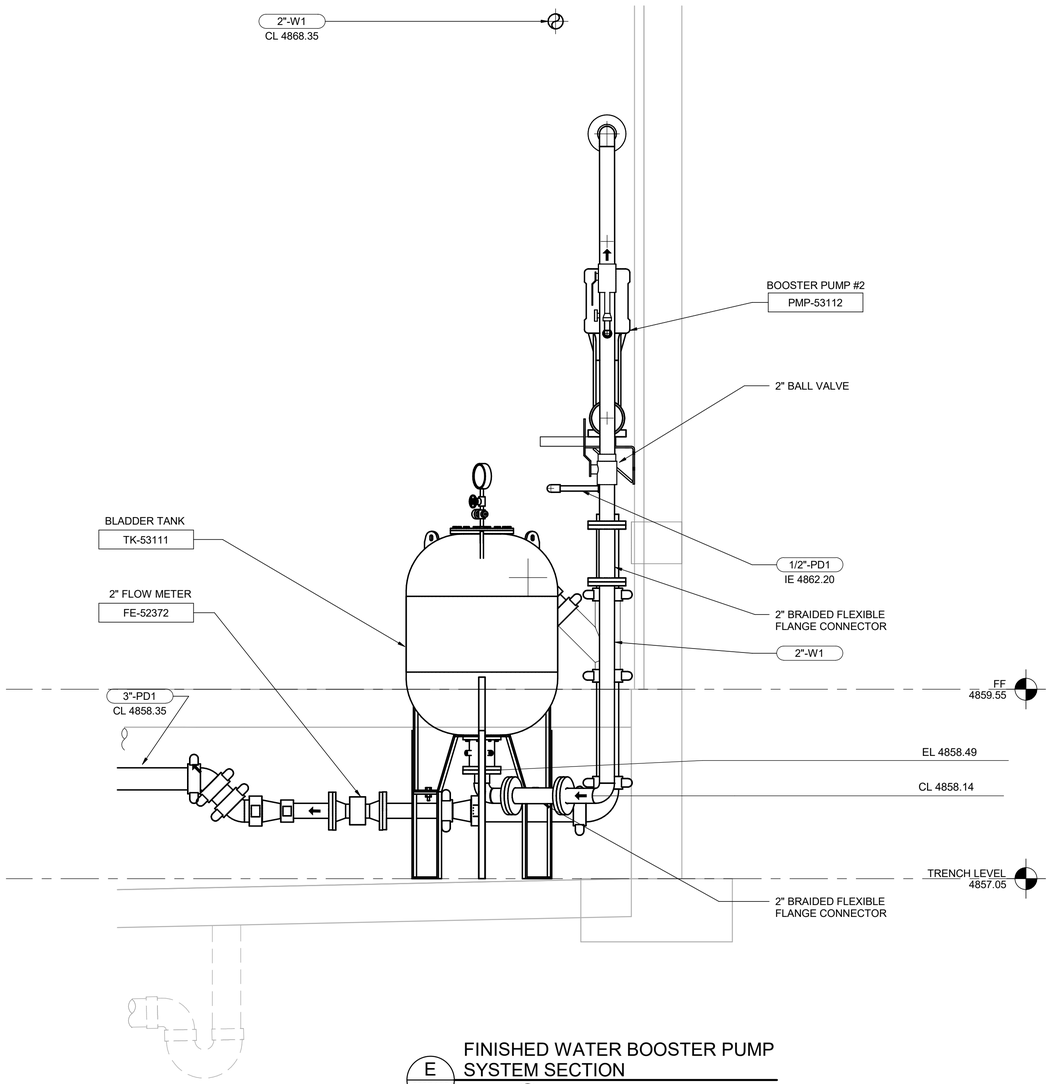

E
052D102

FINISHED WATER BOOSTER PUMP SYSTEM SECTION

SCALE (A)

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATION DETAILS.

SCALE (A) 1 0 1 2 1" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Registered ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D303	TITLE OF SHEET RSPH PROCESS SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 64 OF 113

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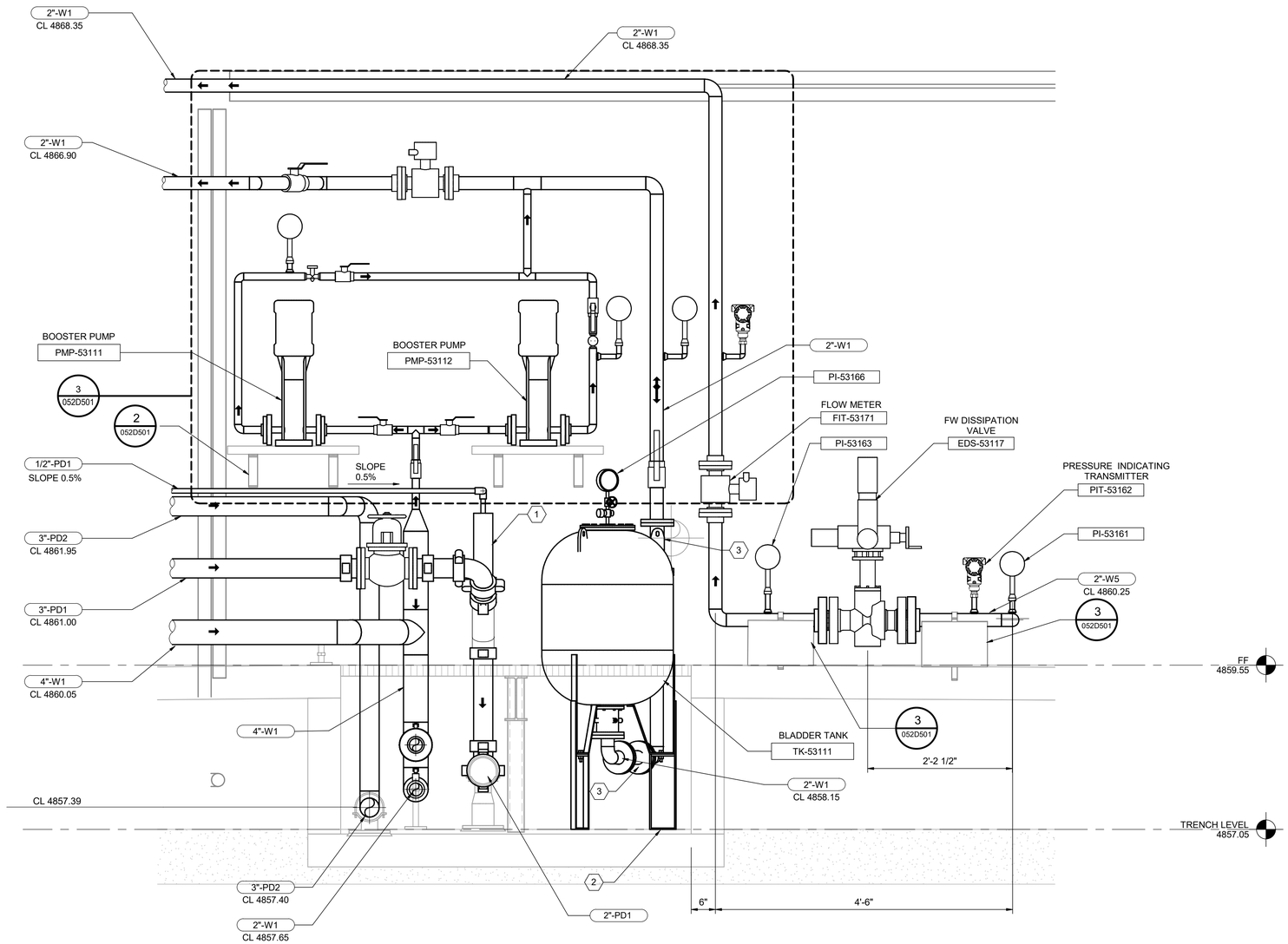
KEYNOTES ⟨X⟩

1. ADD INSECT SCREEN.
2. ADD GROUT PADS TO LEVEL TANK.
3. 2" BRAIDED FLEXIBLE FLANGE CONNECTOR.



F 052D102 FINISHED WATER BOOSTER PUMP SYSTEM SECTION

SCALE (A)

SCALE (A) 1 0 1 2 1" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO CADD C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D304	TITLE OF SHEET RSPH PROCESS SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 65 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_D.rvt
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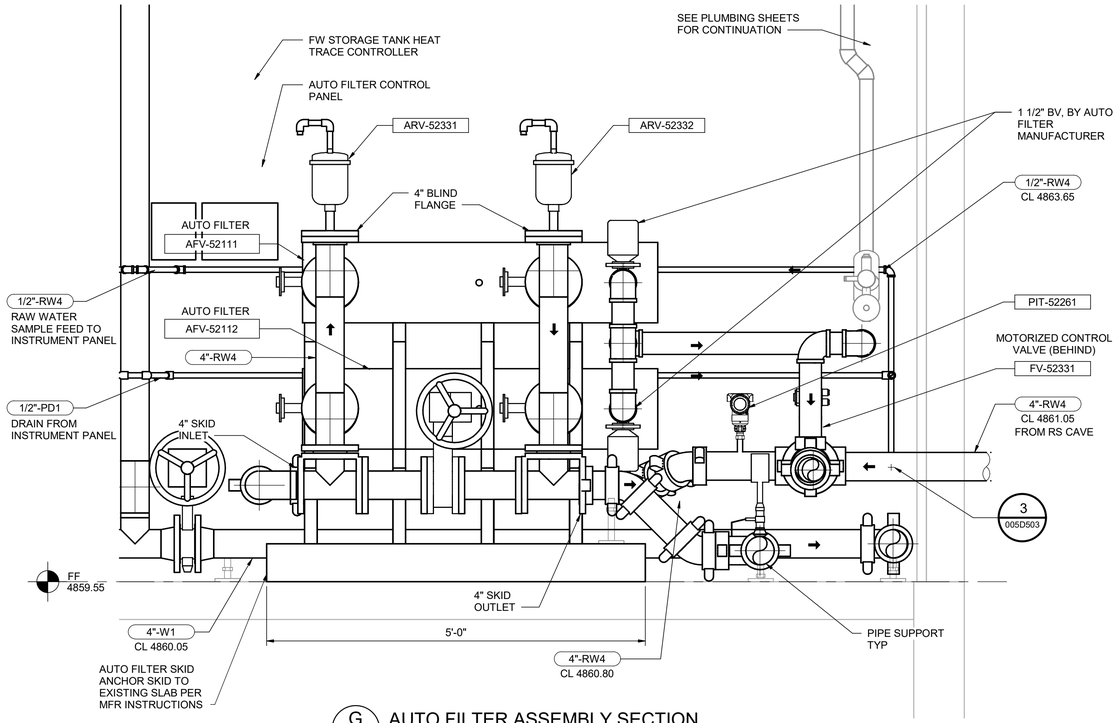
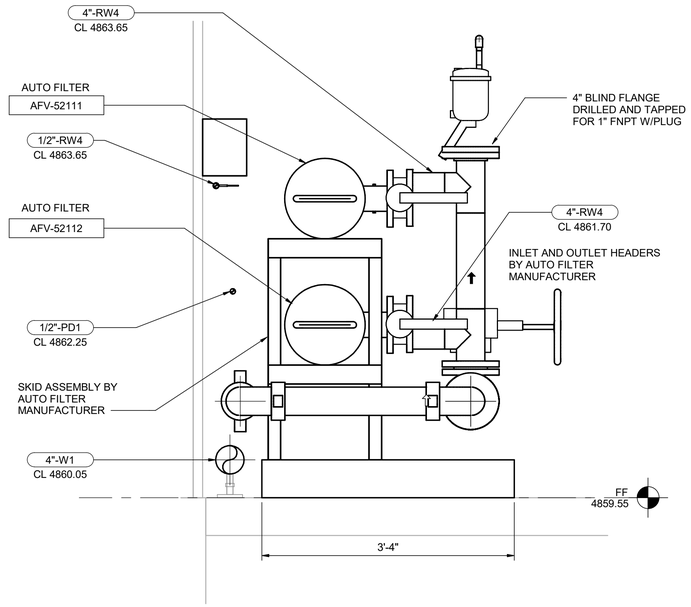
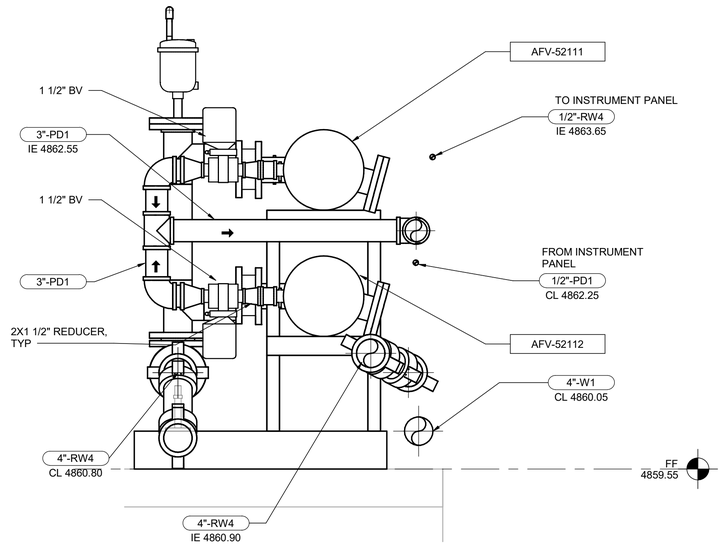


G 052D103 AUTO FILTER ASSEMBLY SECTION
SCALE A



H 052D103 AUTO FILTER ASSEMBLY SECTION
SCALE A



I 052D103 AUTO FILTER BACKWASH PIPING SECTION
SCALE A

SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Designed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D305	TITLE OF SHEET RSPH PROCESS SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 66 OF 113

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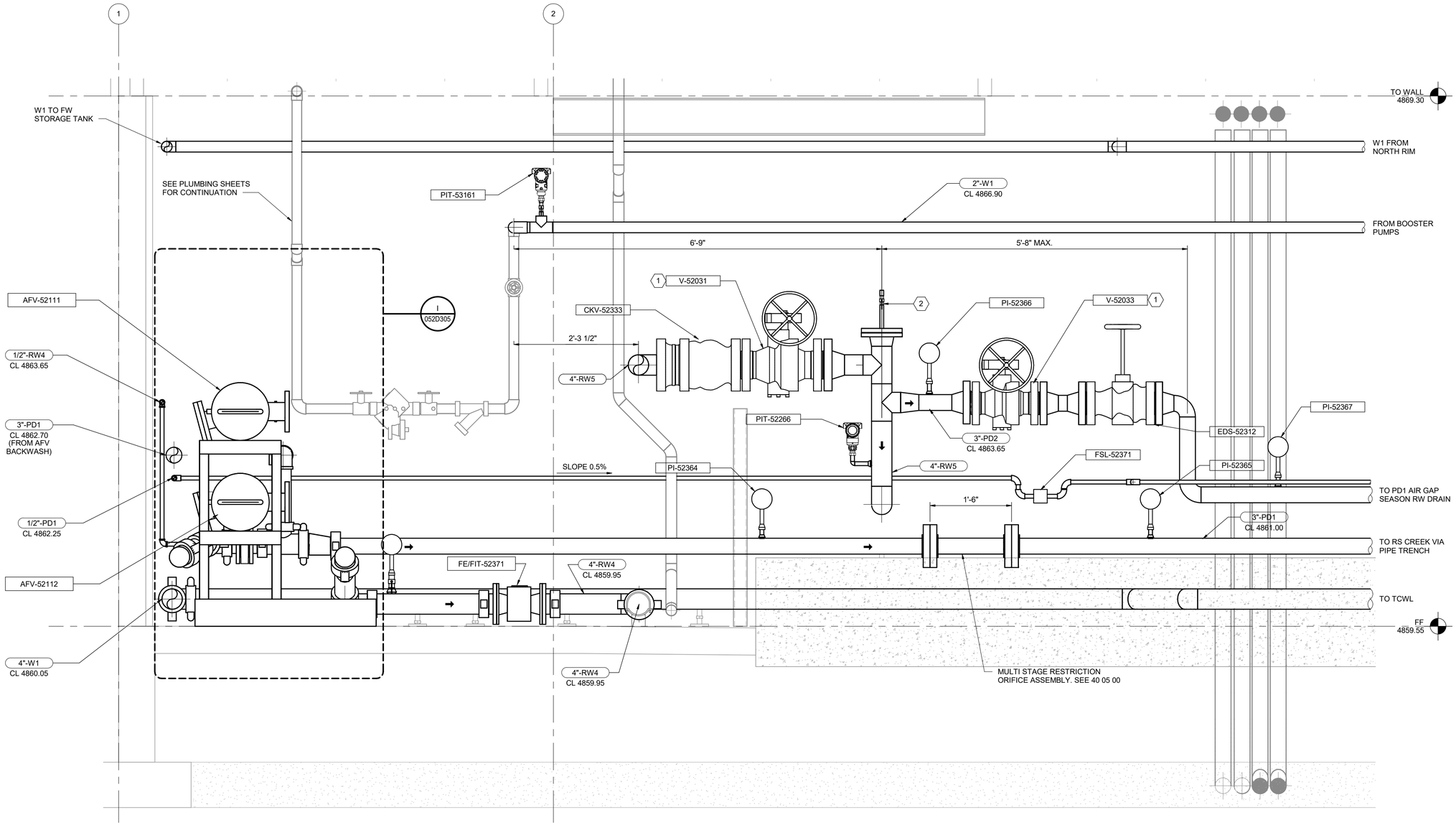
KEYNOTES (X)

1. HIGH PRESSURE ISOLATION BALL VALVE. 1/2" BYPASS PIPING NOT SHOWN FOR CLARITY.
2. TAP BLIND FLANGE AND ADD 1/2" NC BALL VALVE FOR AIR VENTING.



J SOUTH WALL INTERIOR SECTION
SCALE (A)

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not to be Signed ARIZONA, U.S.A.
Nicholas Humphrey

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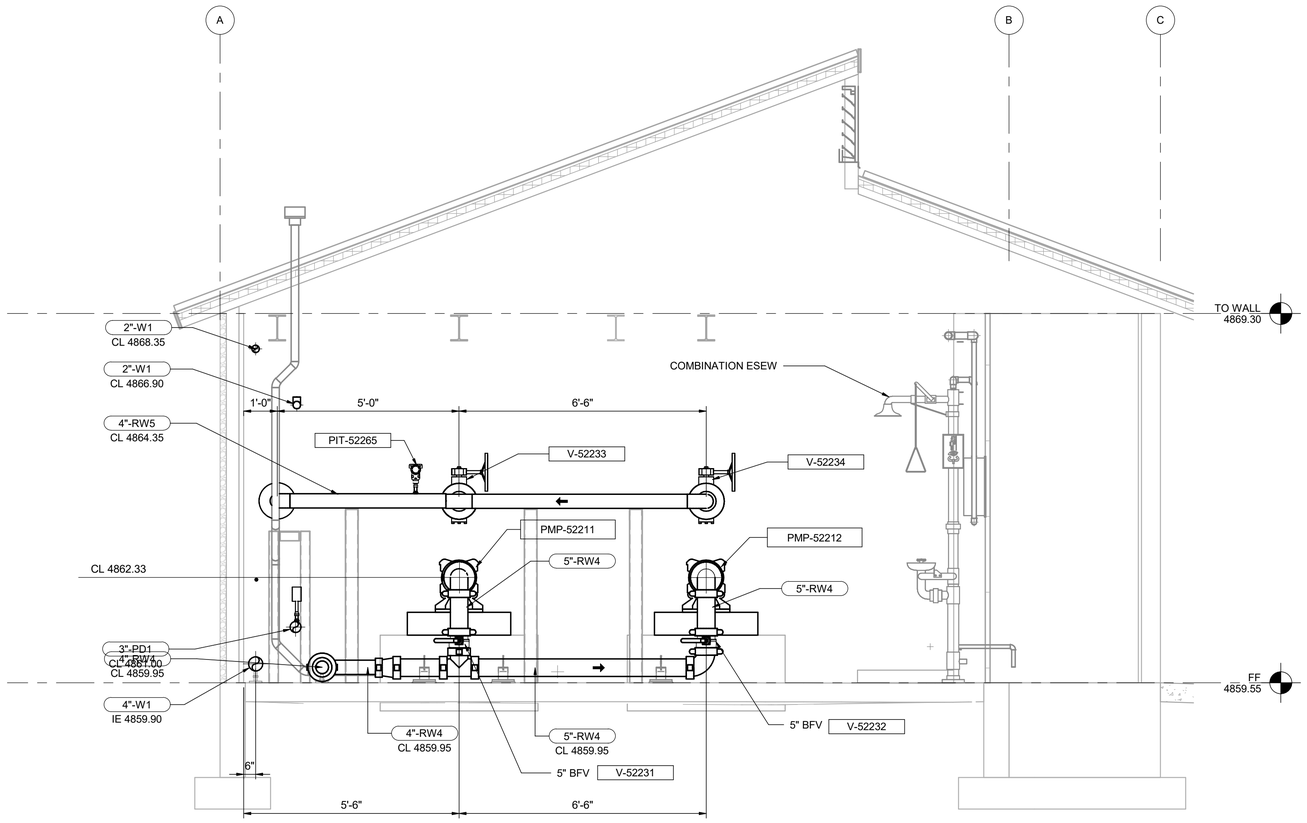

K 052D104 RAW WATER PUMP SECTION
SCALE (A)

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO CADD C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D307	TITLE OF SHEET RSPH PROCESS SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 68 OF 113

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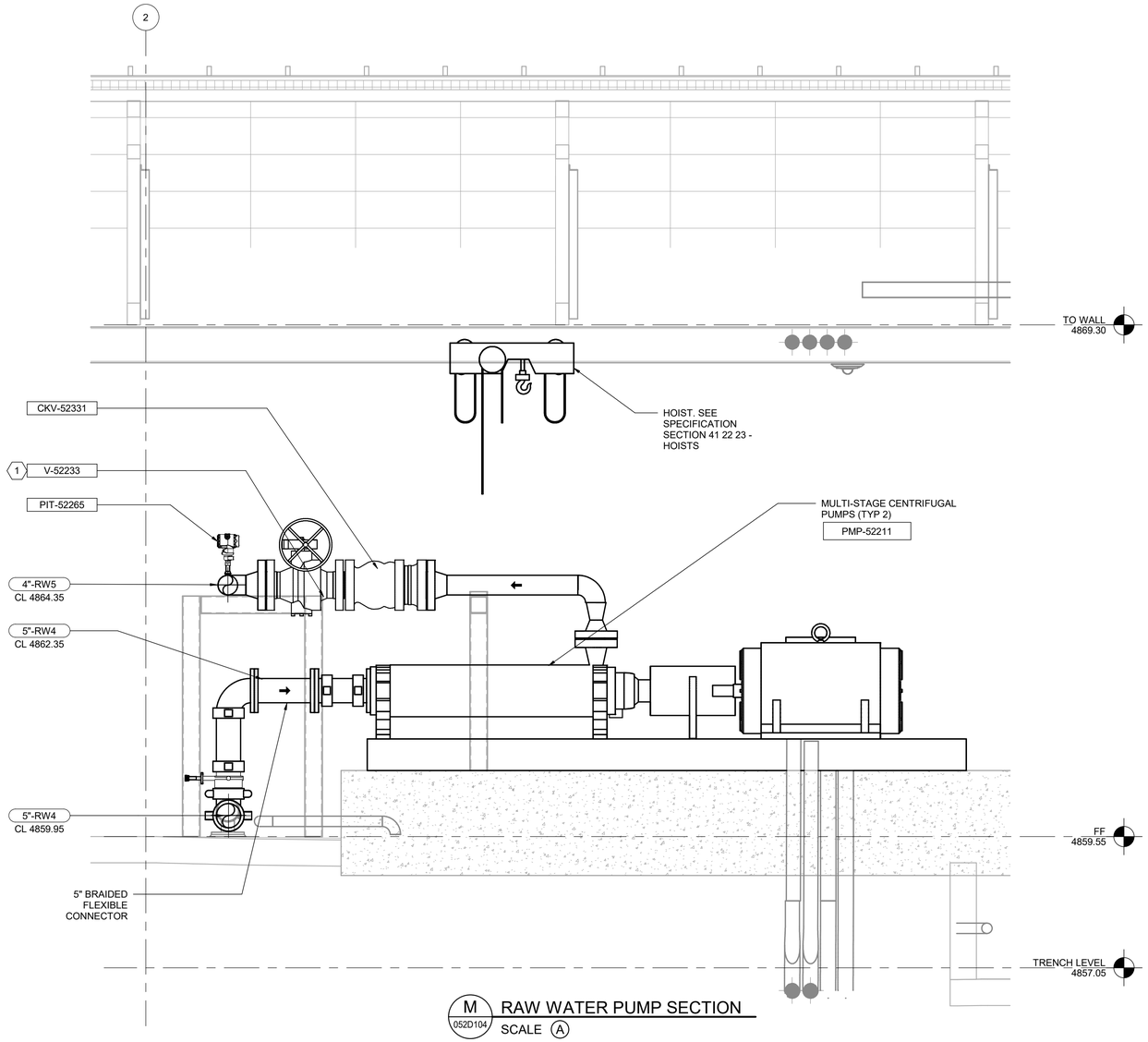


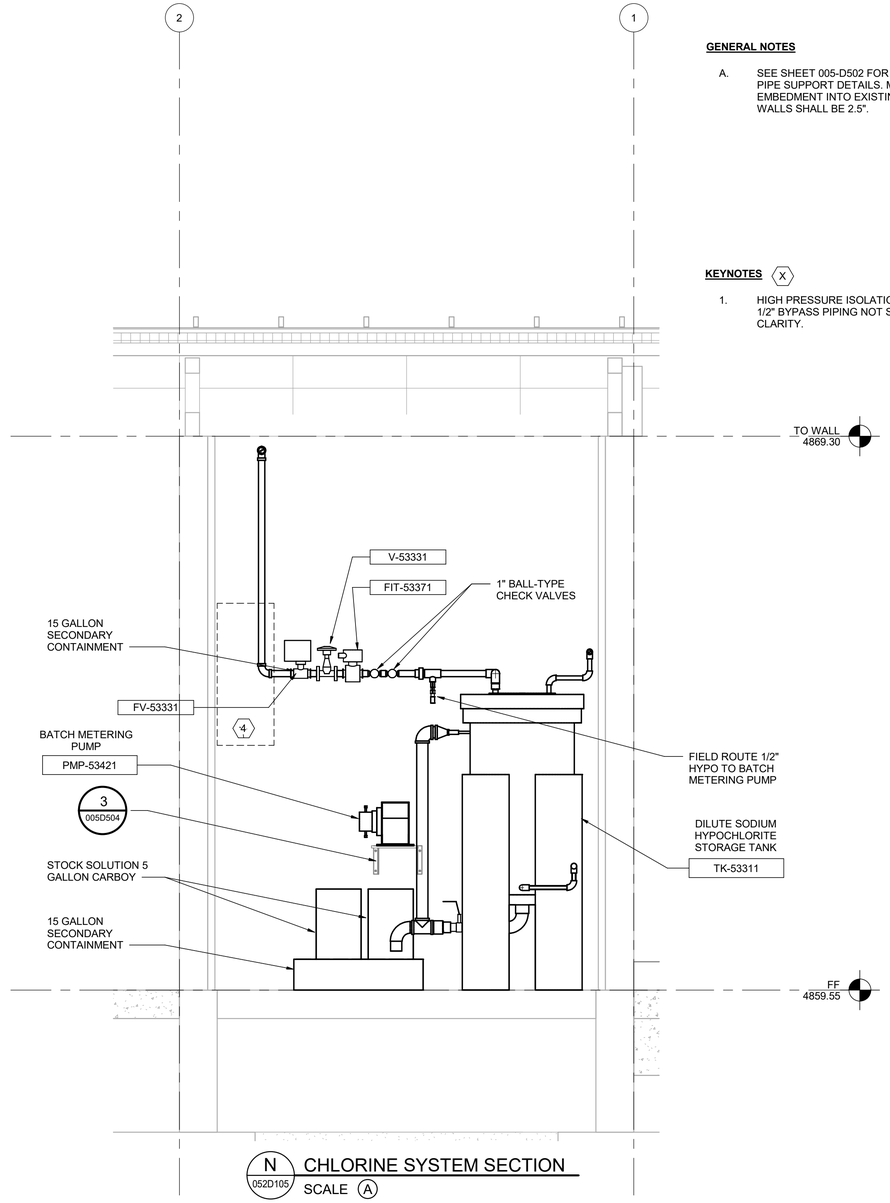
M / 052D104 RAW WATER PUMP SECTION
SCALE A

GENERAL NOTES

A. SEE SHEET 005-D502 FOR WALL MOUNTED PIPE SUPPORT DETAILS. MINIMUM EMBEDMENT INTO EXISTING PRECAST WALLS SHALL BE 2.5".

KEYNOTES X

1. HIGH PRESSURE ISOLATION BALL VALVE. 1/2" BYPASS PIPING NOT SHOWN FOR CLARITY.



N / 052D105 CHLORINE SYSTEM SECTION
SCALE A

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAFTED: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D308	TITLE OF SHEET RSPH PROCESS SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 69 OF 113

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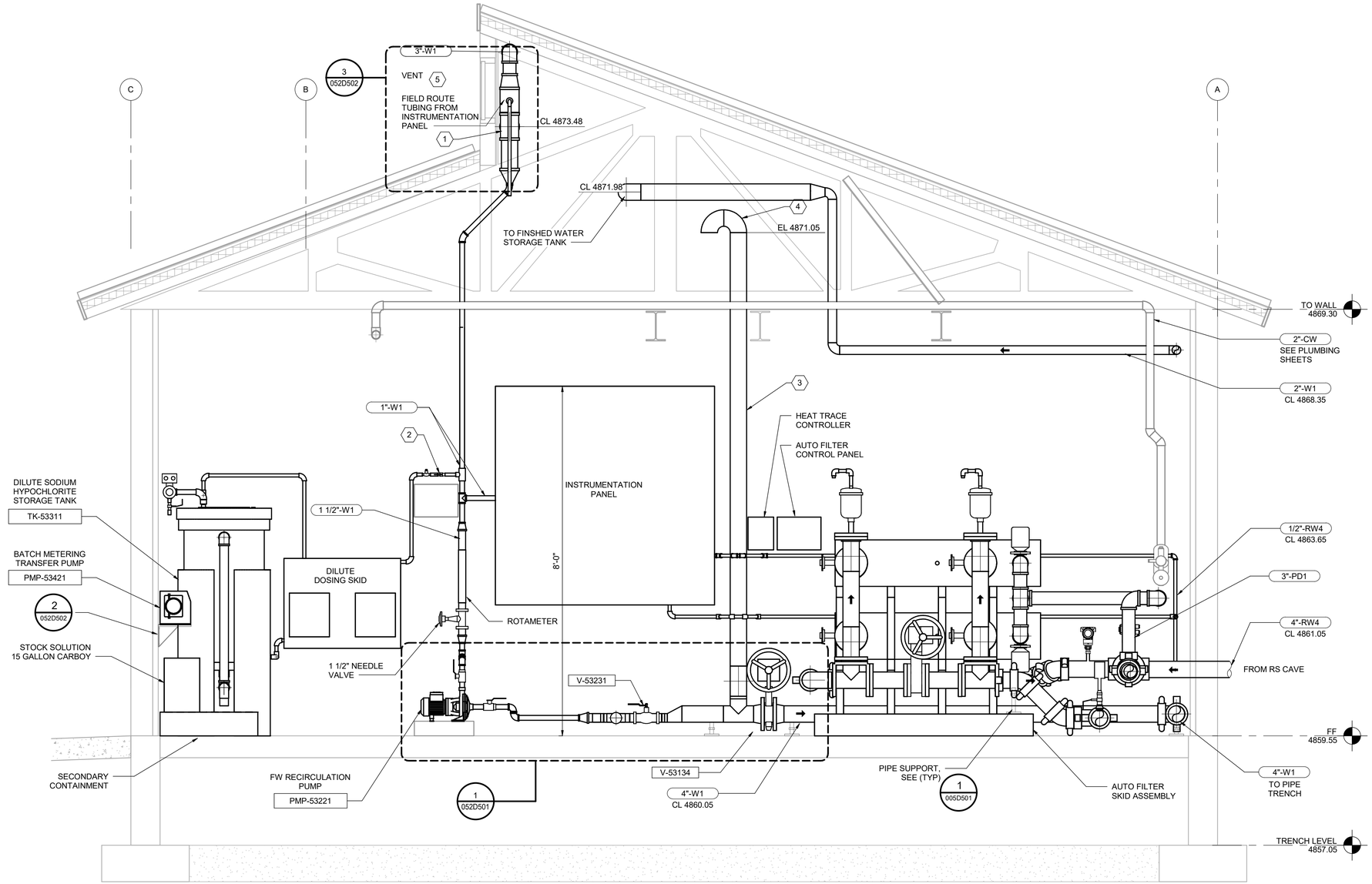

O 052D105 EAST WALL INTERIOR SECTION

SCALE A

GENERAL NOTES

A. SEE SHEET 005-D502 FOR WALL MOUNTED PIPE SUPPORT DETAILS. MINIMUM EMBEDMENT INTO EXISTING PRECAST WALLS SHALL BE 2.5".

KEYNOTES (X)

1. RECIRC PIPE FLOW TRANSITION TO GRAVITY. SEE 052D310.
2. DOSING CONNECTION: TRUE UNION, CHECK VALVE, TRUE UNION ISOLATION BALL VALVE.
3. 4" CLEAR PVC STAND PIPE FOR WATER LEVEL VISUALIZATION IN FINISHED WATER STORAGE TANK. CONTRACTOR TO FIELD MARK GRADUATED LEVELS ON STANDPIPE EVERY 500 GALLONS. CONTRACTOR TO CALL OUT NOTABLE VOLUMES, OPERATING LEVEL, HWL, AND LWL.
4. 4" PVC PRESS-FIT GOOSENECK WITH AFFIXED #24-MESH STAINLESS STEEL SCREEN.
5. 3" PVC PRESS-FIT GOOSENECK WITH AFFIXED #24-MESH STAINLESS STEEL SCREEN.

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D309	TITLE OF SHEET RSPH PROCESS SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 70 OF 113

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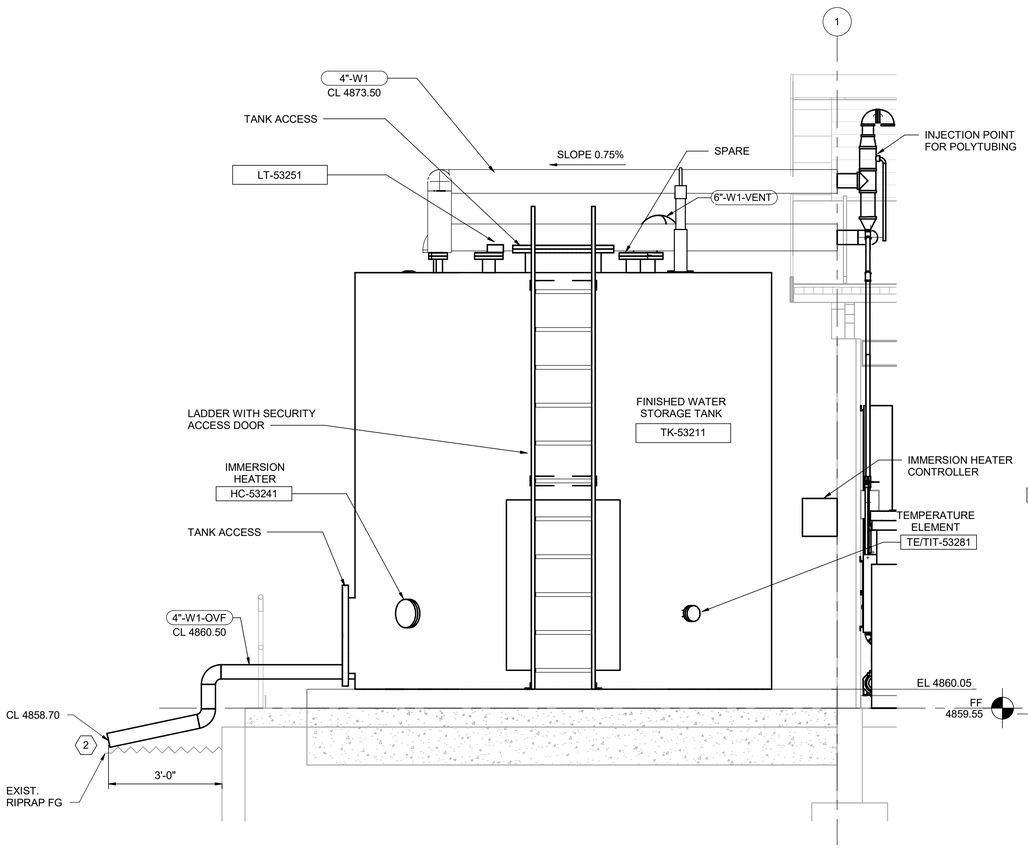
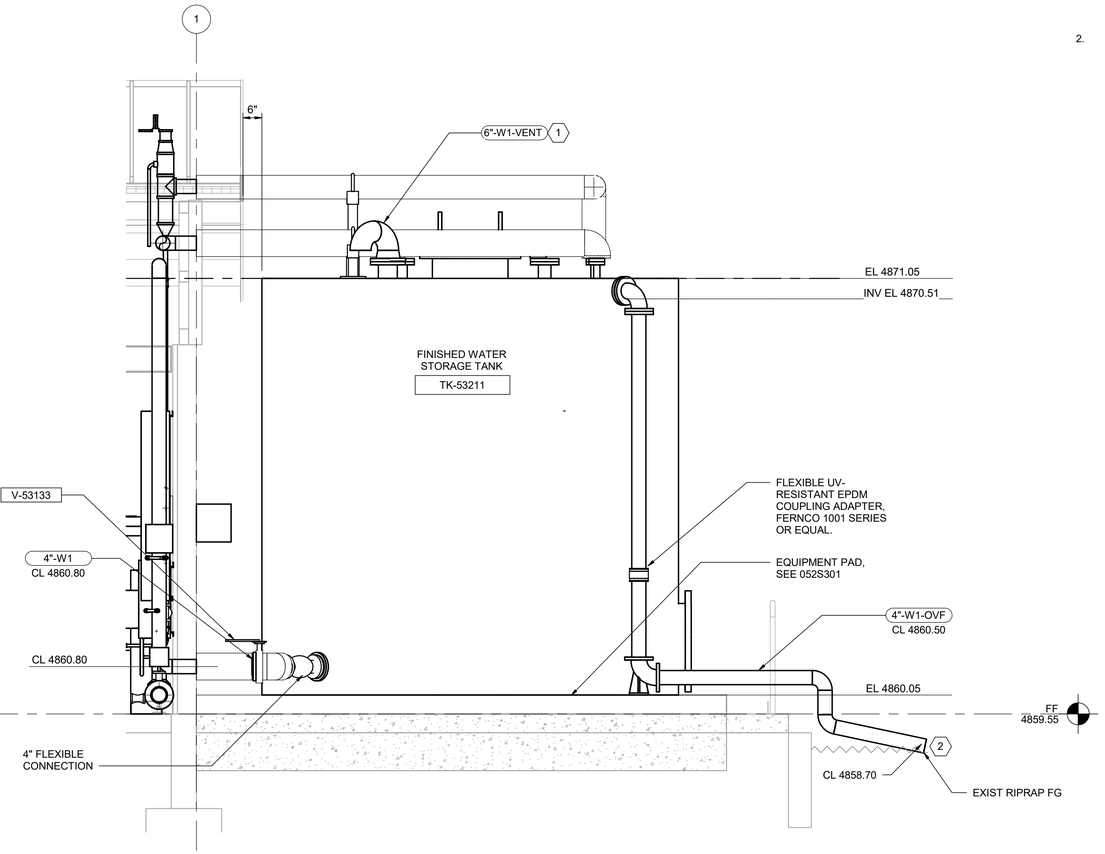
KEYNOTES (X)

1. PROVIDE INSECT SCREEN AND FURNISH IN ACCORDANCE WITH SPECIFICATION SECTION 43 41 45.
2. FINISHED WATER STORAGE TANK OVERFLOW INSTALL #24 MESH STAINLESS STEEL SCREEN AT TERMINATION OF OVERFLOW PIPE. CONTRACTOR TO COORDINATE LOCATION OF OVERFLOW PIPE TERMINATION WITH COR PRIOR TO INSTALLATION. OVERFLOW TERMINATION MUST BE LOCATED 12 TO 24 INCHES ABOVE GROUND LEVEL.



P 052D105 FINISHED WATER STORAGE TANK SECTION
SCALE (A)



Q 052D105 FINISHED WATER STORAGE TANK SECTION
SCALE (A)

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D310	TITLE OF SHEET RSPH PROCESS SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 71 OF 113

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KEY NOTES (X)

1 4" BUTTERFLY VALVE.

2 4"x2" ECCENTRIC REDUCER.

3 BRASS NIPPLE.

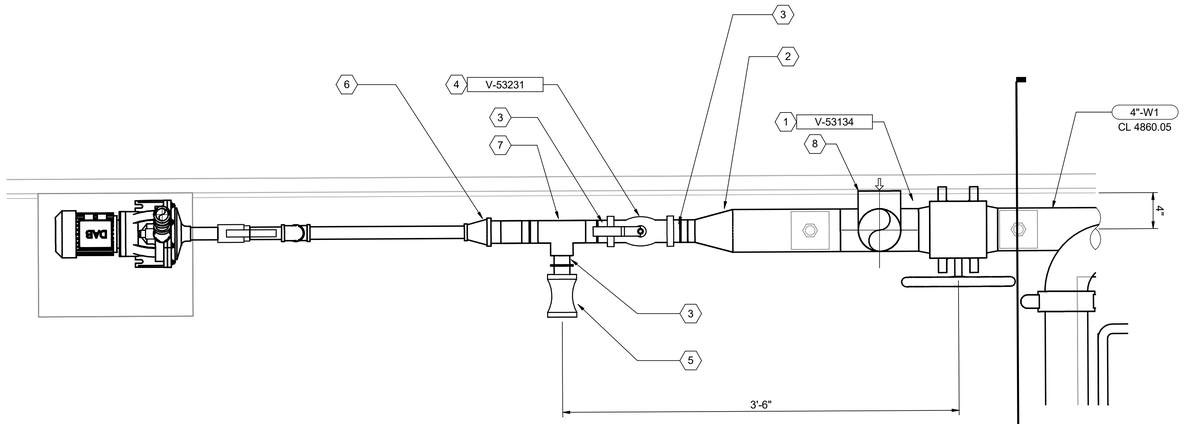
4 FNPT THREADED BRASS BALL VALVE.

5 MALE ADAPTER (MALE QUICK CONNECT). W/CAP OR PLUG

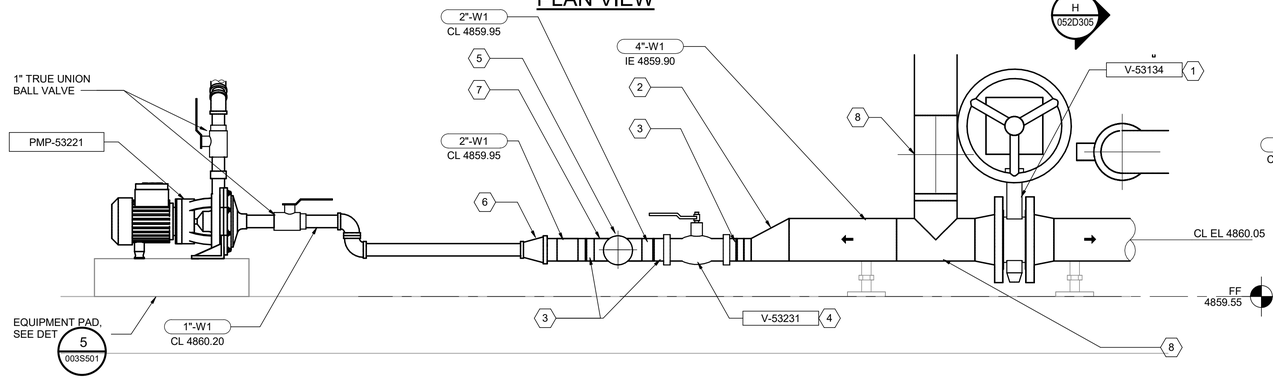
6 2"x1" REDUCER.

7 2"x2"x2" TEE.

8 4"x4"x4" TEE.

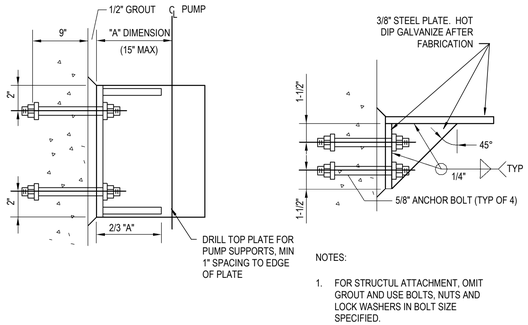


PLAN VIEW



SECTION VIEW

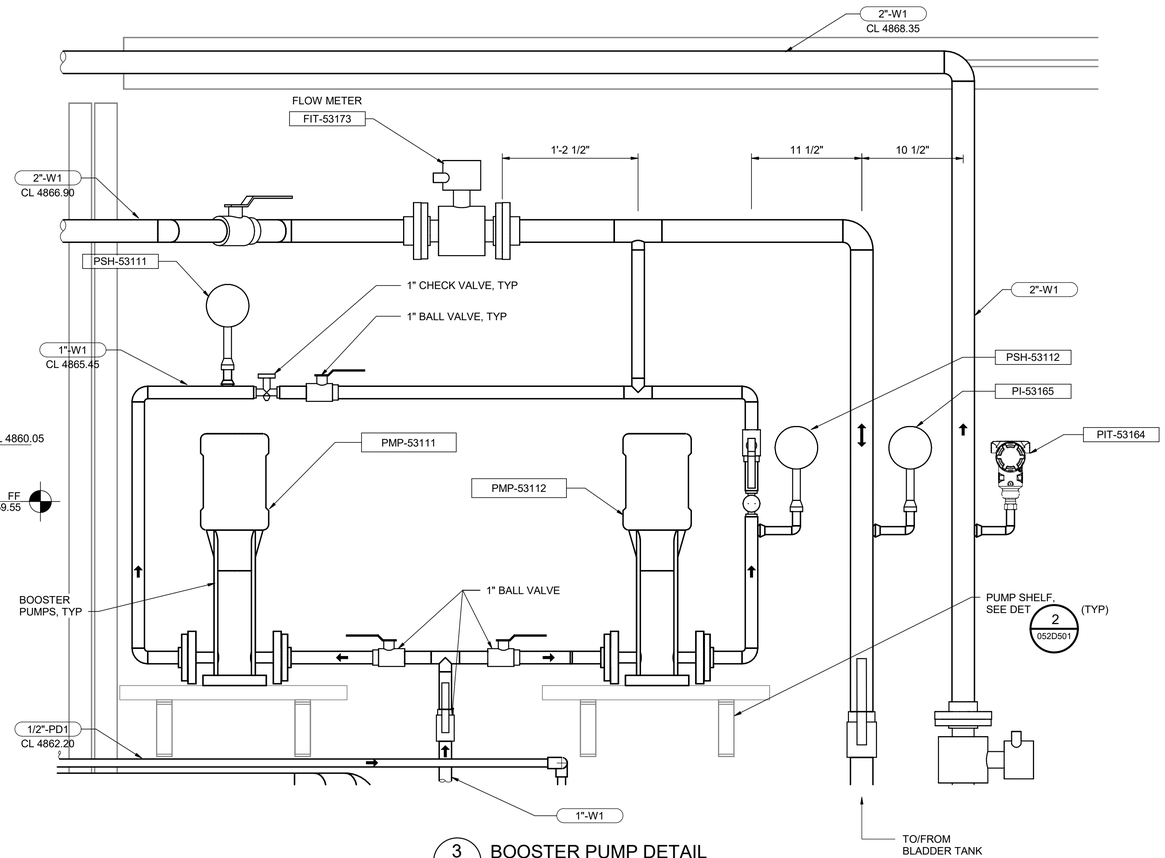
1 052D309 DRAIN CONNECTION - DETAIL
SCALE (A)



NOTES:

1. FOR STRUCTUL ATTACHMENT, OMIT GROUT AND USE BOLTS, NUTS AND LOCK WASHERS IN BOLT SIZE SPECIFIED.

2 052D501 BOOSTER PUMP SHELF DETAIL
NOT TO SCALE



3 052D304 BOOSTER PUMP DETAIL
SCALE (A)

SCALE (A) 12 6 0 12
SCALE OF FEET
1-1/2" = 1'-0"

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Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D501	TITLE OF SHEET RSPH PROCESS DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK
			DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 72 OF 113

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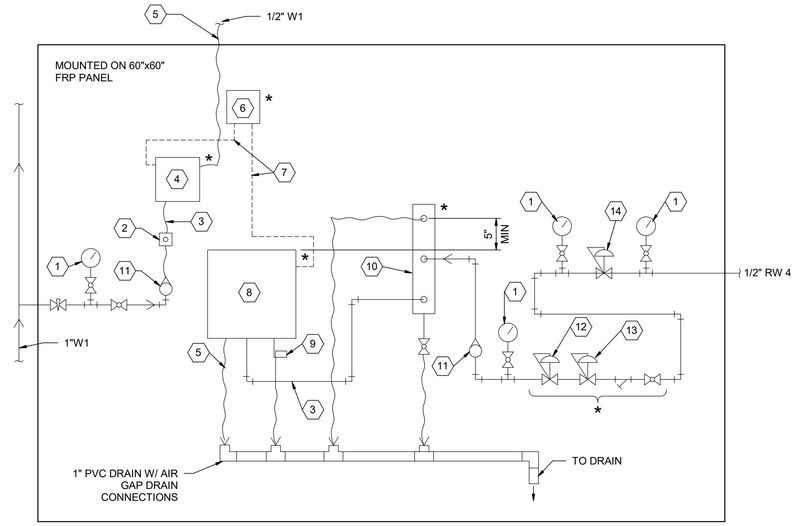


KEYNOTES

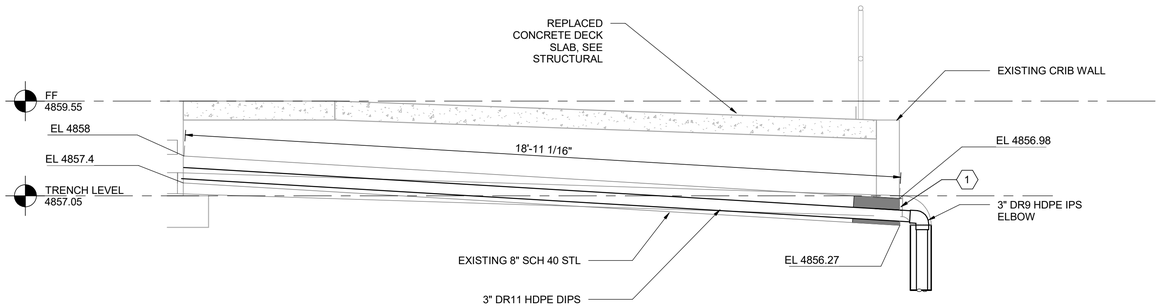
1. PRESSURE GAUGE
2. FLOW SENSOR
3. 1/4" OD POLYETHYLENE INLET TUBING
4. RESIDUAL CHLORINE ANALYZER WITH PH/TEMP PROBE
5. 1/4" OD POLYETHYLENE OUTLET TUBING
6. CONTROLLER
7. MANUFACTURER'S CABLE
8. TURBIDMETER
9. DRAIN TRAP
10. BUBBLE TRAP
11. ROTAMETER
12. ADJUSTABLE PRESSURE REGULATOR
13. RESET PRESURE REGULATOR
14. PRESSURE REDUCING VALVE
15. ROUTE TO RECIRC PUMP DISCHARGE

GENERAL NOTES

1. (*) DENOTES EQUIPMENT AND APPURTENANCES PROVIDED BY PH/CHLORINE/NTU ANALYZER MFR AS PART OF SYSTEM PACKAGE.

1 - INSTRUMENTATION PANEL

NOT TO SCALE

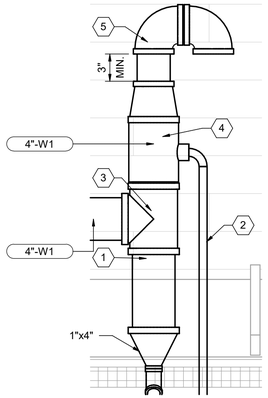


KEYNOTES

1. FILL MIN 8" DEPTH OF INTERSTITIAL SPACE WITH PIPE INSULATION. CAULK WITH SILICONE SEALANT.

2 - SLIPLINING FW PIPELINE TO TCWL

SCALE (A)



KEYNOTES

1. 1" W1 FROM FINISHED WATER RECIRCULATION PUMP.
2. 1/4" POLY TUBING FROM INSTRUMENTATION PANEL.
3. RECIRCULATION PIPE FLOW TRANSITION TO GRAVITY FLOW TO FINISHED WATER STORAGE TANK. SEE SHEET 052D310.
4. REDUCING TEE FOR POLY TUBING.
5. 1" GOOSENECK VENT WITH AFFIXED #24-MESH STAINLESS STEEL MESH SCREEN.

3 - RECIRCULATION LINE AND POLY TUBING DETAIL

NOT TO SCALE

SCALE (A) 2 0 2 4 1/2" = 1'-0"

SCALE OF FEET

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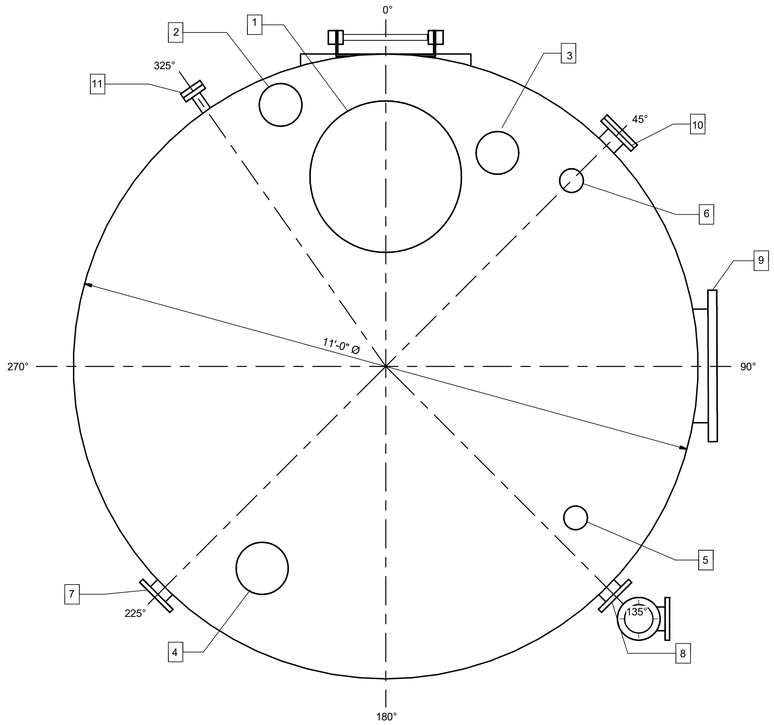


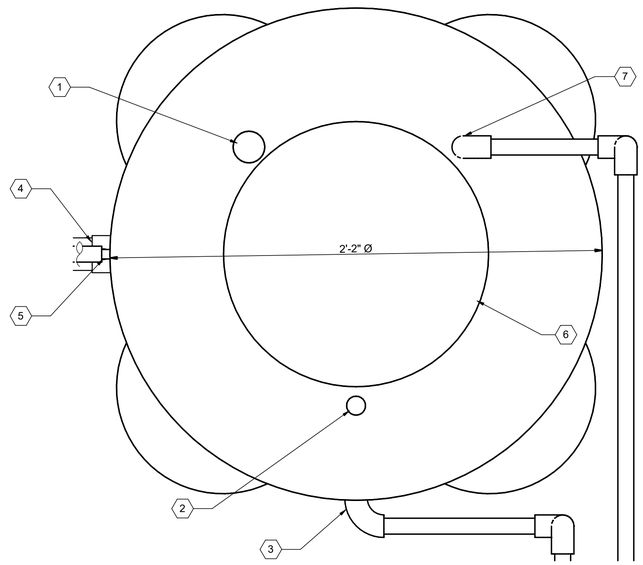
1 TK-53211 FINISHED WATER STORAGE TANK OPENING SCHEDULE

SCALE (A)

ITEM	DESCRIPTION	VERTICAL LOCATION ON TANK	RADIAL DISTANCE FROM CENTER OF TANK TO CENTERLINE OF ITEM (IF LOCATED ON TOP OF TANK)	COMMENT
1	24" MANWAY	TOP OF TANK	3'-6"	BLIND FLANGE
2	4" FLANGE	TOP OF TANK	5'-0"	SPARE
3	4" FLANGE	TOP OF TANK	4'-3"	LEVEL INDICATION
4	6" FLANGE	TOP OF TANK	4'-3"	VENT
5	2" FLANGE	TOP OF TANK	4'-3"	INLET
6	3" FLANGE	TOP OF TANK	4'-7"	RECIRCULATION
7	4" FLANGE	10'-3" (NOTE 1)	N/A	OUTLET
8	4" FLANGE	6" (NOTE 1)	N/A	OVERFLOW
9	24" MANWAY	9'-8" (NOTE 1)	N/A	BLIND FLANGE
10	4" FLANGE	9'-0" (NOTE 1)	N/A	IMMERSION HEATER
11	2" NPT FITTING	9'-0" (NOTE 1)	N/A	TEMPERATURE ELEMENT

NOTES:

1. DISTANCE FROM TOP OF TANK SIDEWALL TO CENTERLINE OF FLANGE.



3 (052D503) TK-53311 DILUTE SODIUM HYPOCHLORITE TANK

SCALE (B)

ITEM	DESCRIPTION	VERTICAL LOCATION ON TANK	HORIZONTAL DISTANCE FROM CENTER OF TANK TO CENTERLINE OF ITEM (IF LOCATED ON TOP OF TANK)	COMMENT
1	1/2" NPT FITTING	TOP OF TANK	7"	INLET
2	NPT BULKHEAD	TOP OF TANK	7"	LEVEL- SIZE FOR INSTRUMENT
3	1/2" NPT FITTING	2" ABOVE BOTTOM OF SIDEWALL	N/A	OUTLET
4	2" FITTING	BOTTOM OF TANK	N/A	DRAIN
5	2" NPT FITTING	2" BELOW TOP OF SIDEWALL	N/A	OVERFLOW
6	14" VENTED CENTER ACCESS	TOP OF TANK	N/A	VENT
7	1/2" NPT FITTING	TOP OF TANK	7"	DILUTE PUMP PRESSURE RELIEF

SCALE (B) 6 3 0 6 — 3" = 1'-0"
SCALE OF FEET

SCALE (A) 1 0 1 2 3 — 3/4" = 1'-0"
SCALE OF FEET

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RAW WATER			
INLET/OUTLET PRESSURE (PSI) AND FLOW (GPM) SETTING TABLE			
DESIGNATION	INLET/OUTLET	NORMAL FLOW	PEAK FLOW
AFV-52111	145/135	150	270
AFV-52112	145/135	150	270
ARV-52331	135/135	150	175
CKV-52331	1,700/1,700	150	175
CKV-52332	1,700/1,700	150	175
CKV-52333	1,700/1,700	150	175
EDS-52312	1,420/20	110	140
FV-52331	150/150	65	65
V-52331	1/0	65	65
V-52131	145/145	150	175
V-52132	145/145	150	175
V-52133	135/135	150	175
V-52134	135/135	150	175
V-52231	135/135	150	175
V-52232	135/135	150	175
V-52233	1,700/1,700	150	175
V-52234	1,700/1,700	150	175
V-52031	1,700/1,700	150	175
V-52032	150/1,700	150	175
V-52033	1,420/1,420	110	140

FINISHED WATER			
INLET/OUTLET PRESSURE (PSI) AND FLOW (GPM) SETTING TABLE			
DESIGNATION	INLET/OUTLET	NORMAL FLOW	PEAK FLOW
EDS-53117	1,510/10	40	80
V-53131	1,510/1,510	40	80
V-53133	4.5/4.5	20	132
V-53134	4.5/4.5	20	132
V-53231	4.5/4.5	5	10
V-53232	5.3/5.3	5	10
V-53135	5.2/5.2	5	10

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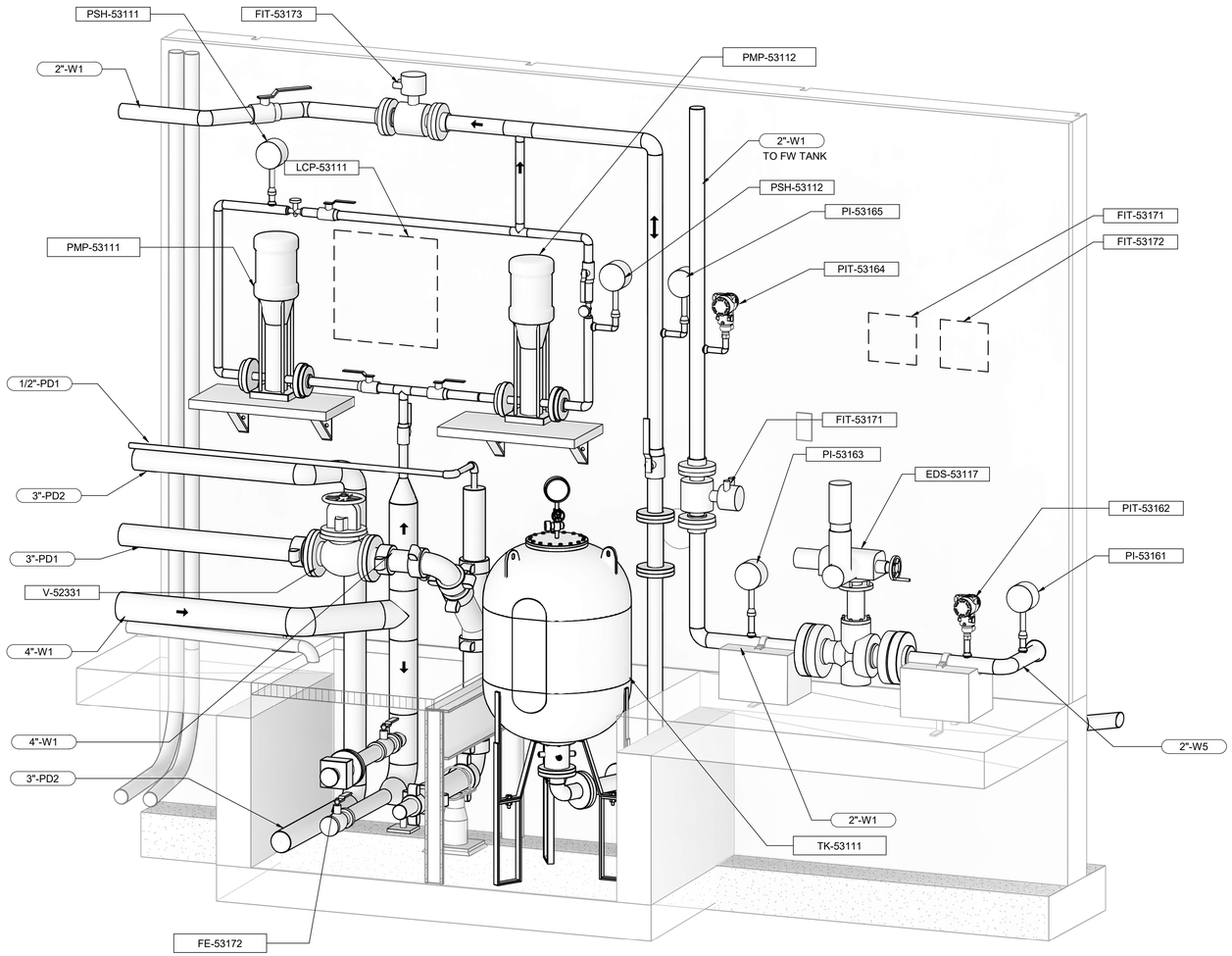

ISOMETRIC VIEW-BLADDER TANK AND BOOSTER PUMP

SCALE Ⓐ

SCALE Ⓐ 1 0 1 2 1" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

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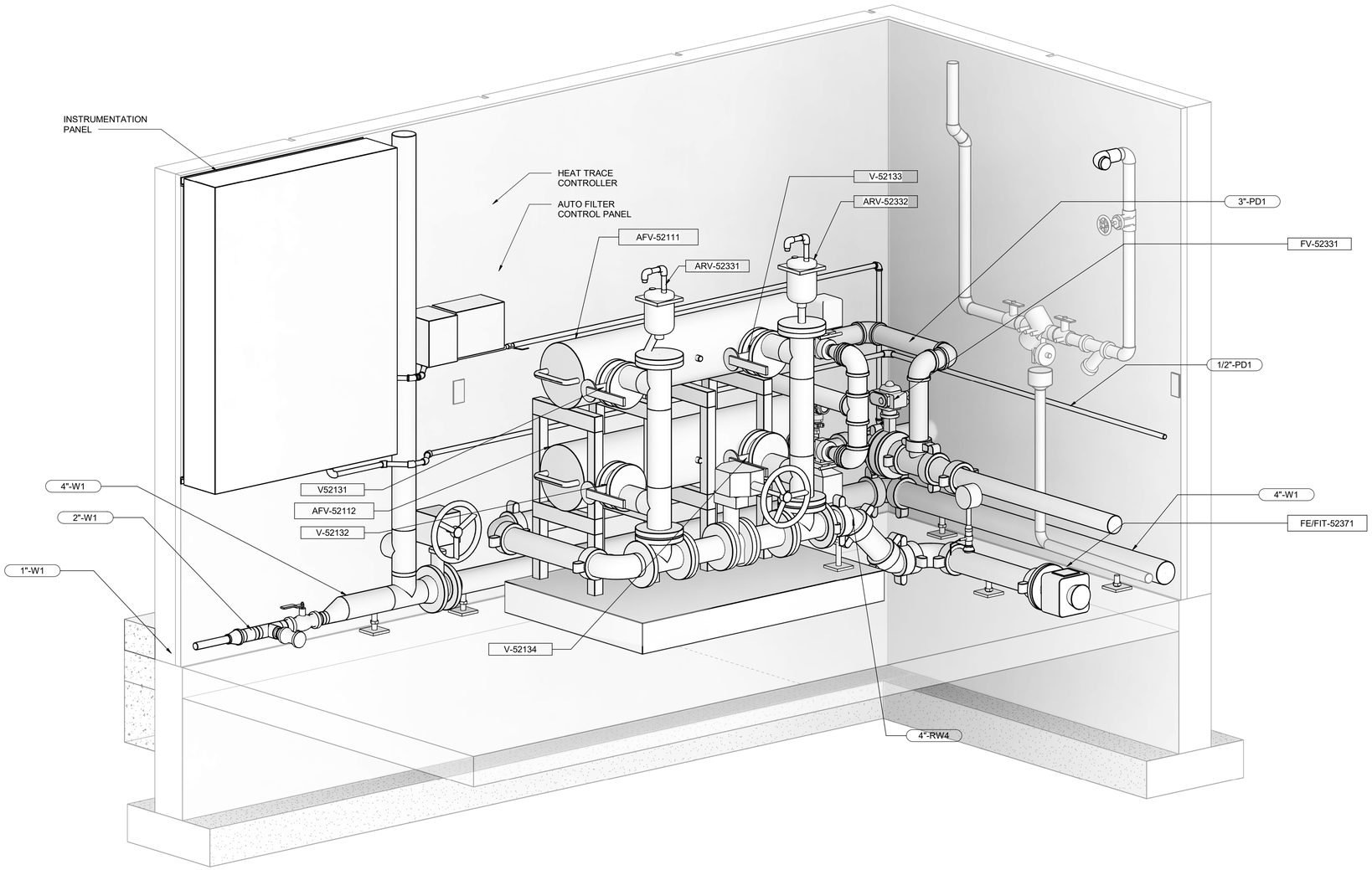

ISOMETRIC VIEW-AUTO FILTERS

SCALE (A)

SCALE (A) 2 0 2 4 1/2" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

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DRAWING NO. 113 / 182752

PMIS/PKG NO. 318719

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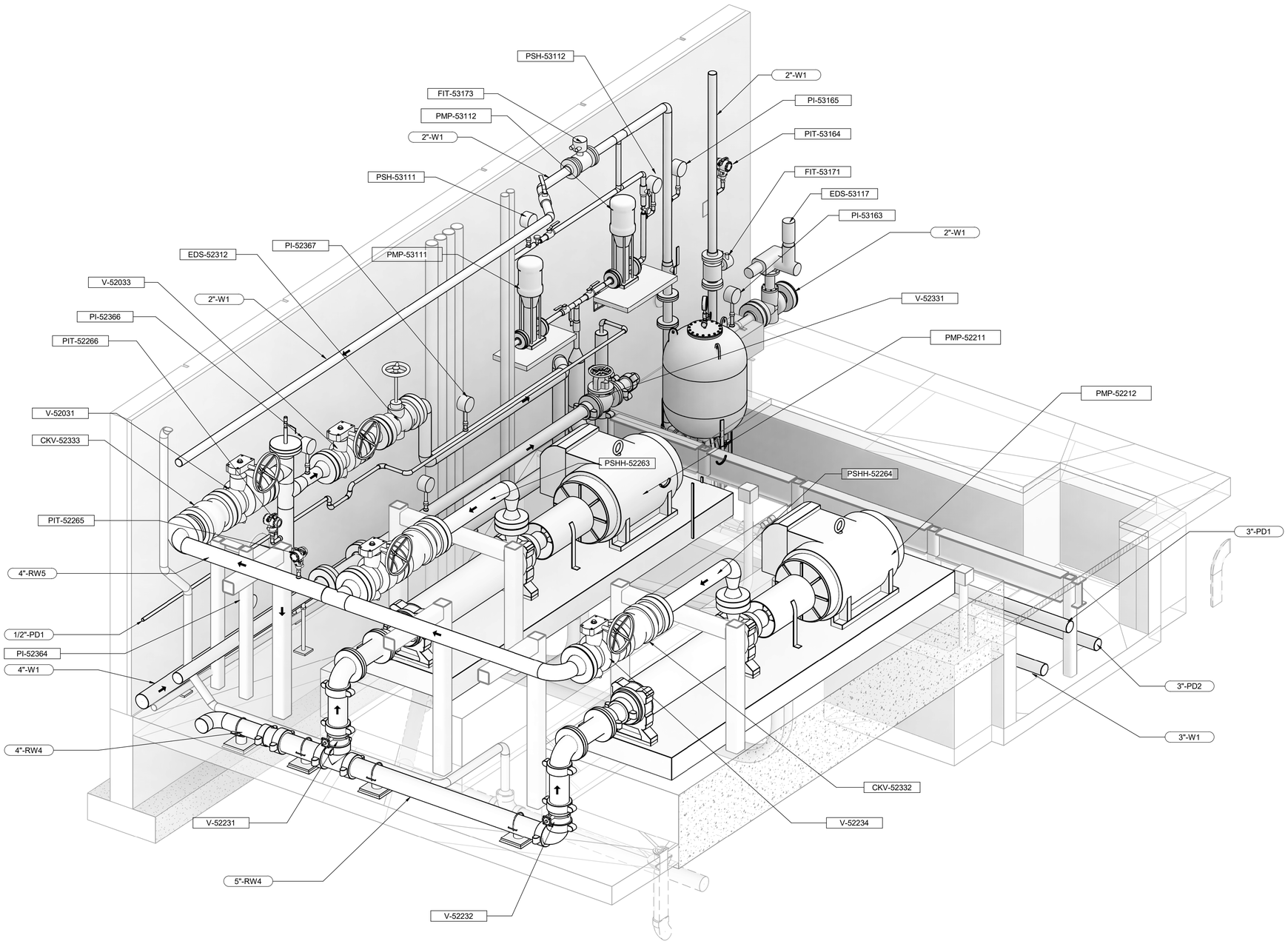

ISOMETRIC VIEW-RAW WATER PUMPS

SCALE (A)

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

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